2026 | A2预估考

WorkBook

Circular Motion · Gravitational Fields · Oscillations

slides · formulae & definitions · paper4 2024–2026

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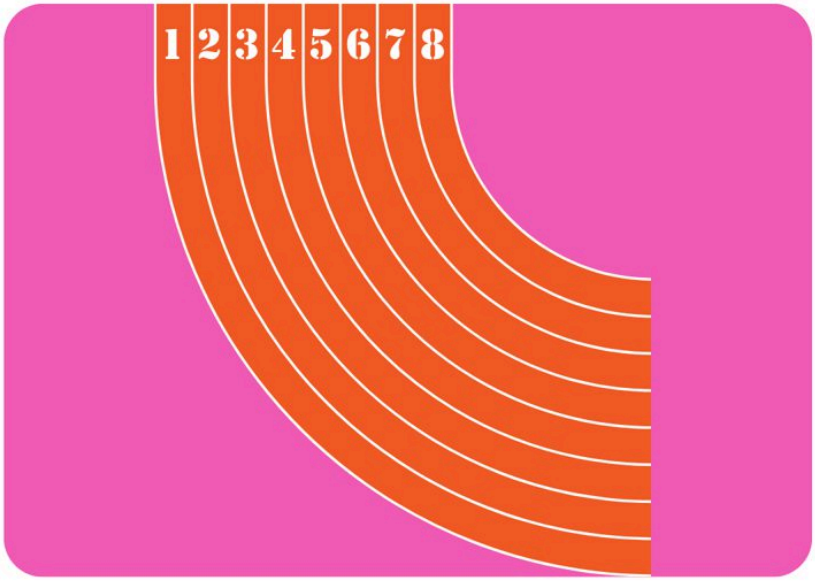
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Chapter 16

Circular Motion



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A Level Physics
Prepared and taught by James Gan

Chapter Outline

16.1 Understanding Circular Motion
16.2 Measuring Angles in Radians
16.3 Constant Speed with Changing Velocity
16.4 Angular Speed
16.5 Centripetal Forces
16.6 Determining Acceleration and Force
16.7 More on Centripetal Forces



16.1 Understanding Circular Motion

Examples of object moving in a circle



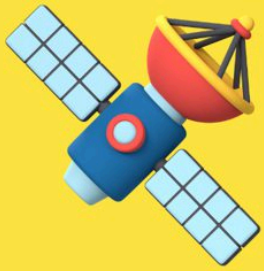
A car moving along a roundabout



The Moon orbiting the Earth



A ferris wheel at an amusement park

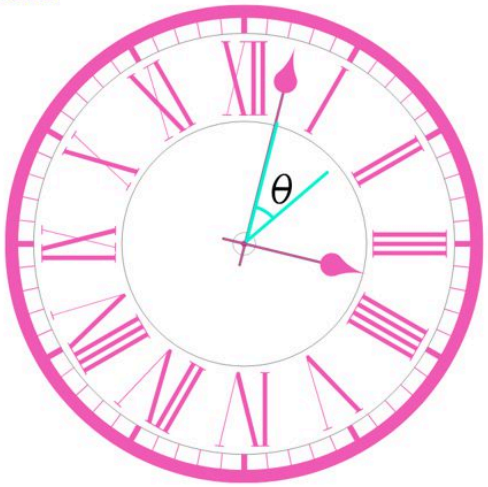


A satellite orbiting a planet

16.1 Understanding Circular Motion

Movement of the Second Hand of a Clock

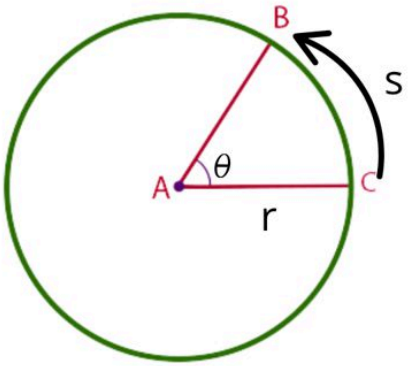
- The second hand completes 1 cycle in 60s.
- 1 cycle = 360°, so the hand moves 6°/s.
- This example shows how we can measure the motion of an object in a circle using **angles**.
- This concept leads to measuring the amount of rotation using **angular displacement**.



16.2 Measuring angles in radians

Angular Displacement

Angular displacement is the **angle** through which an object moves in a circle, measured in radians.

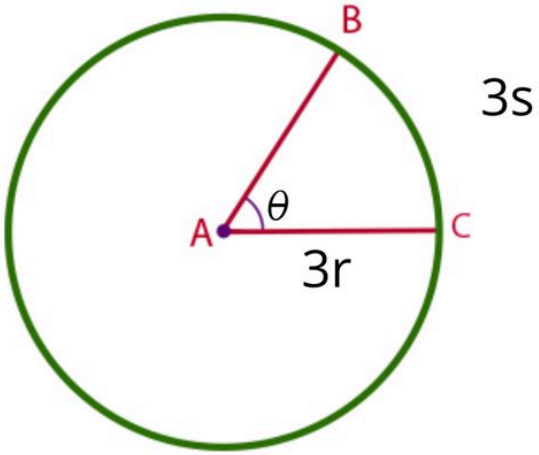


$$\theta = \frac{s}{r}$$

16.2 Measuring angles in radians

Angular Displacement

Angular displacement is the **angle** through which an object moves in a circle, measured in radians.



$$\theta = \frac{3s}{3r} = \frac{s}{r}$$

Angular displacement is simply a **ratio** of the arc length traveled to the radius of the circle.

16.2 Measuring angles in radians

Radians to Degree & v.v

Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.

Formula: $\theta = \frac{s}{r}$

The angular displacement for a full circle = $2\pi r$

A complete circle contains:

$$\frac{2\pi r}{r} = 2\pi \text{ rad}$$

16.2 Measuring angles in radians

Radians to Degree & v.v

Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.

Formula: $\theta = \frac{s}{r}$

The angular displacement for a full circle = $2\pi r$

A complete circle contains:

$$\frac{2\pi r}{r} = 2\pi \text{ rad}$$

2π rad = 360°

π rad = 180°

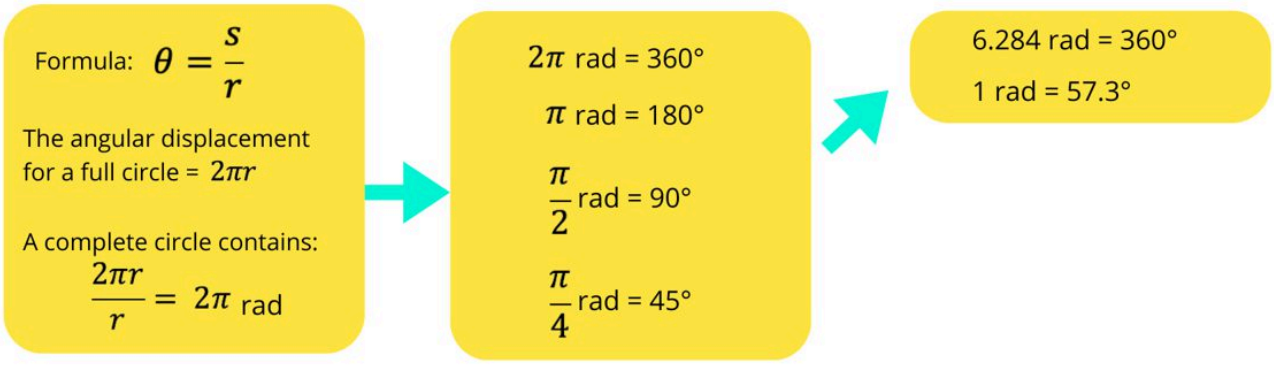
$\frac{\pi}{2}$ rad = 90°

$\frac{\pi}{4}$ rad = 45°

16.2 Measuring angles in radians

Radians to Degree & v.v

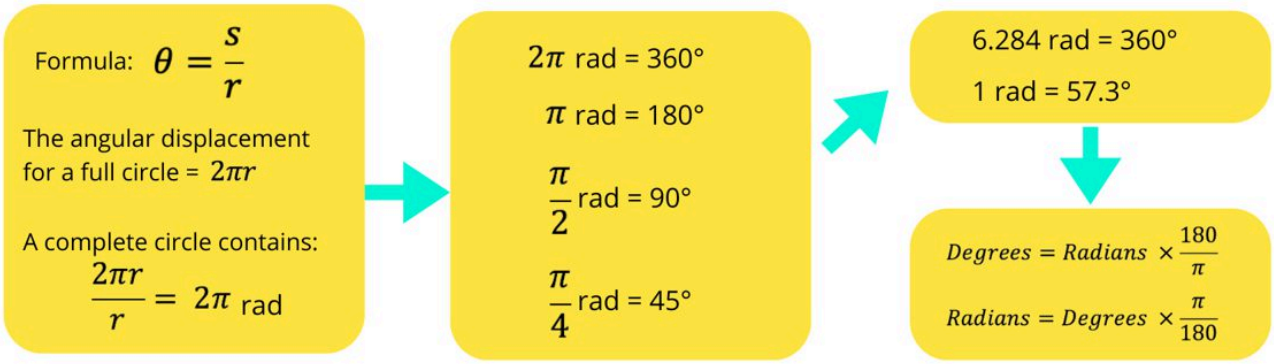
Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.



16.2 Measuring angles in radians

Radians to Degree & v.v

Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.



16.2 Measuring angles in radians

Worked Example

Convert 135° to radian.

16.2 Measuring angles in radians

Worked Example

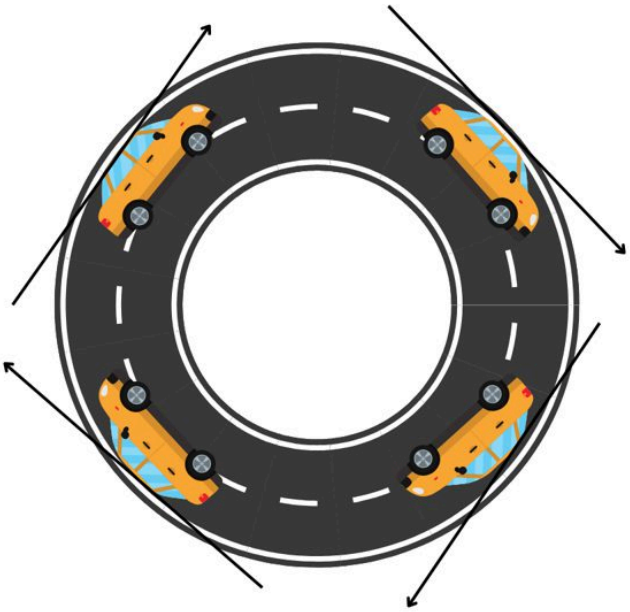
Convert 135° to radian.

$$135° \times \frac{\pi}{180} = \frac{3\pi}{4}\ radians$$

16.3 Constant Speed with Changing Velocity

Key Concept

When an object moves in a circle at constant speed, **its velocity changes** because velocity is a vector quantity that depends on both magnitude (speed) and **direction**.

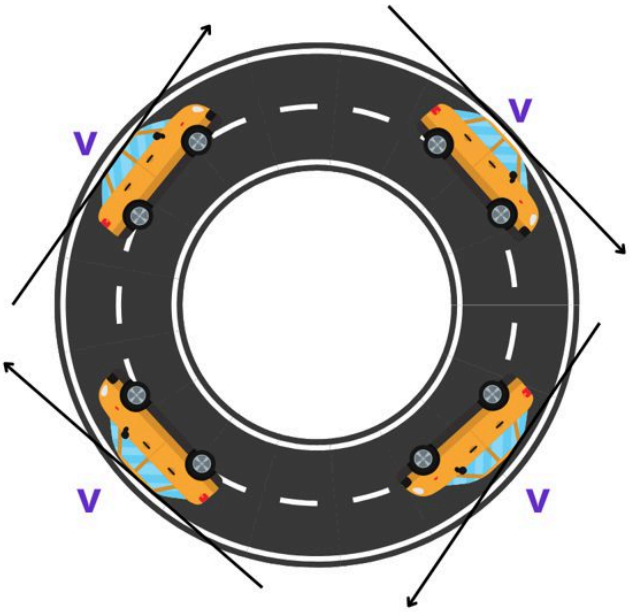


16.3 Constant Speed with Changing Velocity

Why does velocity change?

In circular motion, the direction of motion is always **changing**.

The velocity vector is always **tangential** to the circle at any given point.



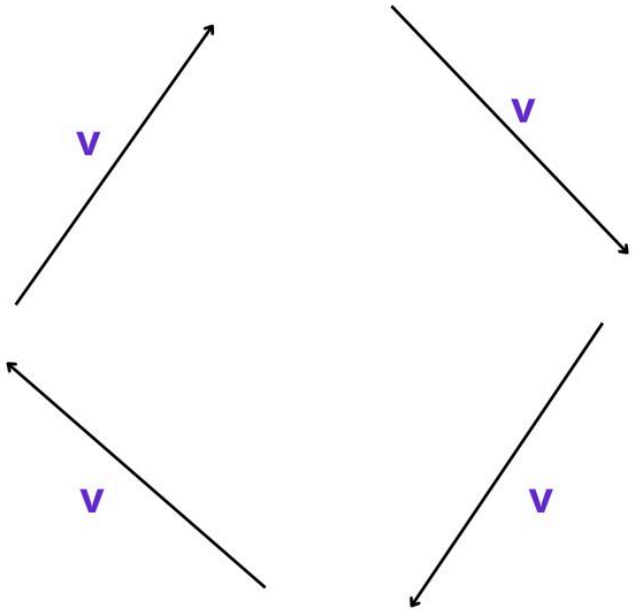
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16.3 Constant Speed with Changing Velocity

Why does velocity change?

The tangent is a straight line that touches the circle at a single point and is perpendicular to the radius.

As the object moves along the circle, **the direction of this tangent—and thus the velocity—keeps changing.**



16.4 Angular Speed

DEFINITION

Angular speed is the rate at which an object moves through an angle in circular motion. Angular speed tells us how quickly an object rotates or revolves, regardless of its position in the circle.

FORMULA RECAP

$\theta = \frac{s}{r}$ *Angular Displacment (unit = radian)*

$s = \frac{d}{t}$ *Speed*

FORMULA

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Unit: Radians per second

16.4 Angular Speed

FORMULA

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Unit: Radians per second

SINGLE REVOLUTION

$$\omega = \frac{2\pi}{t}$$

16.4 Angular Speed

Worked Example

A Pirelli wheel on an F1 car completes one full revolution in 0.5 seconds.
What is its angular speed?



16.4 Angular Speed

Worked Example

A Pirelli wheel on an F1 car completes one full revolution in 0.5 seconds.
What is its angular speed?

$$\omega = \frac{2\pi}{0.5} = 4\pi \; rad/s$$



16.4 Angular Speed

Types of Speed (not in syllabus, just for clarification)

Linear speed (often simply called speed) refers to **how fast an object moves along its circular path**. It is measured in meters per second (m/s).



VS

Angular speed **(ω)** refers to **how fast an object is rotating** (in radians per second).

While angular speed tells you how much angle the object has rotated in a certain time, linear speed tells you how far the point has moved along the circular path in that same time.

16.4 Angular Speed

Relating Angular Speed and Radius to Speed

As an object moves in a circle, the linear speed of a point on the object depends on both the **angular speed (ω)** and the **radius (r)** of the circle.



KEY CONCEPT

The further a point is from the center of the circle, the faster its velocity, even if the angular speed remains constant.

16.4 Angular Speed

Relating Angular Speed and Radius to Speed

As an object moves in a circle, the linear speed of a point on the object depends on both the **angular speed (ω)** and the **radius (r)** of the circle.

FORMULA

$$v = \omega r$$

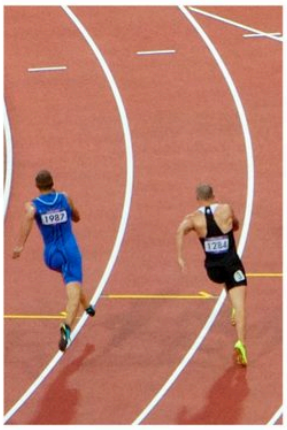
16.4 Angular Speed

Worked Example

Two runners are on a circular track. **One runner is in the inner lane, which has a radius of 50 m**, and **the other runner is in the outer lane, which has a radius of 60 m.** Both runners complete 1 full lap in 60 seconds. Calculate the linear speed for both runners.

FORMULA

$$v = \omega r$$



16.4 Angular Speed

Worked Example

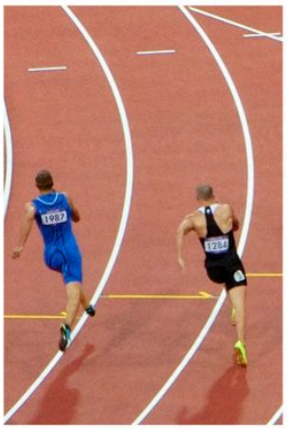
Two runners are on a circular track. **One runner is in the inner lane, which has a radius of 50 m**, and **the other runner is in the outer lane, which has a radius of 60 m.** Both runners complete 1 full lap in 60 seconds. Calculate the linear speed for both runners.

FORMULA

$$v = \omega r$$

CALCULATING ANGULAR SPEED

$$\omega = \frac{2\pi}{60} = \frac{\pi}{30}\ rad/s$$



16.4 Angular Speed

Worked Example

Two runners are on a circular track. **One runner is in the inner lane, which has a radius of 50 m**, and **the other runner is in the outer lane, which has a radius of 60 m.** Both runners complete 1 full lap in 60 seconds. Calculate the linear speed for both runners.

FORMULA

$$v = \omega r$$

CALCULATING ANGULAR SPEED

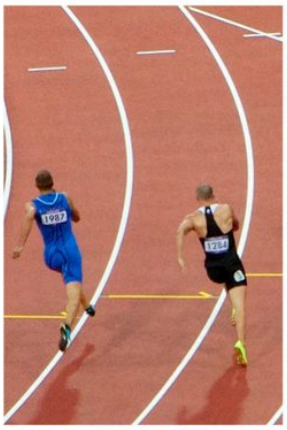
$$\omega = \frac{2\pi}{60} = \frac{\pi}{30}\ rad/s$$

INNER LANE:

$$v = 50 \times \frac{\pi}{30} = 5.24\ m/s$$

OUTER LANE:

$$v = 60 \times \frac{\pi}{30} = 6.28\ m/s$$



16.5 Centripetal Forces

Centripetal Force in Circular Motion

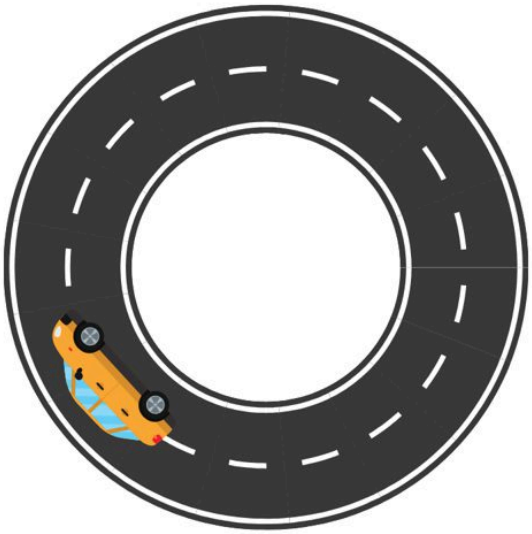
Speed is constant, but velocity changes due to direction change.

Recap

DEFINITION



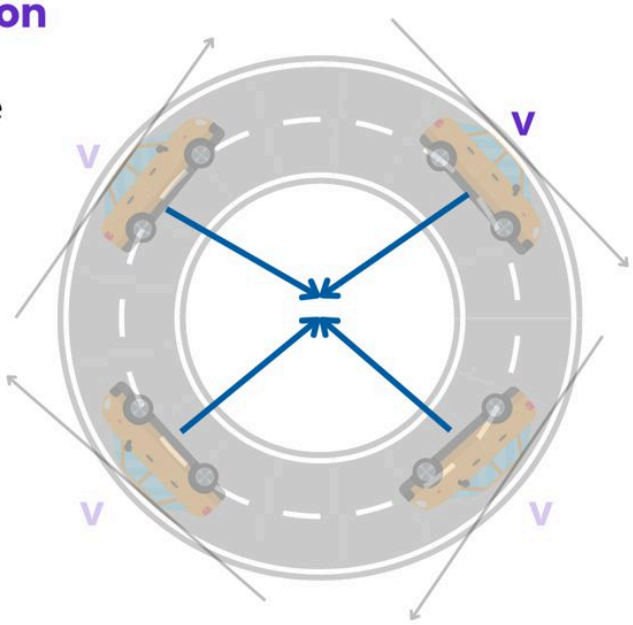
- An object at rest will **stay at rest** unless acted upon by an external force.
- An object in motion will **continue moving at a constant velocity** unless acted upon by an external force.



16.5 Centripetal Forces

Centripetal Force in Circular Motion

- This force is called centripetal force.
- It is always directed towards the center of the circle.
- Without centripetal force, the object would move in a straight line.



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16.5 Centripetal Forces

More examples



Fidget Spinner



Ferris Wheel

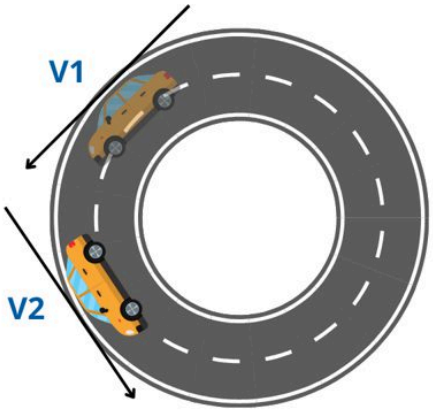
Solar System

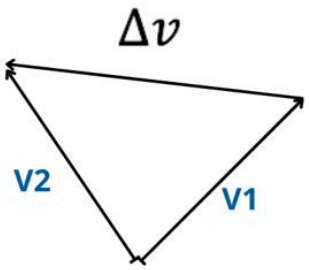


16.5 Centripetal Forces

Introduction to Acceleration in Circular Motion

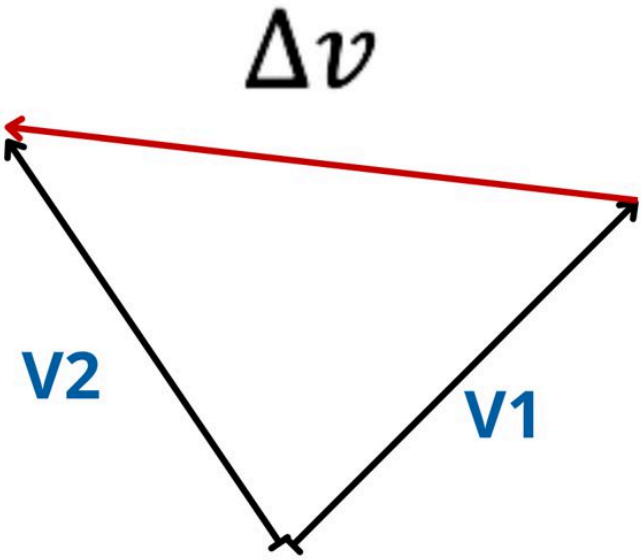
- In circular motion, velocity changes direction constantly.
- Acceleration is the rate of change of velocity.
- The change in velocity can be analyzed using a vector diagram.





16.5 Centripetal Forces

Vector Diagram and Direction of Acceleration



The **acceleration** is directed **towards the center of the circle.**

Newton's Second Law

DEFINITION



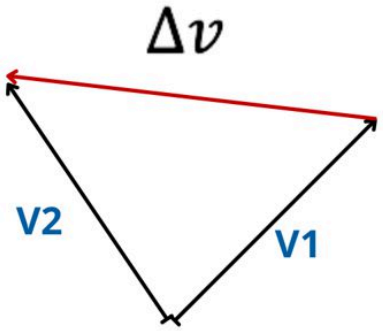
The **acceleration** of an object is
- **directly proportional** to the **net force** acting on it

Since acceleration points towards the center, **the force is also directed towards the center (centripetal force)**.

16.5 Centripetal Forces

Why Speed Remains Constant

- The speed of the object remains constant because the **force is perpendicular to the velocity.**
- The force doesn't cause any change in the object's speed, **only its direction**.
- All the force is used to **keep the object moving in a circle**, not to change its speed.



16.6 Determining Acceleration and Force

DEFINITION

Centripetal Acceleration is the acceleration that keeps an object moving in a circular path, directed towards the center of the circle.

FORMULA

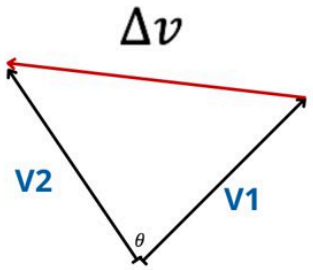
$$a = \frac{v^2}{r}$$

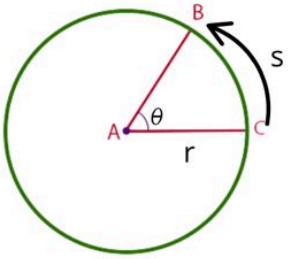
$$a = \omega^2 r$$

16.6 Determining Acceleration and Force

$v = \omega r$ $\omega = \frac{\Delta\theta}{\Delta t}$

Deriving the formula





$\theta = \frac{s}{r}$

$\Delta\theta = \frac{\Delta v}{v}$

$\frac{v\Delta\theta}{\Delta t} = \frac{\Delta v}{\Delta t}$

$\frac{v\Delta\theta}{\Delta t} = a$

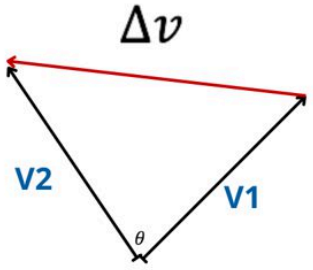
$v \cdot \frac{\Delta\theta}{\Delta t} = a$

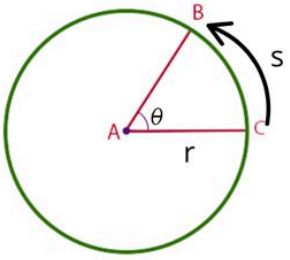
$v \cdot \omega = a$

16.6 Determining Acceleration and Force

$v = \omega r$ $\omega = \frac{\Delta\theta}{\Delta t}$

Deriving the formula





$\theta = \frac{s}{r}$

$\Delta\theta = \frac{\Delta v}{v}$

$\frac{v\Delta\theta}{\Delta t} = \frac{\Delta v}{\Delta t}$

$\frac{v\Delta\theta}{\Delta t} = a$

$v \cdot \frac{\Delta\theta}{\Delta t} = a$

$v \cdot \omega = a$

$v \cdot \left(\frac{v}{r}\right) = a$

$a = \frac{v^2}{r}$

$wr \cdot \omega = a$

$a = \omega^2 r$

16.6 Determining Acceleration and Force

Deducing a formula for centripetal force

NEWTON 2ND LAW

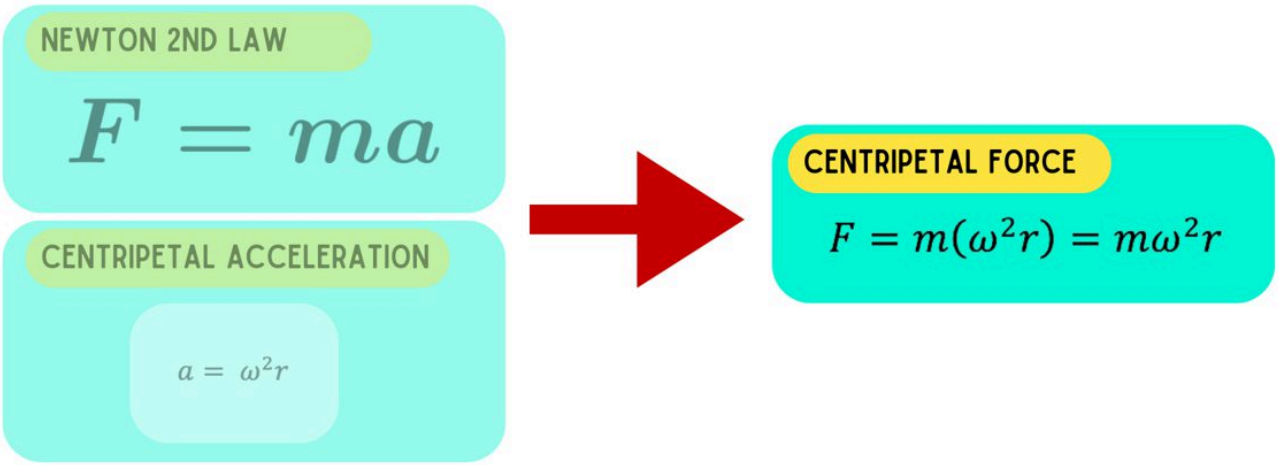
$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \omega^2 r$$

16.6 Determining Acceleration and Force

Deducing a formula for centripetal force



16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

1. Calculate the speed of the car.

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

1. Calculate the speed of the car.

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{5 \times 2\pi}{120} = 0.262\ rad/s$$

$$v = \omega r = 0.262\ (50) = 13.1\ m/s$$

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

2. Calculate the centripetal acceleration

© James Gan

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

2. Calculate the centripetal acceleration

$$a = \frac{v^2}{r} = \frac{13.1^2}{50} = 3.43\ m/s^2$$

© James Gan

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

3. Calculate the centripetal force

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

3. Calculate the centripetal force

$$F = ma = 800\ (3.43) = 2744N$$

16.6 Determining Acceleration and Force

Finding the orbital speed

NEWTON 2ND LAW

$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r}$$

EARTH GRAVITATIONAL PULL

$$w = mg$$

16.6 Determining Acceleration and Force

Finding the orbital speed

NEWTON 2ND LAW

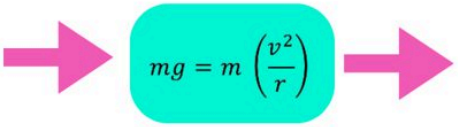
$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r}$$

EARTH GRAVITATIONAL PULL

$$w = mg$$



$$mg = m\left(\frac{v^2}{r}\right)$$

16.6 Determining Acceleration and Force

Finding the orbital speed

NEWTON 2ND LAW

$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r}$$

EARTH GRAVITATIONAL PULL

$$w = mg$$

$$mg = m\left(\frac{v^2}{r}\right)$$

$$mg = \frac{mv^2}{r}$$

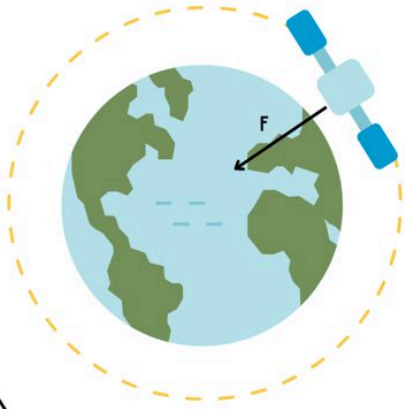
$$g = \frac{v^2}{r}$$

$$9.8 = \frac{v^2}{6400\,000}$$

$$v = 7920\ m/s$$

GRAVITATIONAL ACCELERATION

RADIUS OF THE EARTH IN METER



TO KEEP AN OBJECT IN ORBIT AROUND EARTH, IT MUST TRAVEL AT 8 KM/S. AT THIS SPEED, IT CONTINUOUSLY FALLS AROUND THE PLANET INSTEAD OF HITTING THE GROUND. AIR RESISTANCE AND GRAVITY PREVENT THROWING AN OBJECT THIS FAST, REQUIRING ROCKETS INSTEAD. 🚀

16.7 More on Centripetal Forces

Forces Acting on a Car Moving in a Circle - Vertical Force



These forces are balanced, so the car does not move vertically.

16.7 More on Centripetal Forces

Forces Acting on a Car Moving in a Circle - Horizontal Force



THE FRICTION BETWEEN THE TIRES AND THE ROAD PROVIDES THE CENTRIPETAL FORCE.

This force acts toward the center of the circular path, keeping the car in motion around the curve.

16.7 More on Centripetal Forces

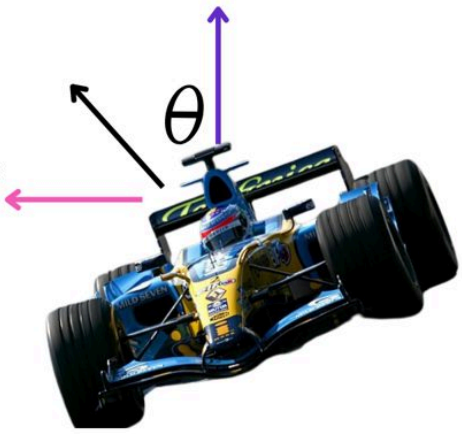
Car Cornering on a Banked Road

On a banked road, the normal force (N) is tilted.



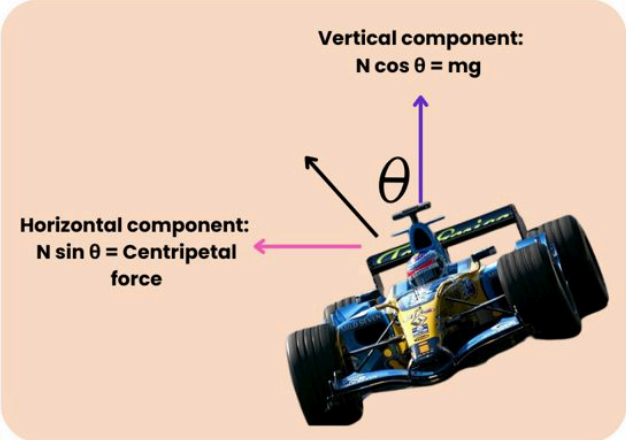
Vertical component:
N cos θ = mg

Horizontal component:
N sin θ = Centripetal force



16.7 More on Centripetal Forces

Key Takeaways



- Unlike a flat road where friction provides centripetal force, on a banked road, the tilted normal force plays a role.
- On a well-designed banking angle, **the car can turn without relying on friction**, useful for high-speed tracks.
- If the angle is too steep or the speed is too low, additional forces (friction or an engine force) may be needed.

16.7 More on Centripetal Forces

An Airplane Banking in the Sky



- When an airplane turns, it tilts its wings.
- The lift force from the wings has a **horizontal component that acts as centripetal force.**
- The vertical component still balances the plane's weight.

Formulae & Definitions — Topic 12 · Circular Motion

Radian 弧度 (rad) — the angle subtended at the centre of a circle by an arc of length equal to the radius. 360° = 2π rad, 1 rad ≈ 57.3°.

Angular displacement 角位移 θ — the angle swept out by an object with respect to the centre of rotation (rad).

Angular velocity 角速度 ω — the rate of change of angular displacement: $\omega = \Delta\theta/t = 2\pi/T$ (rad s^{-1}).

Linear velocity 线速度 v — the instantaneous (tangential) speed at a point on the path: $v = 2\pi r/T = r\theta/t = r\omega$. Its direction is tangential to the circle.

Uniform circular motion 匀速圆周运动 — constant speed but changing velocity (direction changes), so there is always an acceleration, a force and a change in momentum.

Centripetal acceleration 向心加速度 — $a = r\omega^2 = v^2/r$, directed towards the centre of the circle (m s^{-2}).

Centripetal force 向心力 — the resultant of all the real forces directed towards the centre: $F = mr\omega^2 = mv^2/r$ (N). It is at right angles to the velocity, so it changes direction but not speed.

点评 · Insight

重点：向心力不是一个具体性质的力（如重力、摩擦力），而是各具体力在指向圆心方向的合效果。解题关键是先分析向心力的来源——由哪些指向圆心的力（或分量）提供。

1 **(a)** Define the radian.

.....

..... [1]

(b) A circular metal disc spins horizontally about a vertical axis, as shown in Fig. 1.1.

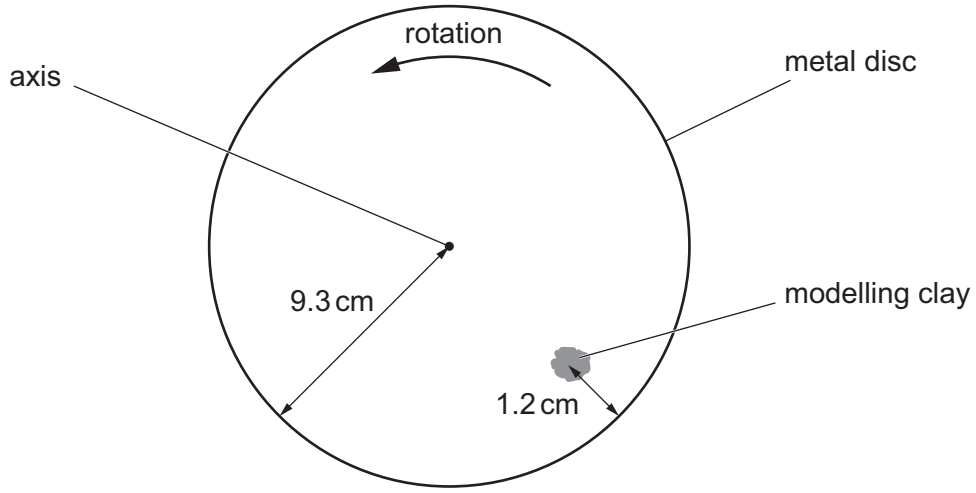


Fig. 1.1 (not to scale)

A piece of modelling clay is attached to the disc.

For the instant when the piece of modelling clay is in the position shown, draw on Fig. 1.1:

(i) an arrow, labelled V, showing the direction of the velocity of the modelling clay [1]

(ii) an arrow, labelled A, showing the direction of the acceleration of the modelling clay. [1]

(c) The metal disc in Fig. 1.1 has a radius of 9.3 cm.
The centre of gravity of the modelling clay is 1.2 cm from the rim of the disc and moves with a speed of 0.68 m s^{-1}.

(i) Calculate the angular speed ω of the disc.

ω = ... rad s^{-1} [2]

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(ii) Calculate the acceleration a of the centre of gravity of the modelling clay.

a = ... $m\,s^{-2}$ [2]

(d) A second piece of modelling clay is attached to the disc in the position shown in Fig. 1.2.

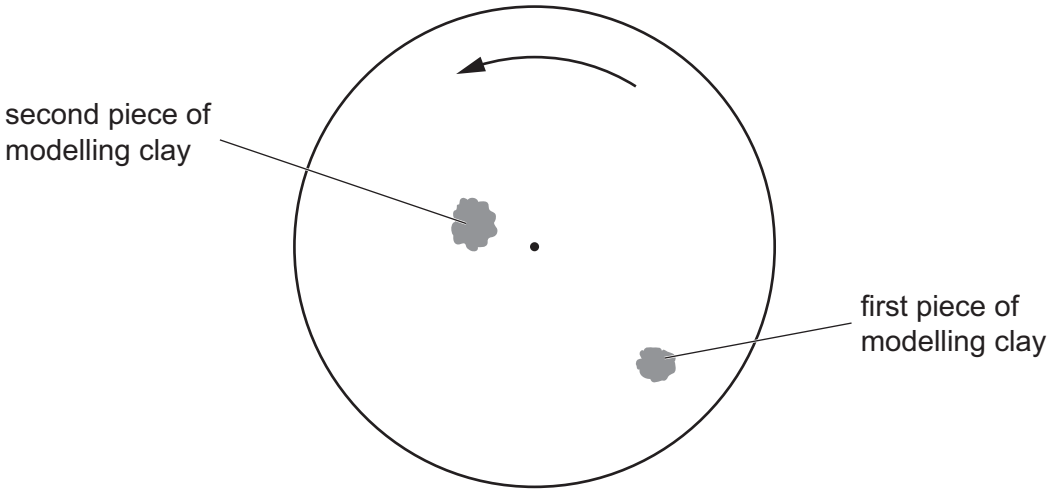


Fig. 1.2

The second piece of modelling clay has a larger mass than the first piece.

By placing **one** tick (✓) in each row, complete Table 1.1 to show how the quantities indicated compare for the two pieces of modelling clay.

Table 1.1

quantity	less for second piece than first piece	same for both pieces	greater for second piece than first piece
angular speed			
linear speed			
acceleration			

[3]

[Total: 10]

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1 A steel ball is placed on the inside surface of a hollow circular cone. The ball moves in a horizontal circle at constant speed, as shown in Fig. 1.1.

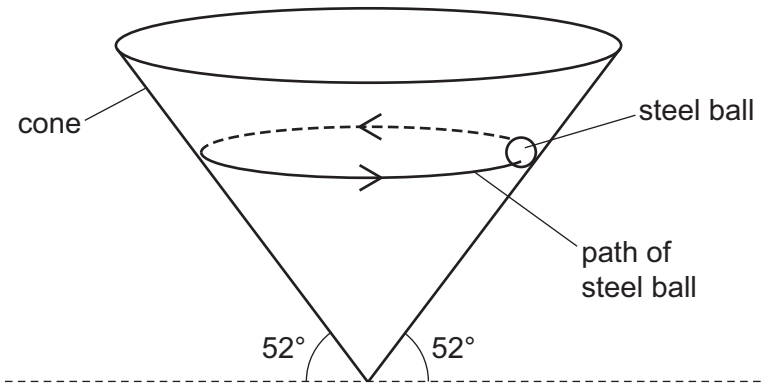


Fig. 1.1

The angle of the side of the cone to the horizontal is 52°. There is no friction between the ball and the cone.

(a) Fig. 1.2 shows a cross-section through the cone and the steel ball.

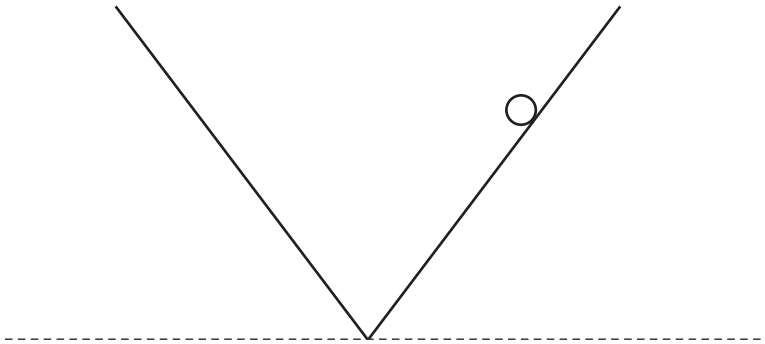


Fig. 1.2

On Fig. 1.2, draw labelled arrows to show the **two** forces acting on the ball. [1]

(b) Describe how the forces acting on the ball cause its acceleration to be centripetal.

.....

.....

.....

..... [2]

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(c) The ball moves in a circle of radius 0.15 m.

Show that the speed of the ball is 1.4 m s^{-1}.

[3]

(d) Calculate the angular speed ω of the ball.

ω = ... rad s^{-1} [2]

(e) The speed of the ball is increased.

Explain why the radius of the circular path of the ball increases.

.....

.....

..... [1]

[Total: 9]

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1 **(a)** Define the radian.

.....

..... [1]

(b) The rear wheel and the pedals of a bicycle are connected by a chain that passes around two cogs (toothed wheels), as shown in Fig. 1.1.

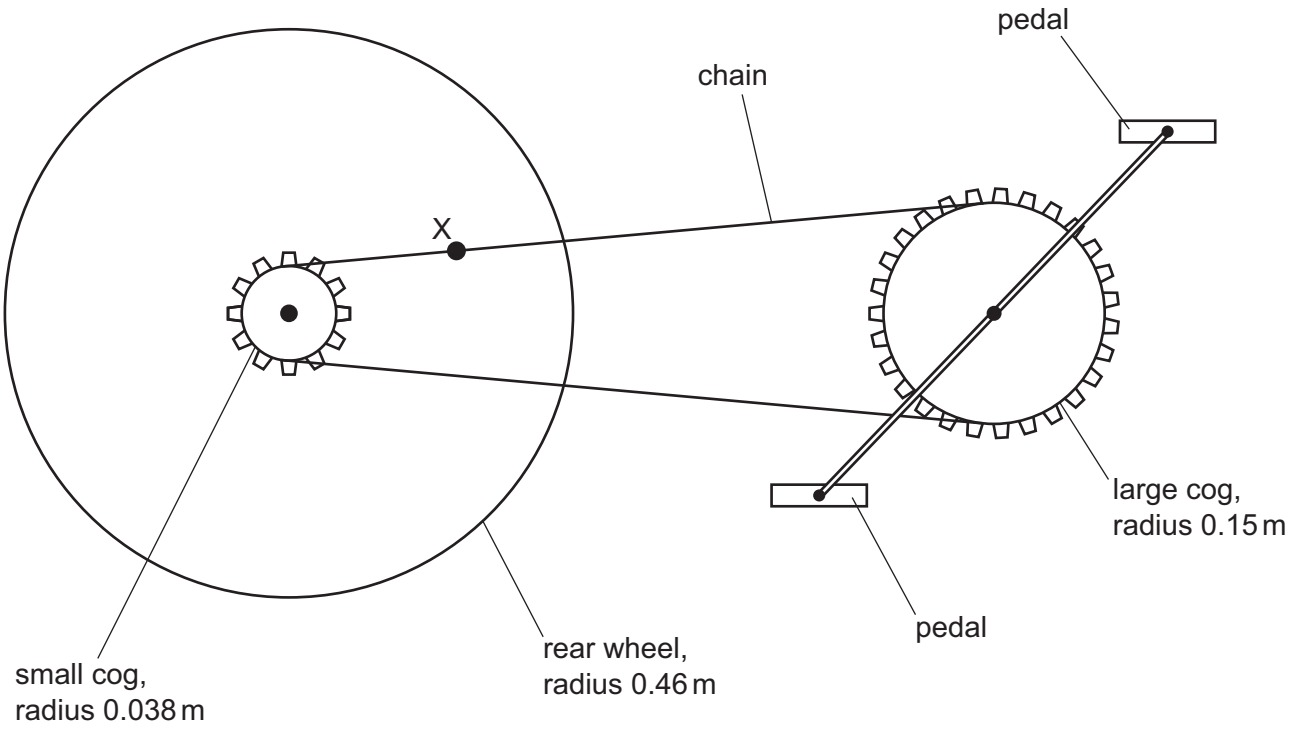


Fig. 1.1 (not to scale)

The small cog has a radius of 0.038 m and is fixed to the rear wheel so that it rotates with it. The large cog has a radius of 0.15 m and is fixed to the pedals so that it rotates with them. The rear wheel has a radius of 0.46 m.

The bicycle is being pedalled so that it moves in a straight line at a constant speed of 17 m s^{-1}.

(i) Calculate the angular speed of the rear wheel.

angular speed = ... rad s^{-1} [2]

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(ii) Calculate the period of rotation of the small cog.

period = ... s [2]

(iii) Show that the distance moved by point X on the chain during one full rotation of the small cog is 0.24 m.

[1]

(iv) Use the information in **(b)(iii)** to determine the angle through which the large cog rotates during one full rotation of the small cog.

angle = ... rad [2]

(c) The chain of the bicycle in **(b)** is moved onto a smaller cog fixed to the rear wheel. The speed of the bicycle does not change.

Explain, without calculation, the effect of this change on the angular speed of the pedals.

.....

.....

..... [2]

[Total: 10]

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1 **(a)** In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

(b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius *R*.

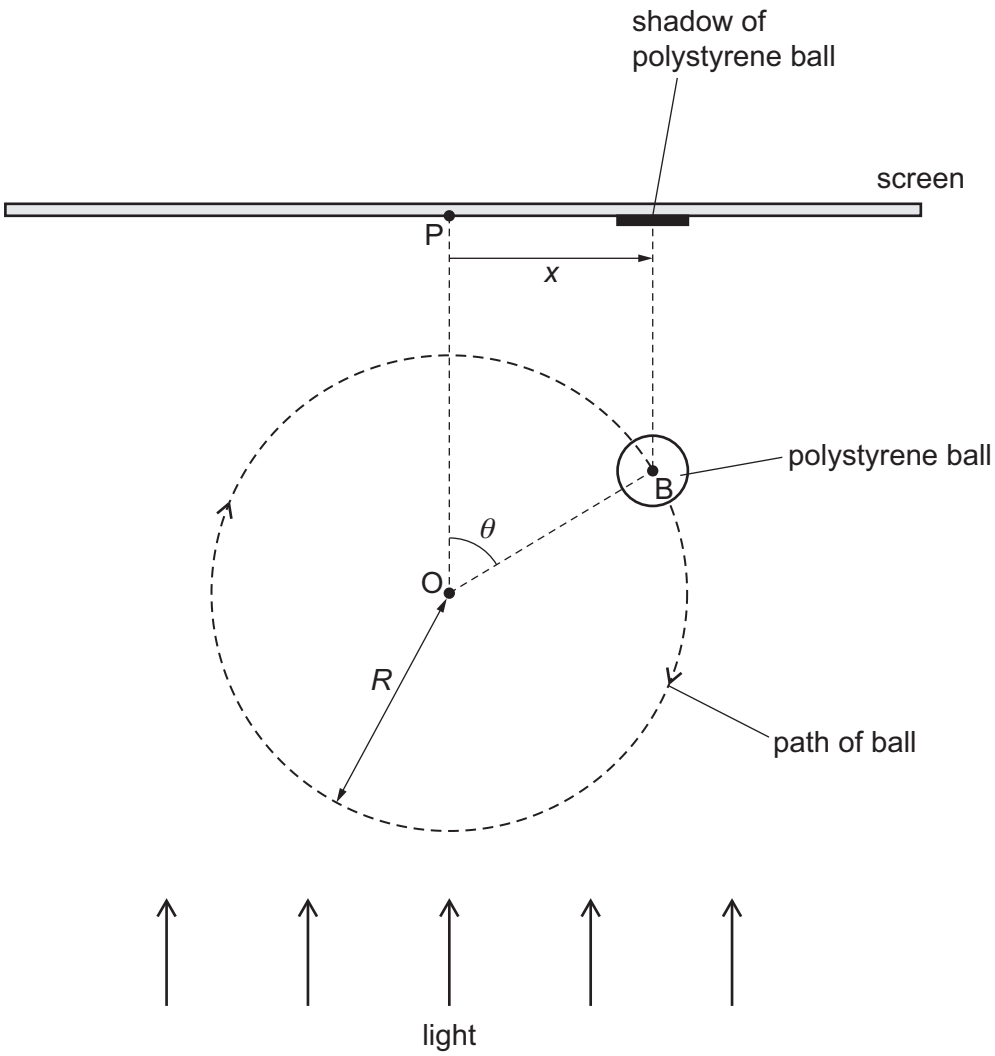


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω.

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(i) State an expression, in terms of R and ω, for the speed v of the ball.

v = ... [1]

(ii) Determine an expression, in terms of v and ω, for the centripetal acceleration of the ball.

centripetal acceleration = ... [2]

(c) The ball in **(b)** is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

(i) Determine an expression, in terms of R and θ, for the displacement x of the shadow from P.

x = ... [1]

(ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t.

θ = ... [1]

(iii) Use your answers in **(c)(i)** and **(c)(ii)** to show that x is given by

$$x = R\sin\omega t.$$

[1]

(iv) Explain, with reference to the equation in **(c)(iii)**, why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]

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(d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1}.

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

(i) the amplitude

amplitude = ... m [1]

(ii) the period

period = ... s [2]

(iii) the maximum acceleration.

maximum acceleration = ... $m\,s^{-2}$ [2]

(e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

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1 The Earth may be considered as a uniform sphere of radius 6.37×10^6 m.

Cambridge is at a point on the Earth's surface that has a latitude of 52.2° north of the Equator, as shown in Fig. 1.1.

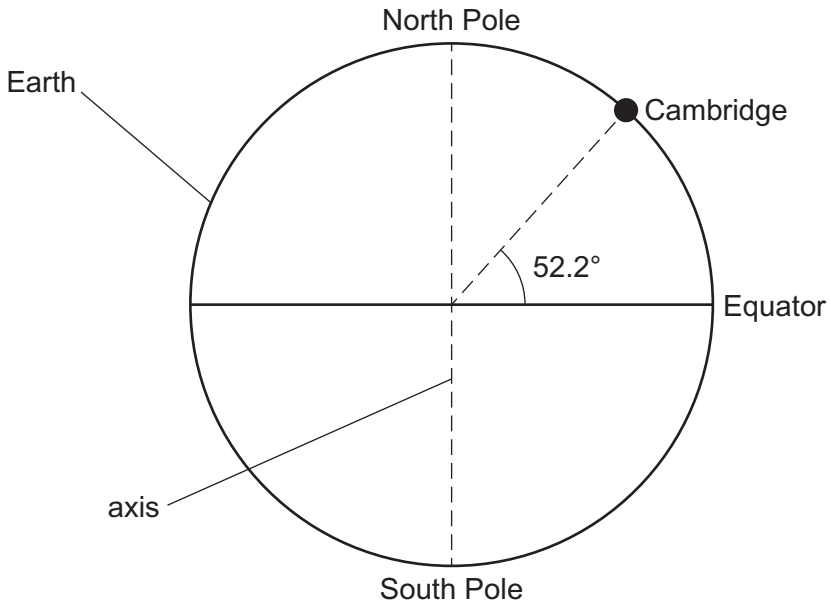


Fig. 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

(a) (i) Show that the radius of the circle around which Cambridge moves is 3.90×10^6 m.

[1]

(ii) Calculate the speed at which Cambridge moves around the circle.

speed = ... $m\,s^{-1}$ [3]

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(b) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

(i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = ... N [2]

(ii) On Fig. 1.2, draw an arrow to show the direction of the resultant force that acts on the student.

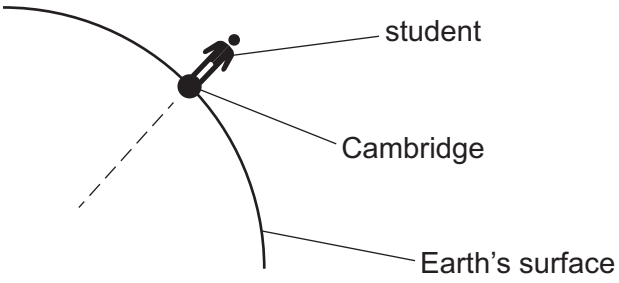


Fig. 1.2 (not to scale)

[1]

(iii) On Fig. 1.3, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in **(b)(ii)**.

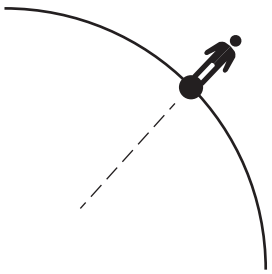


Fig. 1.3 (not to scale)

[2]

[Total: 9]

1 **(a)** In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

(b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius *R*.

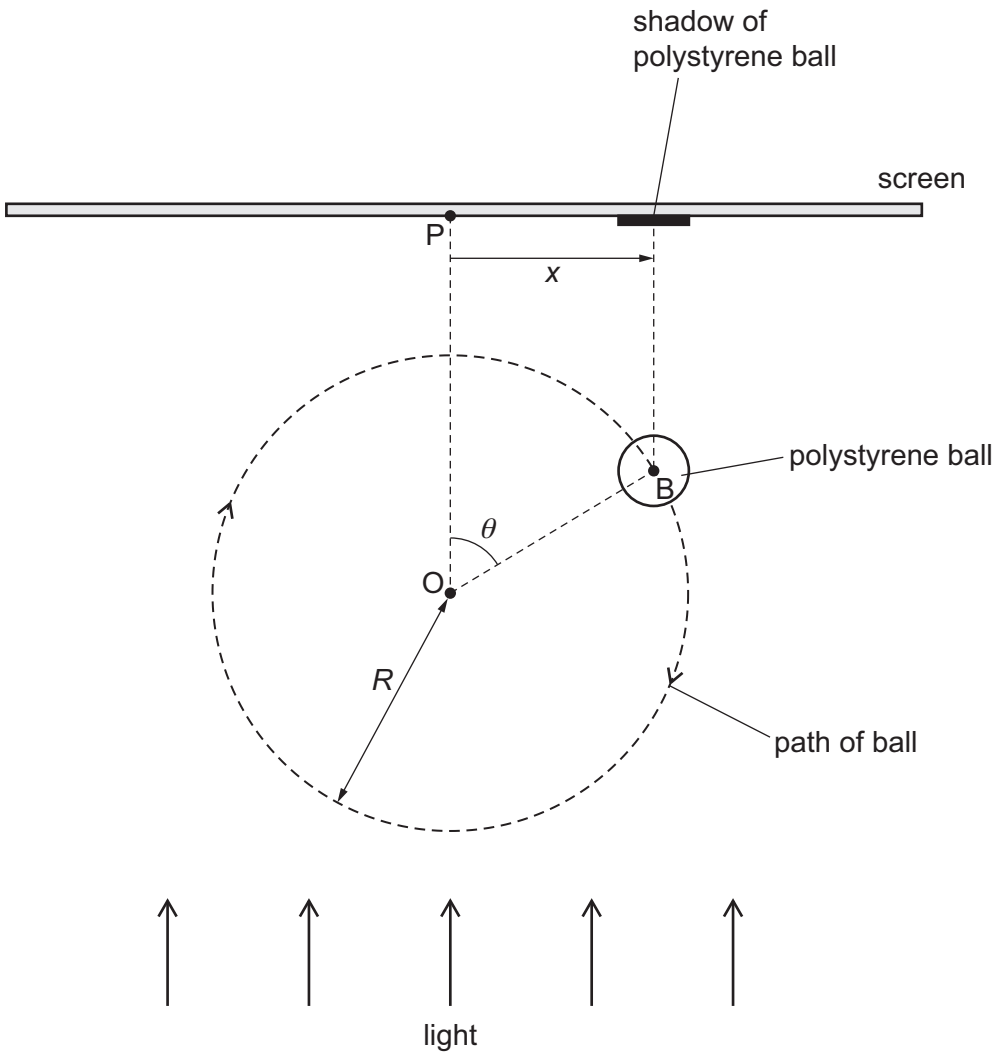


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω.

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(i) State an expression, in terms of R and ω, for the speed v of the ball.

v = ... [1]

(ii) Determine an expression, in terms of v and ω, for the centripetal acceleration of the ball.

centripetal acceleration = ... [2]

(c) The ball in **(b)** is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

(i) Determine an expression, in terms of R and θ, for the displacement x of the shadow from P.

x = ... [1]

(ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t.

θ = ... [1]

(iii) Use your answers in **(c)(i)** and **(c)(ii)** to show that x is given by

$$x = R\sin\omega t.$$

[1]

(iv) Explain, with reference to the equation in **(c)(iii)**, why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]

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(d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1}.

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

(i) the amplitude

amplitude = ... m [1]

(ii) the period

period = ... s [2]

(iii) the maximum acceleration.

maximum acceleration = ... $m\,s^{-2}$ [2]

(e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

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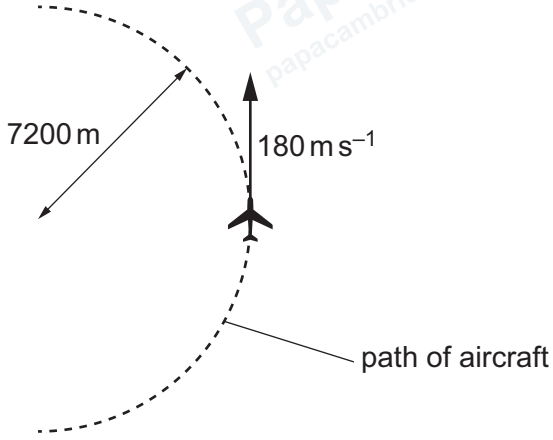
1 **(a)** With reference to velocity and acceleration, describe uniform circular motion.

.....

.....

..... [2]

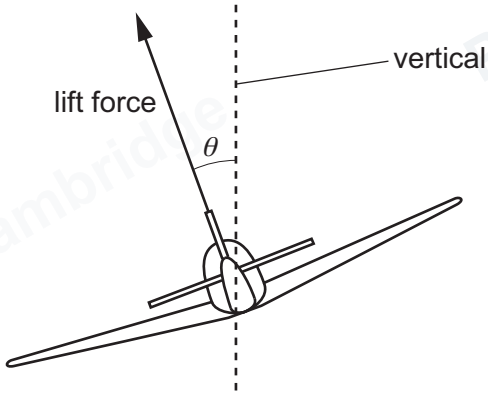
(b) An aircraft is travelling in a horizontal circle of radius 7200 m at a speed of 180 m s^{-1}, as shown in the view from above in Figure 1.1.



not to scale

Figure 1.1

In order to move in a circle, the aircraft has to tilt its wings so that the lift force from the wings is not vertical, as shown in the view from behind in Figure 1.2.



not to scale

Figure 1.2

(i) With reference to the forces acting on the aircraft, explain why the circular motion of the aircraft requires the lift force to be in the direction shown rather than vertically upwards.

.....

.....

..... [2]

(ii) On Figure 1.2, draw an arrow from the aircraft to represent the direction of the resultant force on the aircraft. [1]

(iii) Calculate the centripetal acceleration of the aircraft.

acceleration = ... $m\,s^{-2}$ [2]

(iv) Determine the angle θ at which the aircraft is tilted. Explain your reasoning.

θ = ... ° [3]

[Total: 10]

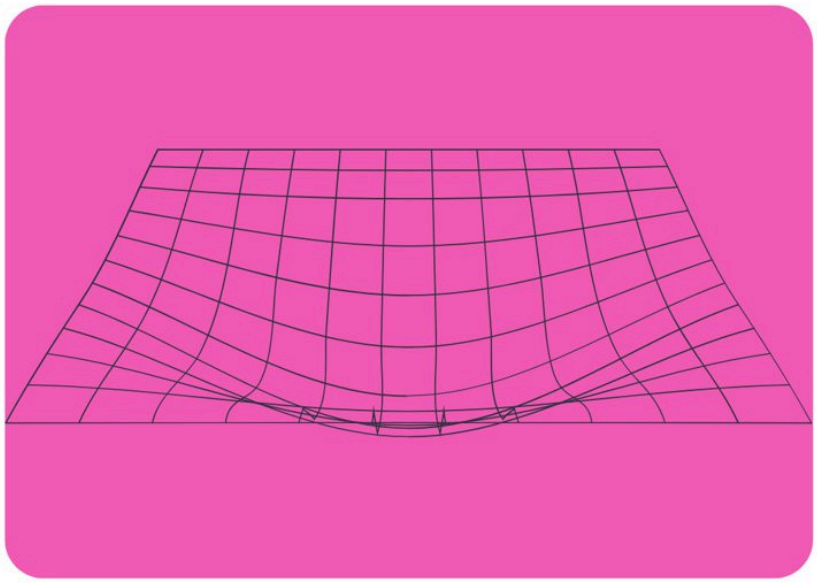
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13 · Gravitational Fields

slides → formulae & definitions → paper4 2024–2026

Chapter 17

Gravitational Fields



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A Level Physics
Prepared and taught by James Gan

Chapter Outline

17.1 Newton's law of gravitation

17.2 Gravitational Field Strength (g)

17.3 Gravitational Potential

17.4 Motion in Gravitational Orbits

17.5 Calculating Orbital Periods

17.6 Earth's Orbital Mechanics



17.1 Newton's law of gravitation

Newton's law of gravitation

Definition

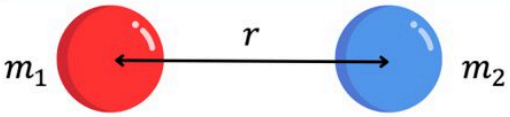
Every object with mass **attracts** every other mass with a **force** that is:

- Directly proportional to the product of their masses
- Inversely proportional to the square of the distance between them

Formula

$$F = G\frac{m_1 m_2}{r^2}$$

Diagram



Gravitational Constant

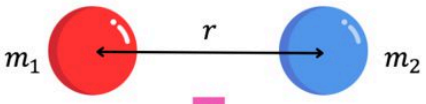
$6.674 \times 10^{-11}\ Nm^2/kg^2$

Why G is needed: It sets the scale of the gravitational force, converting the proportional relationship into an exact, measurable value in Newtons. Without G , we would only know how masses and distance affect the force but not the actual numerical force.

17.1 Newton's law of gravitation

Newton's law of gravitation

When applying Newton's law of gravitation, the distance ***r*** is measured from the center of mass of one object to the center of mass of the other.



Since we are calculating the distance from an object's center of mass, we can essentially treat each object as a **point-like mass**.



17.1 Newton's law of gravitation



Newton's Third Law

The gravitational force that object 1 exerts on object 2 is equal in magnitude and opposite in direction to the force that object 2 exerts on object 1.

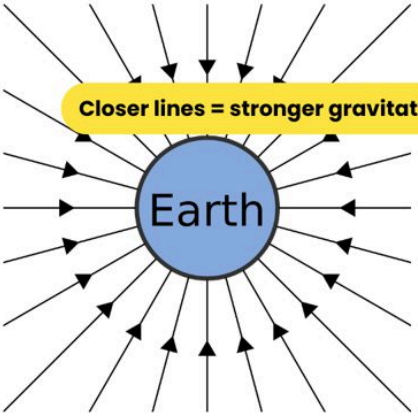
Both objects pull on each other with the same force, regardless of their size.

However, because ***Force = Mass × Acceleration***, the apple moves more noticeably than the Earth because of its much smaller mass. The Earth, with its enormous mass, experiences an extremely small acceleration, making its movement negligible.

17.1 Newton's law of gravitation

Visualizing Gravitational Fields

- Gravitational fields are represented by **field lines**.
- Arrows on the lines indicate the **direction of the gravitational force (towards the mass)**.



Closer lines = stronger gravitational field.

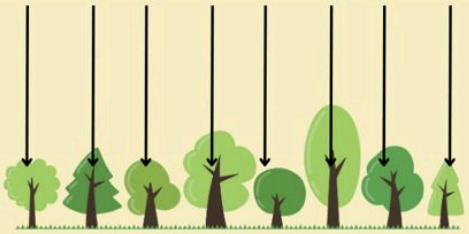
Wider spaced lines = Weaker gravitational field.

17.1 Newton's law of gravitation

Uniform vs. Radial Gravitational Fields

Near Earth's Surface

- The gravitational field is uniform.
- Field lines are parallel and equally spaced.
- Gravity pulls straight down with the same strength everywhere.



Far From Earth

- The gravitational field is radial.
- Field lines spread out from Earth's center.
- Gravity points toward Earth's center and weakens with distance.



17.1 Newton's law of gravitation

The Inverse Square Law

- Gravitational force decreases with the square of the distance between objects.
- If the distance ***r*** is doubled, the force becomes 4 times weaker:

$$F \propto \frac{1}{r^2}$$



This explains why gravity is stronger near planets and weaker in space.

17.2 Gravitational Field Strength (g)

Gravitational Field Strength

Definition

Gravitational field strength ***(g)*** is the **force per unit mass** experienced by an object in a gravitational field.

Unit

m/s^2 , N/kg

In Relation to Weight

Gravitational field strength is directly related to the weight of an object. **The weight is the force exerted by gravity on an object**:

$$w = mg$$

17.2 Gravitational Field Strength (g)

Equation for Gravitational Field Strength

$$w = mg$$

$$F = mg$$

$$g = \frac{F}{m}$$

$$F = G\frac{m_1 m_2}{r^2}$$

$$g = \frac{G\frac{mM}{r^2}}{m} = \frac{GmM}{r^2 m} = \frac{GM}{r^2}$$

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17.2 Gravitational Field Strength (g)

Equation for Gravitational Field Strength

$$w = mg$$

$$F = mg$$

$$g = \frac{F}{m}$$

$$F = G\frac{m_1 m_2}{r^2}$$

$$g = \frac{G\frac{mM}{r^2}}{m} = \frac{GmM}{r^2 m} = \frac{GM}{r^2}$$

Final Formula

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Equation for Gravitational Field Strength

$$w = mg$$

$$F = mg$$

$$g = \frac{F}{m}$$

$$F = G\frac{m_1 m_2}{r^2}$$

$$g = \frac{G\frac{mM}{r^2}}{m} = \frac{GmM}{r^2 m} = \frac{GM}{r^2}$$

Final Formula

$$g = \frac{GM}{r^2}$$

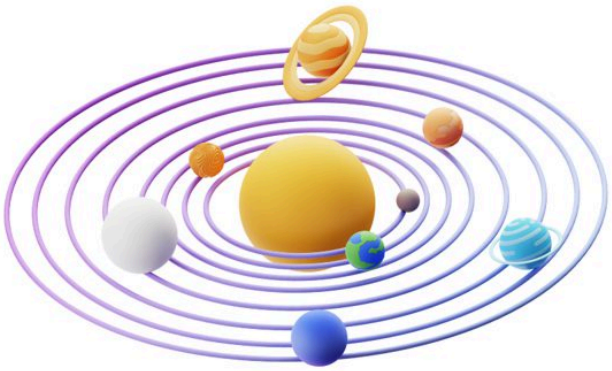
Why is this derivation important?

- By deriving this equation, we can calculate the **gravitational field strength** at any point around a mass, such as Earth.
- This helps predict how objects will move in a gravitational field, which is crucial for satellite orbits, object motion, and space missions.
- The mass ***m*** of an object placed in the field does not appear in the equation for gravitational field strength because ***g* is the strength of the field itself, independent of the object's mass.**

17.2 Gravitational Field Strength (g)

Gravitational Field Strength as a Vector Quantity

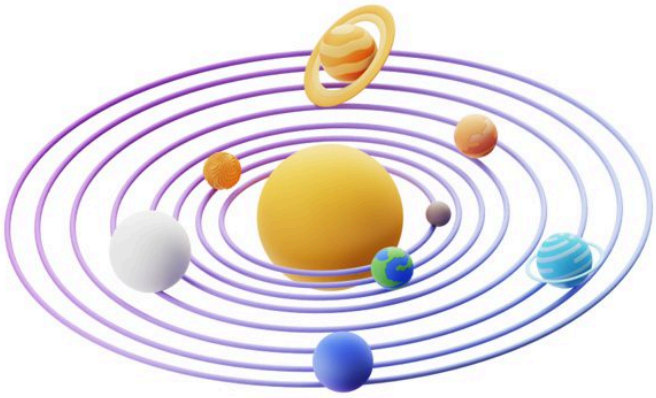
- Gravitational field strength is a **vector quantity**, meaning it has both magnitude and direction.
- The direction of the gravitational field strength is always toward the **center** of the mass creating the field.



17.2 Gravitational Field Strength (g)

Inverse square law

- The field strength **decreases** as the distance **increases** from the source of gravity.



- This means that as you move farther from the source mass, the gravitational field strength weakens.
- This follows from the equation:

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Acceleration Due to Gravity (Free Fall)

Acceleration of free fall is the acceleration experienced by an object in a gravitational field when no other forces (like air resistance) are acting on it.



It is equal to the gravitational field strength at that location.

Near Earth's surface, the acceleration due to gravity is approximately:

$$g = 9.8\ m/s^2$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 10 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 10 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

$$g = \frac{(6.67\ x\ 10^{-11}) \times (5.98 \times 10^{24})}{(6370 + 10)^2} = 9.81\ N/kg$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 500 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 500 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

$$g = \frac{(6.67\ x\ 10^{-11}) \times (5.98\ \times\ 10^{24})}{(6370 + 500)^2} = 8.44\ N/kg$$

17.3 Gravitational Potential

Limitations of *mgh* for Gravitational Potential Energy

- The equation PE = *mgh* is useful ***only*** near Earth's surface, where g is constant.
- However, as an object moves further from Earth, **g decreases** because of the inverse square law.
- This means that using **h = 0 at Earth's surface is an oversimplification**, as potential energy should be measured relative to a more fundamental reference point.
- To properly account for the change in **g** with distance, we need a more general formula for gravitational potential energy.

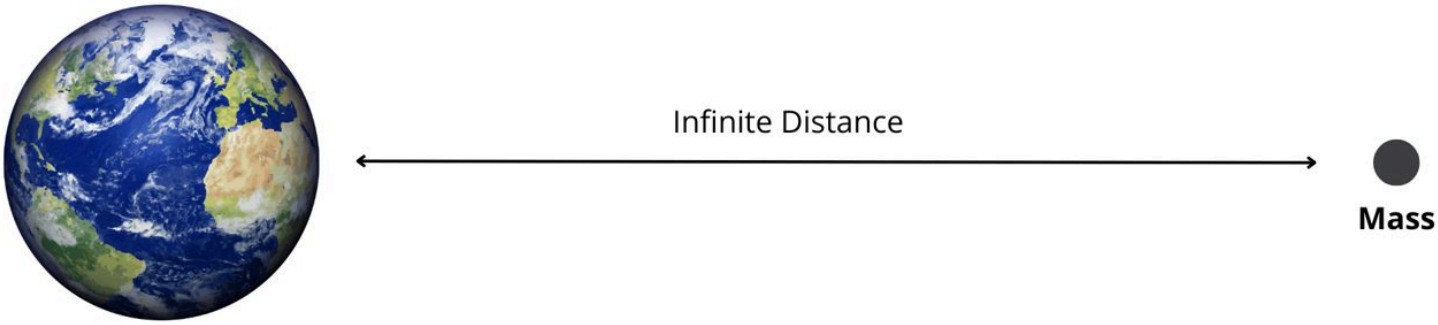


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17.3 Gravitational Potential

Defining Gravitational Potential Energy at Infinity

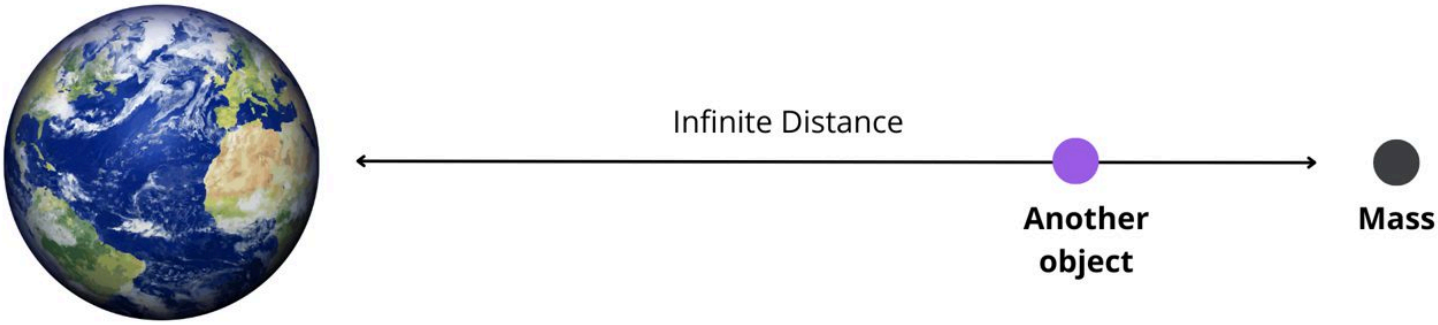
- Imagine a mass very, very far away from everything—so far that gravity has no effect on it.
- We say that at this infinite distance, **the mass has zero potential energy.**



17.3 Gravitational Potential

Defining Gravitational Potential Energy at Infinity

- This makes calculations easier and more universal because any object closer than infinity will have **negative potential energy (since gravity is pulling it back)**.



17.3 Gravitational Potential

Gravitational Potential

Definition

Gravitational potential (Φ) at a point is the work done per unit mass to bring a small test mass from infinity to that point.

Formula

$$\Phi = -\frac{GM}{r}$$

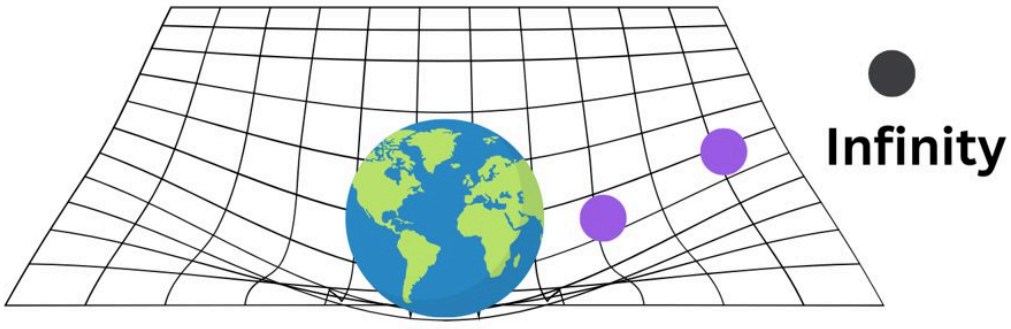
Key Idea

Describes how much potential energy each kg of mass would have at a certain point in a gravitational field.

17.3 Gravitational Potential

Formula

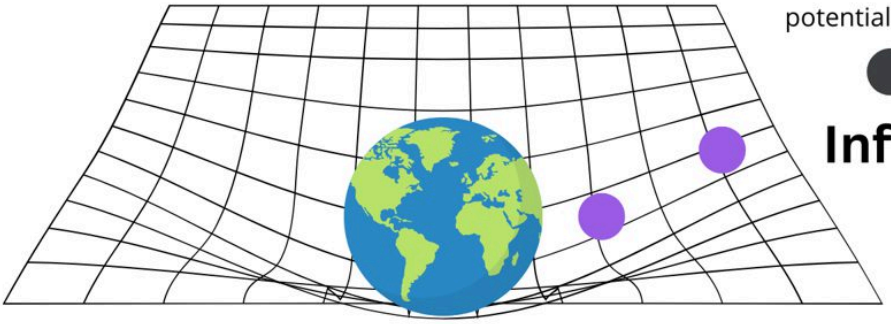
$$\Phi = -\frac{GM}{r}$$



17.3 Gravitational Potential

Why is GP Always Negative?

Zero at infinity: By definition, gravitational potential is zero at infinity.



Infinity

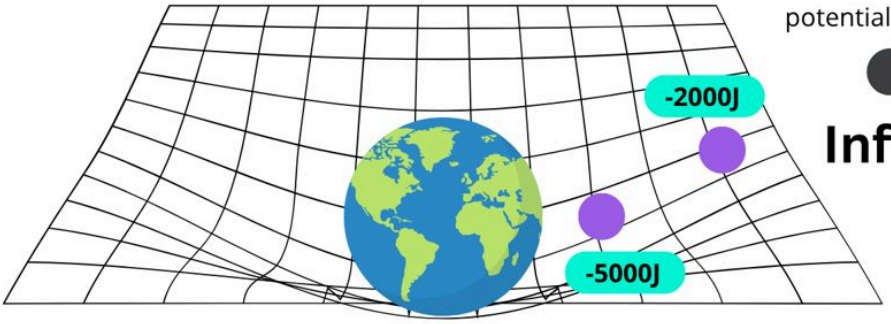
Idea: Objects in a gravitational field are "trapped":

- To move a mass from infinity to a point near a planet, gravity does work on it, pulling it inward.
- This means the object loses potential energy, making the value negative.

17.3 Gravitational Potential

Why is GP Always Negative?

Zero at infinity: By definition, gravitational potential is zero at infinity.



Infinity

Closer = More Negative: The closer an object is to the planet, the more energy is needed to escape, making the stored energy more negative.

More negative = deeper energy well = more energy required to escape.

17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

Compare the gravitational potential at:

1. The surface of the Earth
2. The surface of the Sun

Info:

$$r_E = 6.37 \times 10^6\ m$$

$$M_E = 5.97 \times 10^{24}\ kg$$

$$r_S = 6.96 \times 10^8\ m$$

$$M_S = 1.99 \times 10^{30}\ kg$$

$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$

Answer:

17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

Compare the gravitational potential at:

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$$r_S = 6.96 \times 10^8\ m$$

$$M_S = 1.99 \times 10^{30}\ kg$$

$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$

Answer:

Earth:

$$\Phi = -\frac{GM}{r}$$

$$= -\frac{(6.674 \times 10^{-11}) \times (5.97 \times 10^{24})}{6.37 \times 10^6}$$

$$= -6.26 \times 10^7\ J/kg$$



17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

Compare the gravitational potential at:

1. The surface of the Earth
2. The surface of the Sun

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Answer:

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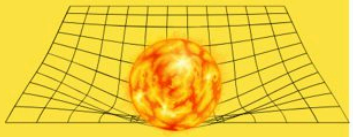


Sun:

$$\Phi = -\frac{GM}{r}$$

$$= -\frac{(6.674 \times 10^{-11}) \times (1.99 \times 10^{30})}{6.96 \times 10^8}$$

$$= -1.91 \times 10^9\ J/kg$$



17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

A **500 kg satellite** is launched from the Earth's surface into a circular orbit at an altitude of **500 km** above the Earth. Calculate the energy required to move the satellite from the Earth's surface to its orbital height.

Info:

$$r_E = 6.37 \times 10^6\ m$$

$$M_E = 5.97 \times 10^{24}\ kg$$

$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$

Answer:

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17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

A **500 kg satellite** is launched from the Earth's surface into a circular orbit at an altitude of **500 km** above the Earth. Calculate the energy required to move the satellite from the Earth's surface to its orbital height.

Info:

$r_E = 6.37 \times 10^6\ m$

$M_E = 5.97 \times 10^{24}\ kg$

$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$

Answer:

$$\Delta \emptyset = \frac{GM}{r_1} - \frac{GM}{r_2}$$

$$\Delta \emptyset = GM\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$= 6.674 \times 10^{-11} \times 5.97 \times 10^{24} \times \left(\frac{1}{6.37 \times 10^6} - \frac{1}{6.87 \times 10^6}\right)$$

$$= 2.19 \times 10^8\ J$$

17.3 Gravitational Potential

Comparing Gravitational Field Strength and Gravitational Potential

	Gravitational Field Strength	Gravitational Potential
Definition	Defined as the force per unit mass experienced by a small test mass in the field.	Defined as the work done per unit mass to bring a small test mass from infinity to a point in the field.
Formula	$g = \frac{GM}{r^2}$	$\Phi = -\frac{GM}{r}$
Types	Vector Quantity	Scalar Quantity
Units	N/kg	J/kg

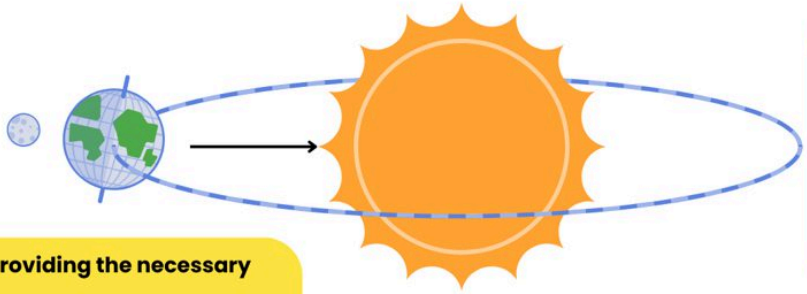
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17.4 Motion in Gravitational Orbits

Deriving $v^2 = \frac{GM}{r}$

Step 1: Understanding the Forces Acting on an Orbiting Object

- An object in a circular orbit experiences:



Centripetal Force (needed to keep the object moving in a circle):

$$F = \frac{mv^2}{r}$$

Gravitational Force (providing the necessary centripetal force):

$$F = G\frac{m_1 m_2}{r^2}$$

17.4 Motion in Gravitational Orbits

Deriving $v^2 = \frac{GM}{r}$

Step 2: Equating Gravitational Force to Centripetal Force

- Since gravity is the only force keeping the object in orbit, we set the two equations equal, and simplify:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

cancel out the m

$$\frac{GM}{r^2} = \frac{v^2}{r}$$

multiply by r

$$v^2 = \frac{GM}{r}$$

This means that all satellites at the same orbital radius around Earth will have the same speed, regardless of their mass, ensuring predictable and stable orbits for objects like the ISS, GPS satellites, and space debris.

17.4 Motion in Gravitational Orbits

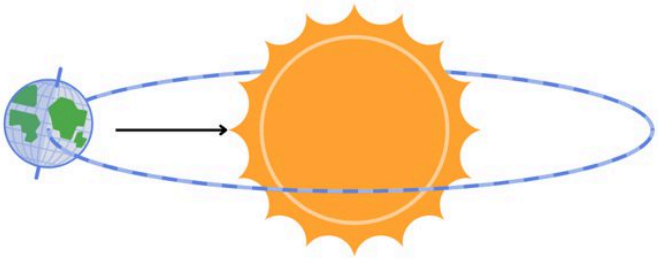
Worked Example

Calculate the orbital speed of the Earth around the Sun.

mass of the sun: $M = 2 \times 10^{30}\ kg$

Average distance between Earth and Sun: $r = 1.5 \times 10^{11}\ m$

$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$



$$v^2 = \frac{GM}{r}$$

17.4 Motion in Gravitational Orbits

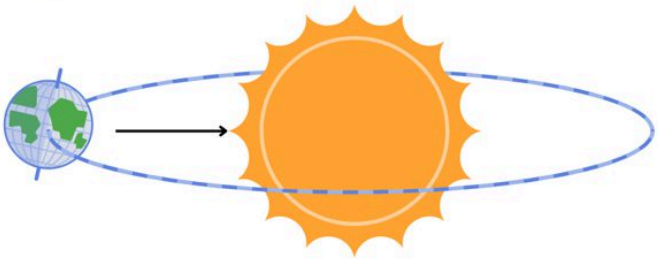
Worked Example

Calculate the orbital speed of the Earth around the Sun.

mass of the sun: $M = 2 \times 10^{30}\ kg$

Average distance between Earth and Sun: $r = 1.5 \times 10^{11}\ m$

$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$



$$v^2 = \frac{GM}{r}$$

$$v = \sqrt{\frac{6.674 \times 10^{-11} \times 2 \times 10^{30}}{1.5 \times 10^{11}}}$$

$$= 2.97 \times 10^4 = 29.7\ km/s$$

17.5 Calculating Orbital Periods

Orbital Period

Definition

Orbital period is the time taken for a satellite to complete one full orbit around a mass.

Deriving the formula

$$s = \frac{d}{t}$$

$$v = \frac{2\pi r}{T}$$

$$v^2 = \frac{GM}{r}$$

$$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r}$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$T = \sqrt{\frac{4\pi r^3}{GM}}$$

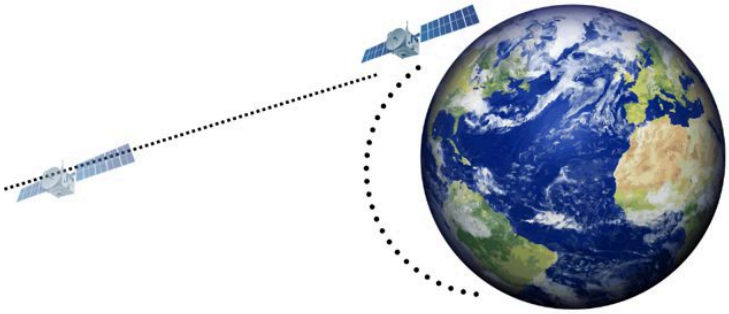
Importance

This formula is important because it allows us to predict how long an object will take to orbit a planet or star, helping in satellite deployment, space missions, and understanding planetary motion.

17.6 Earth's Orbital Mechanics

Orbital Speed and Radius

- For an object to maintain a stable orbit, it must travel at the correct speed for its altitude.
- From $v = \sqrt{\frac{GM}{r}}$, a smaller orbital radius requires a higher speed to counteract gravity.
- If an object moves too fast, it may escape orbit; if too slow, it may fall back to Earth.



17.6 Earth's Orbital Mechanics

Uses of Artificial Satellites



Communication: Enabling global connectivity, TV broadcasts, and internet services.



Navigation: GPS satellites provide location tracking.



Weather Monitoring: Satellites track storms, climate, and environmental changes.



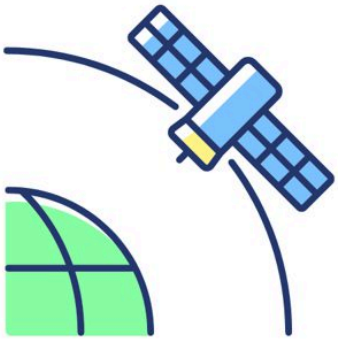
Military and Surveillance: Used for reconnaissance and national security.

17.6 Earth's Orbital Mechanics

Geostationary Orbits

A geostationary satellite orbits Earth once every 24 hours, staying fixed over a single location.

Usage: Ideal for telecommunications, weather observation, and broadcasting.



17.6 Earth's Orbital Mechanics

Determining Satellite Distance Using Orbital Period

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

Step 1: Rearrange our equation so that r (distance) becomes the subject.

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

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17.6 Earth's Orbital Mechanics

Determining Satellite Distance Using Orbital Period

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

Step 1: Rearrange our equation so that r (distance) becomes the subject.

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

Step 2: A geostationary satellite has an orbital period of 24 hours or 86,400 seconds.

$$M_E = 5.97 \times 10^{24}\ kg$$
$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$
$$T = 86400s$$

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$
$$= \sqrt[3]{\frac{(6.674 \times 10^{-11}) \times (5.97 \times 10^{24}) \times (86400)^2}{4\pi^2}}$$

Distance from the centre of the Earth ~ 42300km

Formulae & Definitions — Topic 13 · Gravitational Fields

Newton's law of gravitation 万有引力定律 — the force between two point masses is proportional to the product of their masses and inversely proportional to the square of their separation: $F = GMm/r^2$.

Universal gravitational constant — $G = 6.67 \times 10^{-11}$ N m² kg⁻².

Uniform sphere — the whole mass may be taken as a point mass at its centre (r measured from the centre).

Gravitational field 引力场 — a region of space where a mass experiences a force; field lines show the direction of the force (the closer the lines, the stronger the field).

Field strength 场强 g — the force per unit mass: $g = F/m = GM/r^2$ (N kg⁻¹).

Near the Earth's surface — the field is (approximately) uniform and $g \approx 9.81$ N kg⁻¹ (height ≪ radius).

Effect of Earth's rotation — at the equator $mg = GMm/R^2 - mR\omega^2$ (the rotation term is tiny); at the poles $mg = GMm/R^2$.

Gravitational potential 引力势 φ — the work done per unit mass in bringing a mass from infinity to the point: $\varphi = -GM/r$ (J kg⁻¹). All values are negative (zero at infinity).

Gravitational potential energy 引力势能 — $GPE = m\varphi = -GMm/r$ (J). For a large distance use $\Delta GPE = m\varphi(\text{final}) - m\varphi(\text{initial})$; near the surface (g constant) $\Delta GPE = mgh$.

Field–potential relation — $g = -d\varphi/dr = -$(gradient of the φ–r graph); the minus sign shows φ decreases in the direction of g.

Circular orbit 圆轨道 — the centripetal force is provided by the gravitational force: $GMm/r^2 = mv^2/r \rightarrow v = \sqrt{(GM/r)}$.

First cosmic speed 第一宇宙速度 — the minimum orbital speed at the Earth's surface (r = R), $v \approx 7.9$ km s⁻¹.

Kepler's third law — r^3/T^2 = constant (∝ the central mass M).

Total energy of a satellite — $E = GPE + KE = -GMm/r + \frac{1}{2}mv^2$ (constant for a given orbit).

Escape speed 第二宇宙速度 — $v_E = \sqrt{(2GM/r)}$: the minimum speed to escape to infinity (KE + GPE = 0).

Geostationary satellite 同步卫星 — equatorial orbit, moves west → east with period T = 24 h, so it stays over the same point of the Earth.

点评 · Insight

天体圆轨道问题的统一思路是 "the centripetal force is provided by the gravitational force"：列 $GMm/r^2 = mr\omega^2 = mv^2/r$ 求解。$r^3/T^2 \propto M$ 说明轨道中心天体质量正比于 r^3/T^2。

1 **(a)** Explain why the gravitational potential near to a point mass is negative.

.....

.....

.....

..... [2]

(b) A planet may be assumed to be a uniform sphere. It has gravitational potential ϕ at distance r from the centre of the planet.

The variation with $\frac{1}{r}$ of ϕ is shown in Fig. 1.1.

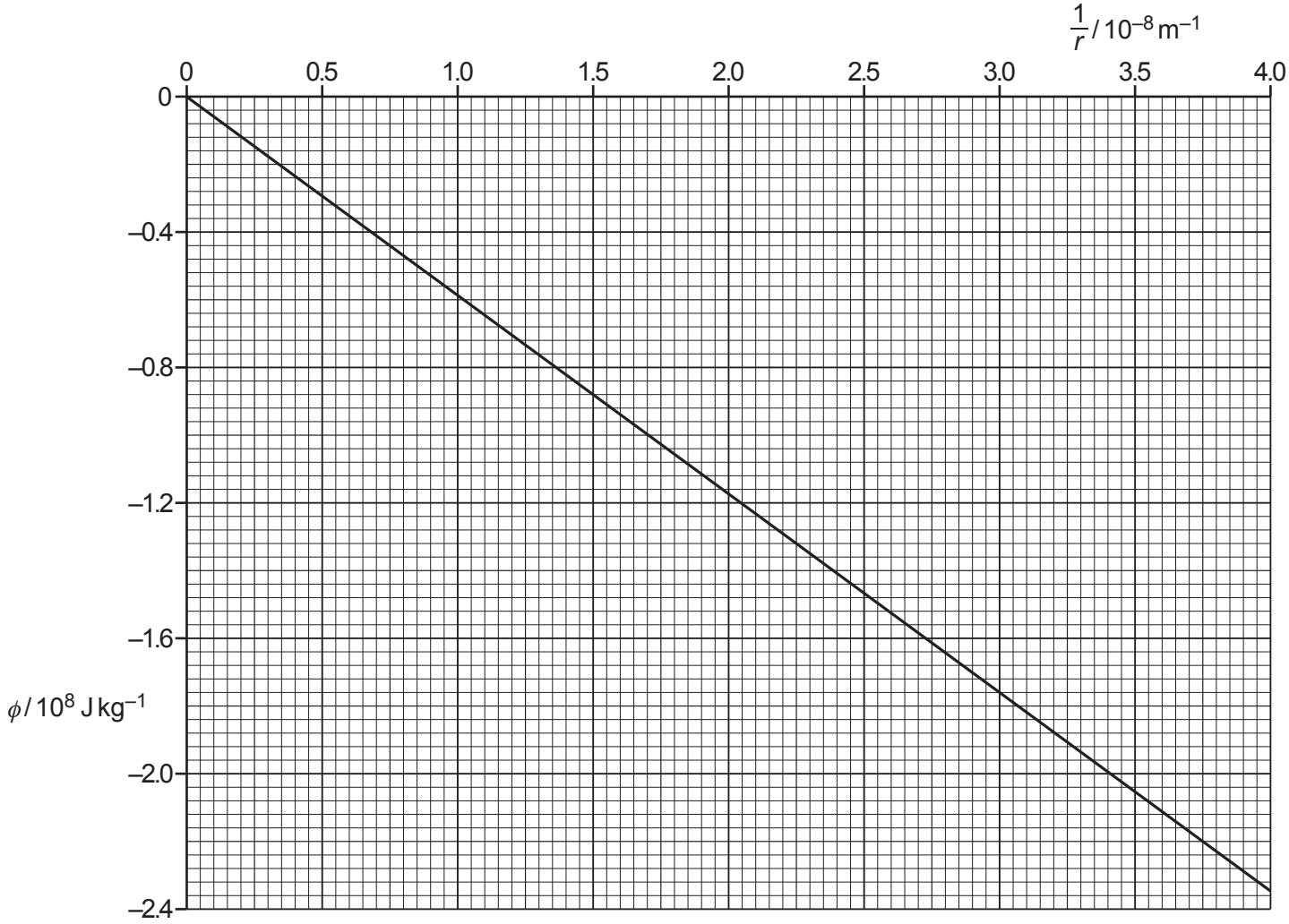


Fig. 1.1

(i) Show that the mass of the planet is 8.8×10^{25} kg.

[2]

(ii) The period of rotation of the planet is 0.72 Earth days.

A satellite in orbit around the planet remains above the same point on the surface of the planet.

Use the mass of the planet in **(b)(i)** to determine the radius R of the orbit of the satellite.

R = ... m [3]

(iii) The speed of the satellite in **(b)(ii)** is 8400 m s^{-1}. The mass of the satellite is 1200 kg.

Determine the additional energy required to move the satellite from its orbit to infinity.

energy required = ... J [3]

[Total: 10]

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A satellite X, of mass M, orbits a planet at a constant distance $4R$ from the centre of the planet, as shown in Fig. 1.1.

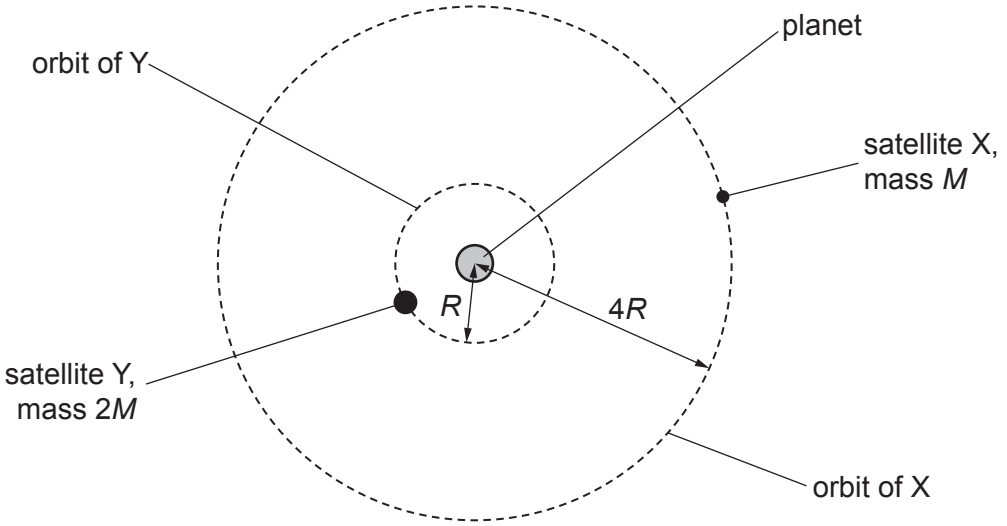


Fig. 1.1 (not to scale)

A second satellite Y, of mass $2M$, orbits the planet with orbital radius R.

The gravitational potential at X due to the planet is $-\Phi$. The planet is a uniform sphere.

(i) Explain why the gravitational potential at X is negative.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ, for the gravitational potential at Y due to the planet.

gravitational potential = ... [2]

(iii) Complete Table 1.1 by giving expressions, in terms of some or all of M, R and Φ, for the quantities indicated for each of the satellites X and Y.

Table 1.1

	satellite X	satellite Y
gravitational field strength at satellite due to planet		
gravitational potential energy of satellite		

[4]

[Total: 10]

1 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) A planet may be considered as a uniform sphere.

A satellite is in circular orbit of period T around the planet at a height h above the surface. The height of the orbit can be adjusted by use of the satellite's rocket engines.

Fig. 1.1 shows the variation with h of $T^{\frac{2}{3}}$.

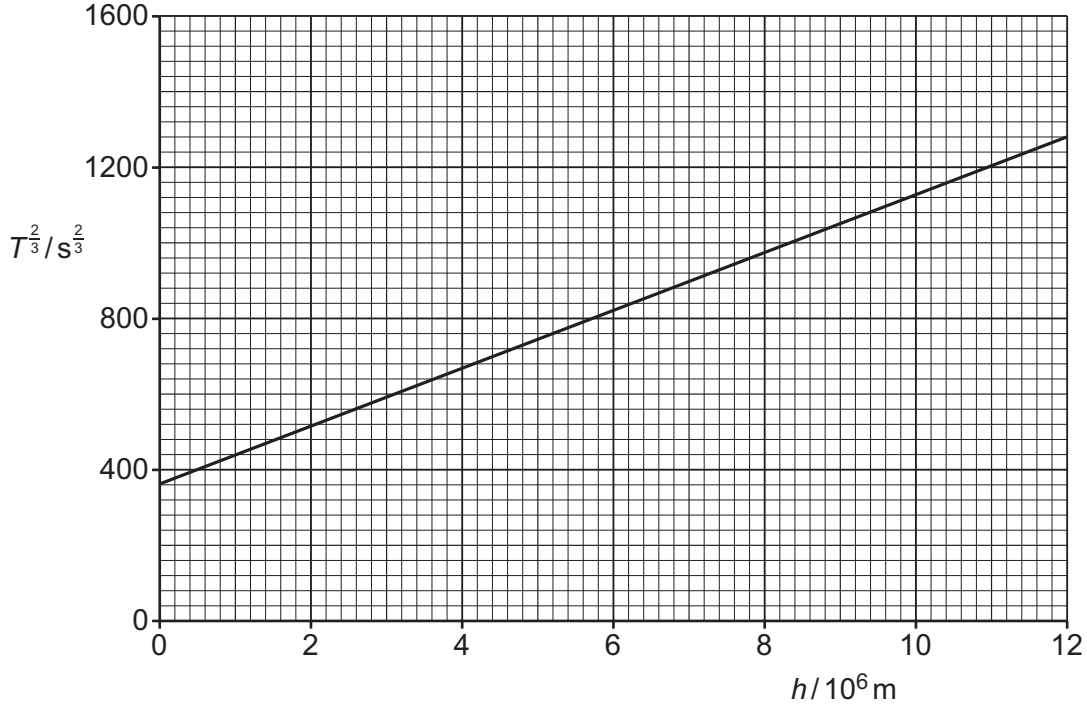


Fig. 1.1

(i) By reference to forces, explain why the orbit of the satellite is circular.

.....

.....

..... [2]

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(ii) Use Newton's law of gravitation to show that h and T are related by

$$(h + B)^3 = \frac{GA}{4\pi^2} T^2$$

where G is the gravitational constant and A and B are constants that depend on the properties of the planet.

[3]

(iii) Use the gradient and intercept of the line in Fig. 1.1 to determine values for A and B. Give units with your answers.

A = ... unit .....

B = ... unit .....

[5]

[Total: 12]

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2 The Sun may be considered as a uniform sphere with a mass of 1.99×10^{30} kg and a surface temperature of 5780 K.

A probe with a mass of 2.63 kg moves in a straight line towards the Sun.
When it is at a distance x from the centre of the Sun, the probe measures the gravitational field strength g due to the Sun and the radiant flux intensity F of radiation from the Sun.

(a) Define gravitational field.

.....

..... [1]

(b) For the position of the probe where $x = 1.47 \times 10^{11}$ m:

(i) calculate g

g = ... N kg^{-1} [2]

(ii) determine the gravitational potential energy E_P of the probe.

E_P = ... J [2]

(c) **(i)** Show that, for any particular value of x, the numerical values of g and F are related by

$$g = \frac{4\pi GM}{L} F$$

where M is the mass of the Sun, L is the luminosity of the Sun and G is the gravitational constant.

[3]

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(ii) Fig. 2.1 shows the variation of g with F.

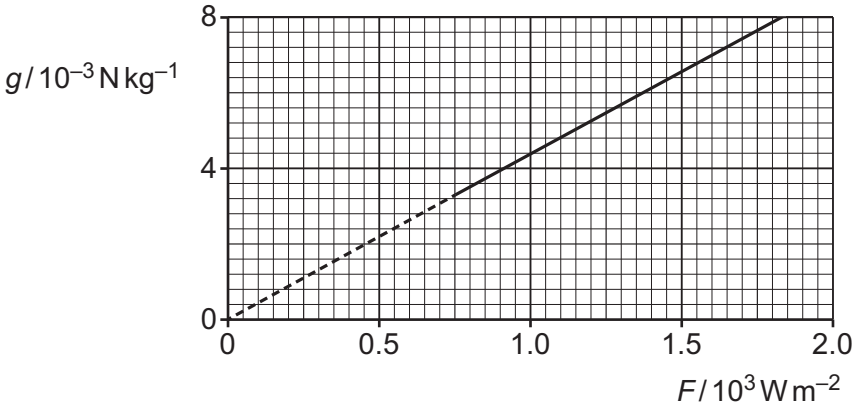


Fig. 2.1

Determine a value for the luminosity L of the Sun. Give a unit with your answer.

L = ... unit ..... [2]

(iii) Use your answer in **(c)(ii)** to determine the radius r of the Sun.

r = ... m [2]

[Total: 12]

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1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) Mars is a planet that may be considered to be an isolated uniform sphere of radius 3.4×10^6 m.

A satellite of mass 122 kg is in orbit around Mars at a constant height of 1.7×10^6 m above the surface of the planet.

The height of the orbit is increased to 6.8×10^6 m above the surface. This increases the gravitational potential energy of the satellite by 5.1×10^8 J.

(i) Show that the mass of Mars is 6.4×10^{23} kg.

[3]

(ii) Calculate the gravitational potential ϕ at the surface of Mars. Give a unit with your answer.

ϕ = ... unit ..... [2]

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(c) The satellite in **(b)** is moved to an orbit in which the satellite remains at the same point above the surface of Mars.

(i) The orbit has a period of 25 hours.

State what can be deduced from this about the rotation of Mars on its axis.

.....

..... [1]

(ii) State **one** other feature of this orbit.

.....

..... [1]

[Total: 9]

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[3]

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1 **(a)** Define gravitational field.

.....

..... [1]

(b) The gravitational field strength g at a distance x from the centre of a uniform spherical planet of mass M is given by the expression

$$g = \frac{GM}{x^2}$$

where G is the gravitational constant and distance x is greater than the radius of the planet.

(i) Describe the pattern of the field lines outside the planet that represent the gravitational field due to the planet.

.....

.....

..... [2]

(ii) Explain why, for small changes in vertical height near the surface of the planet, g may be assumed to be constant.

.....

.....

..... [2]

(c) Assume that the Earth is a uniform sphere. For the Earth, the product GM is equal to $3.99 \times 10^{14}\,m^3\,s^{-2}$.

(i) Determine a value, to three significant figures, for the radius R of the Earth.

R = ... m [2]

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(ii) Calculate the gravitational potential at the Earth's surface. Give a unit with your answer.

gravitational potential = ... unit ..... [2]

(d) Explain why the gravitational potential energy of two point masses is always negative.

.....

.....

..... [2]

[Total: 11]

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1 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) A binary star consists of star A, of mass 4.0×10^{30} kg, and star B, of mass 2.0×10^{30} kg, separated by a distance of 3.3×10^{12} m. The stars are both in circular orbit around their common centre of gravity X, as shown in Fig. 1.1.

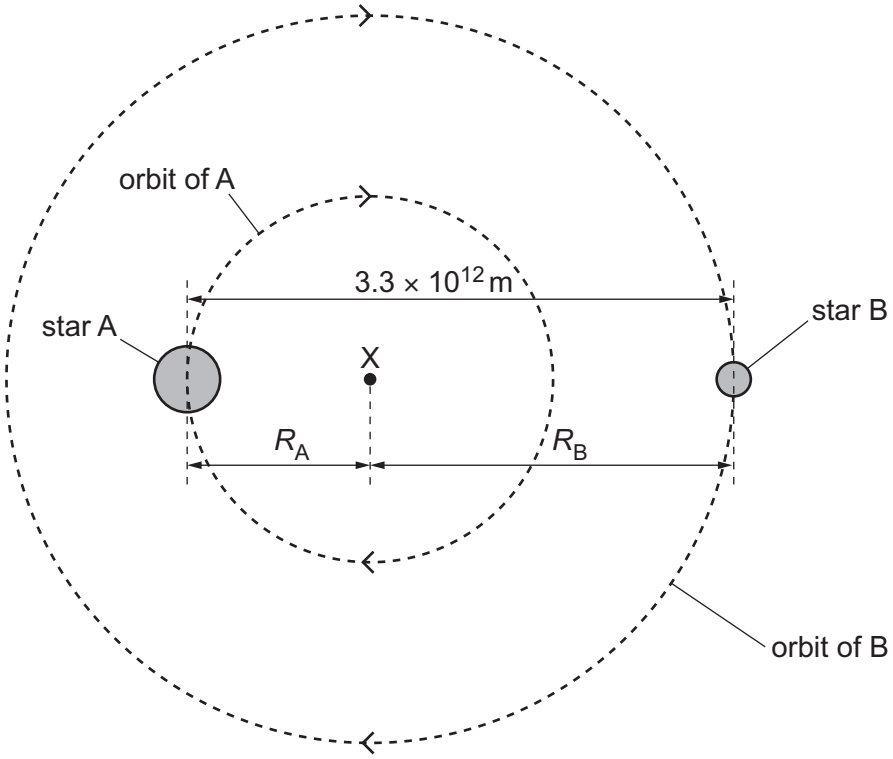


Fig. 1.1

The radius R_B of the orbit of star B is double the radius R_A of the orbit of star A.

(i) Use Newton's law of gravitation to calculate the magnitude of the gravitational force exerted by each star on the other.

force = ... N [2]

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(ii) Calculate the centripetal acceleration of star A.

acceleration = ... $m\,s^{-2}$ [1]

(iii) Use your answer in **(b)(ii)** to determine the period of the orbit of star A.

period = ... s [3]

(iv) By placing a tick (✓) in each row, complete Table 1.1 to show how the quantities indicated for star B compare with the same quantities for star A.

Table 1.1

	B less than A	B equal to A	B greater than A
centripetal acceleration			
linear speed			
period			

[3]

[Total: 11]

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2 **(a)** The magnitude of the gravitational potential on the surface of a planet of radius R is ϕ. The planet can be considered to be an isolated sphere.

On Fig. 2.1, sketch the variation of the gravitational potential with distance x from the centre of the planet for values of x between R and $4R$.

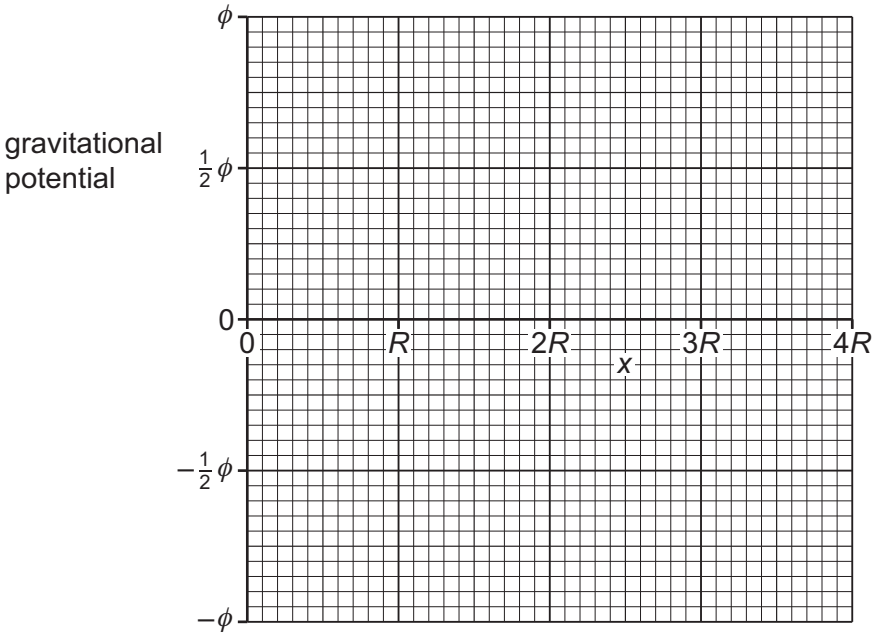


Fig. 2.1

[3]

(b) A satellite is in a geostationary orbit above the Earth. At time t = 0, the magnitude of the gravitational potential due to the Earth at the location of the satellite is ϕ.

On Fig. 2.2, sketch the variation of the gravitational potential due to the Earth at the location of the satellite for values of t between t = 0 and t = 24 hours.

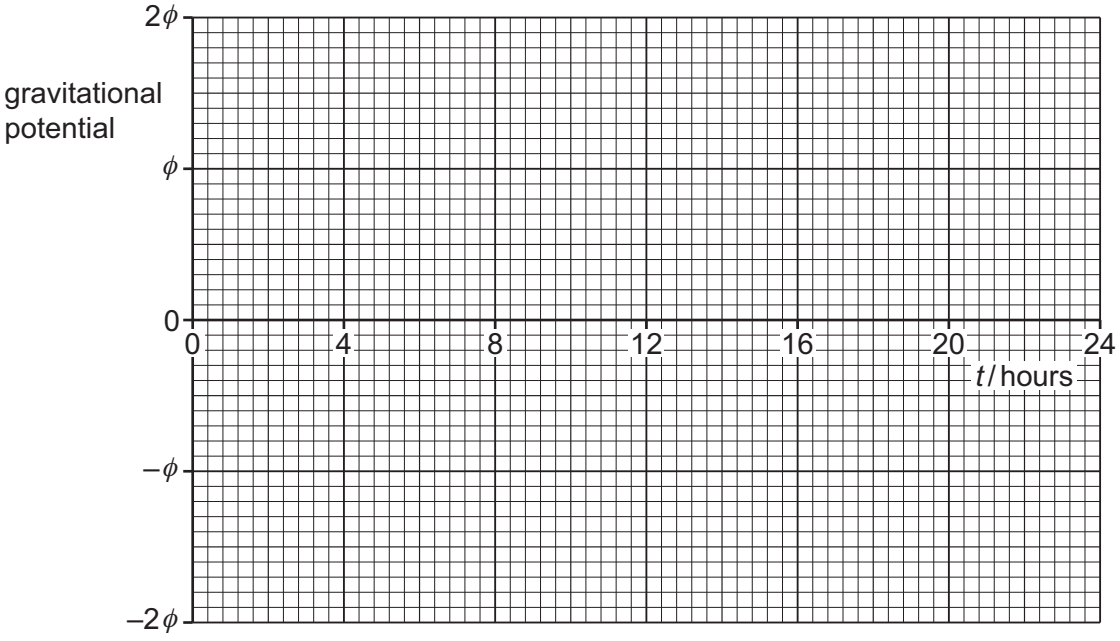


Fig. 2.2

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(c) The electric potential difference (p.d.) between two parallel plates is V, as shown in Fig. 2.3.

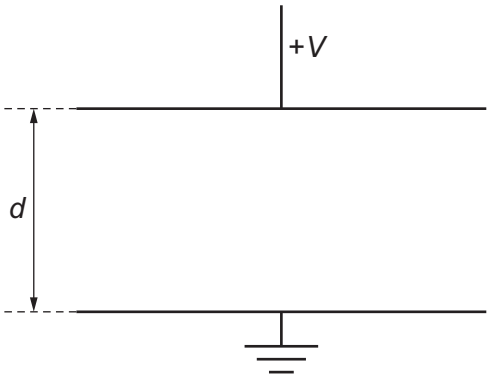


Fig. 2.3

The distance between the plates is d. The region between the plates is a vacuum.

On Fig. 2.4, sketch the variation of the electric potential with distance from the positive plate.

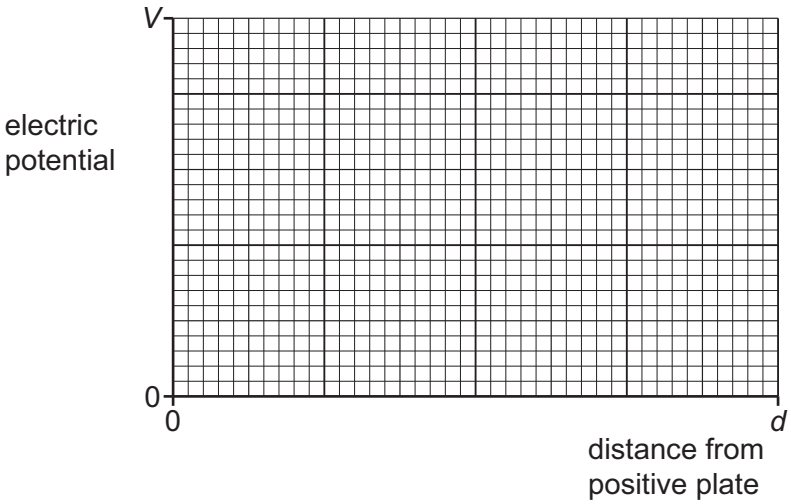


Fig. 2.4

[2]

[Total: 7]

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3 **(a)** Define gravitational field at a point.

.....

..... [1]

(b) Fig. 3.1 shows an isolated point mass of mass *M*.

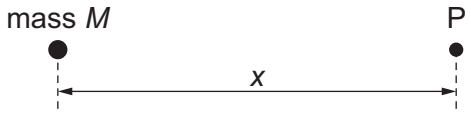


Fig. 3.1

Point P is at distance *x* from the point mass.

(i) By considering the force exerted by the point mass on a test mass of mass *m* placed at P, derive an equation for the gravitational field strength *g* at P, in terms of *M* and *x*. Identify any other symbols you use.

[2]

(ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

(iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]

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(c) Two identical isolated uniform spheres X and Y each have radius R. The centres of the spheres are separated by distance L, as shown in Fig. 3.3.

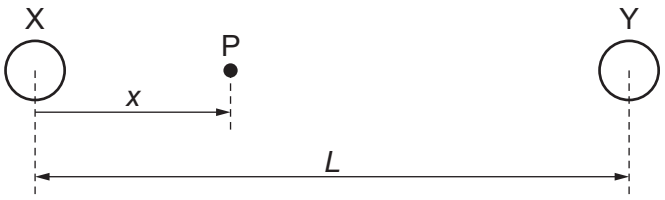


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.
The gravitational field strength at the surface of each sphere is g_0.

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

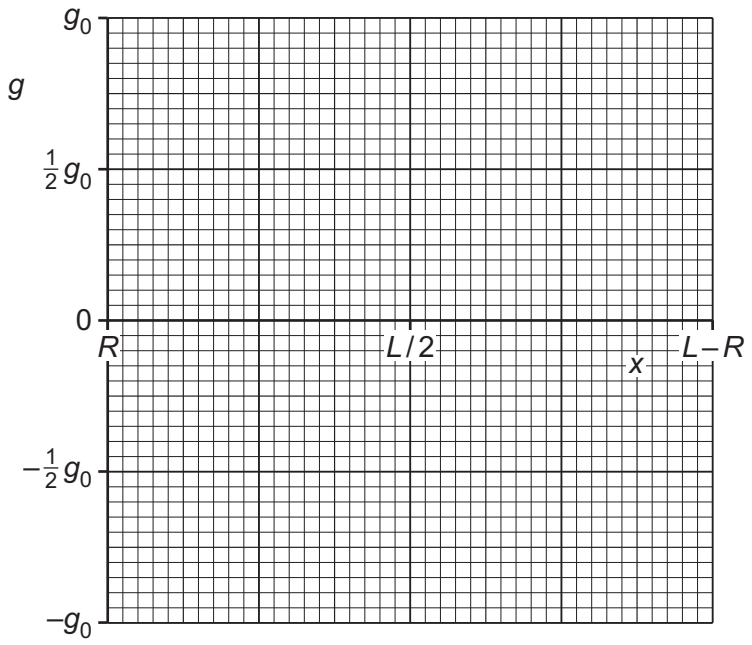


Fig. 3.4

[3]

[Total: 9]

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.....

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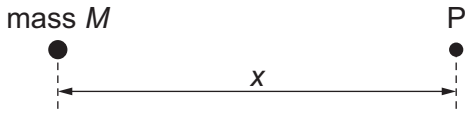


Fig. 3.1

Point P is at distance *x* from the point mass.

(i) By considering the force exerted by the point mass on a test mass of mass *m* placed at P, derive an equation for the gravitational field strength *g* at P, in terms of *M* and *x*. Identify any other symbols you use.

[2]

(ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

(iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]

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(c) Two identical isolated uniform spheres X and Y each have radius R. The centres of the spheres are separated by distance L, as shown in Fig. 3.3.

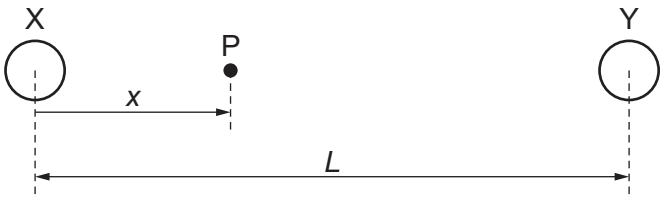


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.
The gravitational field strength at the surface of each sphere is g_0.

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

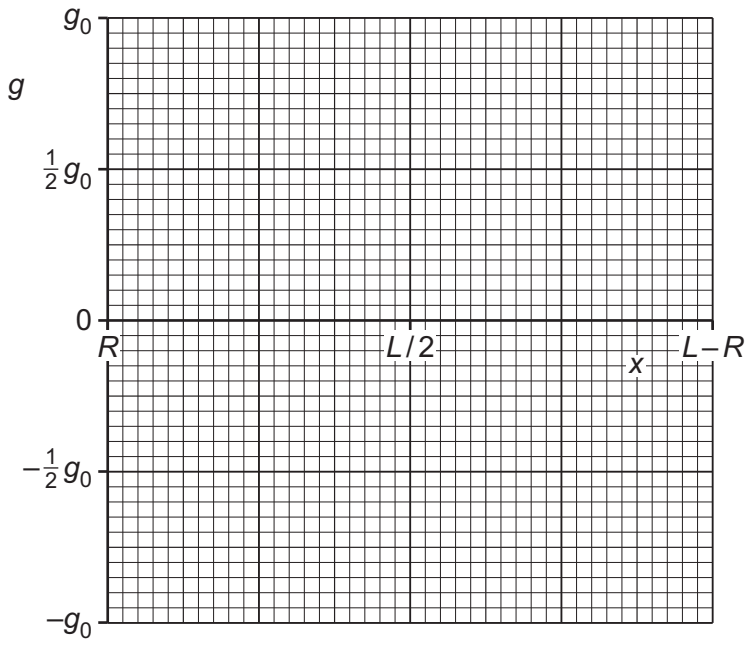


Fig. 3.4

[3]

[Total: 9]

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1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

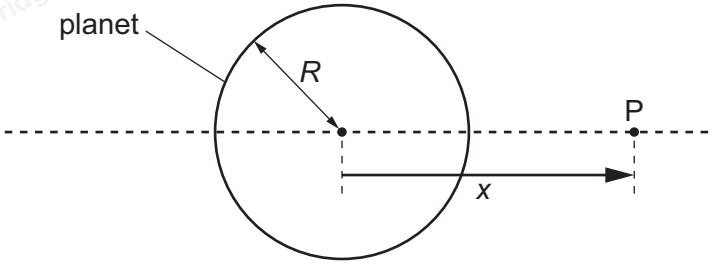


Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

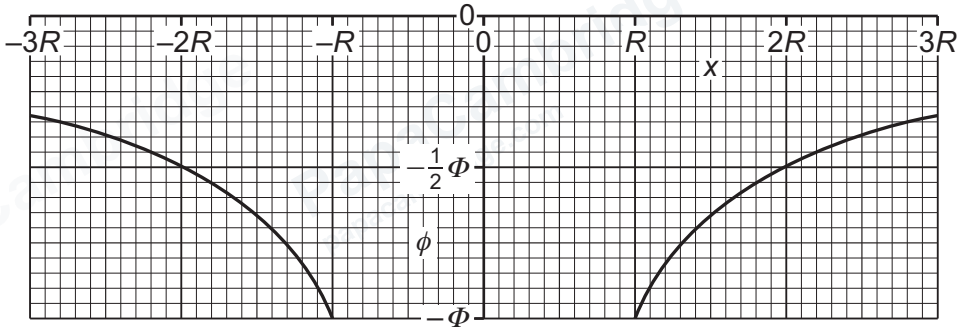


Figure 1.2

(i) Explain why ϕ is negative for all values of x.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use.

M = ... [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet.

g_0 = ... [1]

(iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

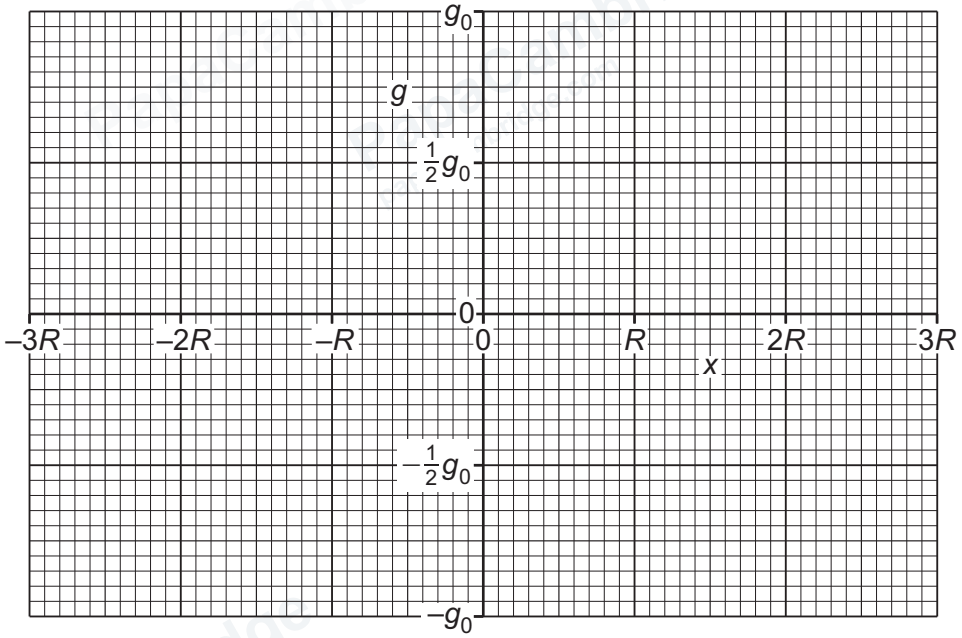


Figure 1.3

[3]

[Total: 10]

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

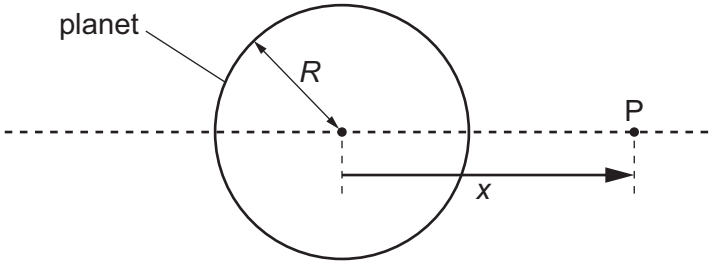


Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

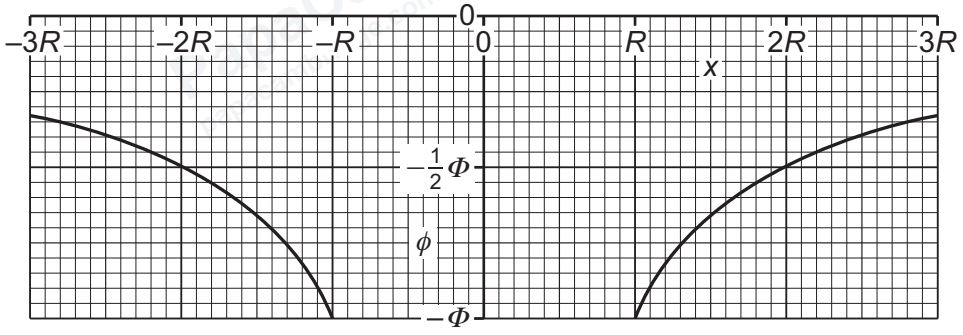


Figure 1.2

(i) Explain why ϕ is negative for all values of x.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use.

M = ... [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet.

g_0 = ... [1]

(iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

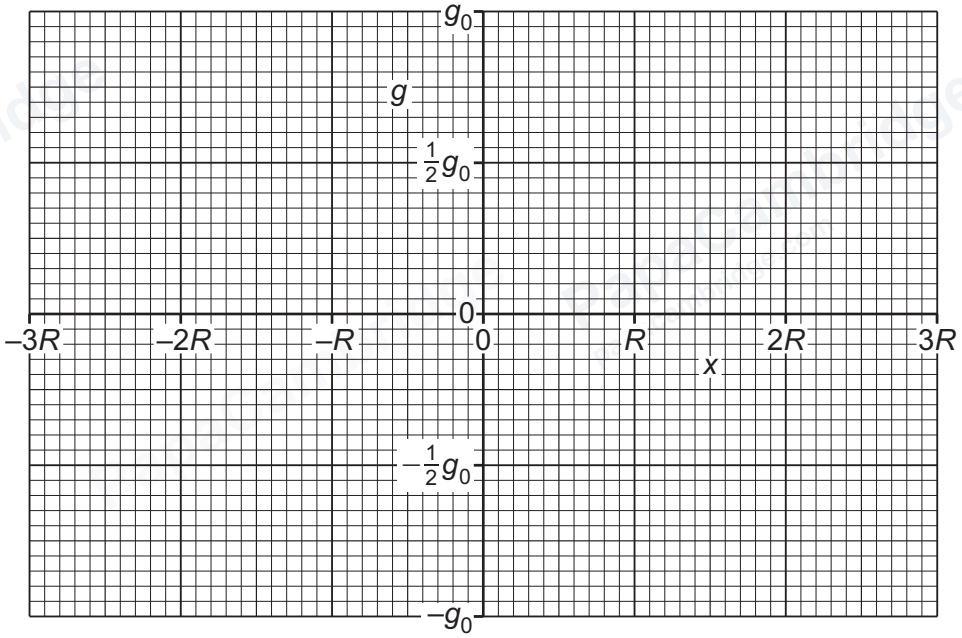


Figure 1.3

[3]

[Total: 10]

2 **(a)** **(i)** Define gravitational field at a point.

.....

..... [1]

(ii) Figure 2.1 shows a horizontal section through a small piece of ground at the Earth's surface and the air at a small distance above it.

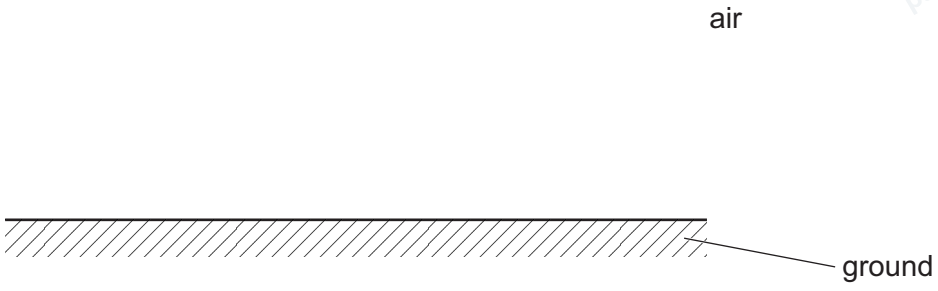


Figure 2.1

On Figure 2.1, draw four field lines to represent the gravitational field of the Earth in the air above the ground. [2]

(b) The Earth may be considered to be an isolated uniform sphere of radius 6.37×10^6 m.

(i) By considering the gravitational field strength at the Earth's surface, calculate a value for the mass of the Earth. Give your answer to three significant figures.

mass = ... kg [2]

(ii) Determine, to three significant figures, the gravitational potential at the Earth's surface. Give a unit with your answer.

gravitational potential = ... unit ..... [2]

(iii) A satellite of mass 96.0 kg is in circular orbit at a height of 1.93×10^6 m above the Earth's surface.

Determine the gravitational potential energy of the satellite.

gravitational potential energy = ... J [2]

[Total: 9]

2 A satellite is in geostationary orbit around the Earth. The orbit is circular.

(a) (i) State what is meant by a geostationary orbit.

.....

.....

..... [1]

(ii) State **two** properties of the orbit.

1 .....

.....

2 .....

.....

[2]

(b) The orbit has radius R and period T.

(i) Determine an expression, in terms of R and T, for the speed v of the satellite in its orbit.

v = ... [1]

(ii) Use Newton's law of gravitation to show that the mass M of the Earth is given by

$$M = \frac{4\pi^2 R^3}{GT^2}$$

where G is the gravitational constant.

[2]

(c) The Earth may be considered to be a sphere of radius 6370 km.
Geostationary satellites orbit at a height of 35 900 km above the surface of the Earth.

Determine a value, to three significant figures, for the mean density of the Earth.

density = ... $kg\,m^{-3}$ [4]

[Total: 10]

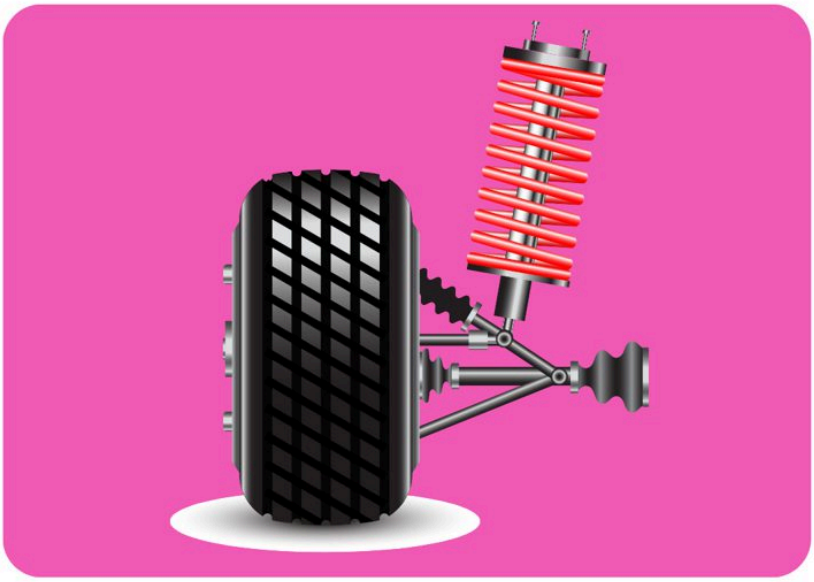
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17 · Oscillations

slides → formulae & definitions → paper4 2024–2026

Chapter 18

Oscillations



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A Level Physics
Prepared and taught by James Gan

Chapter Outline

18.1 Types of Oscillations
18.2 Oscillations in Action
18.3 Characteristics of Oscillatory Motion
18.4 Fundamentals of Simple Harmonic Motion (SHM)
18.5 Visualising Simple Harmonic Motion
18.6 Angular Frequency
18.7 More Equations
18.8 Conversion of energy in SHM
18.9 Damped Oscillations
18.10 Resonance

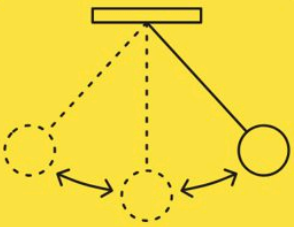


18.1 Types of Oscillations

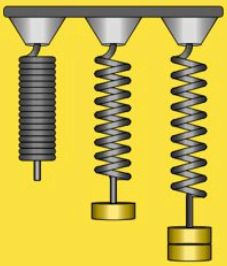
Oscillations

Definition: Oscillations are repetitive back-and-forth movements about a central point.

Examples:



A pendulum swinging



A mass bouncing on a spring



A guitar string vibrating



A seesaw moving up and down

18.1 Types of Oscillations

Oscillations in Real Life



Timekeeping: Clocks use pendulums or quartz oscillations to keep time.



Musical Instruments: Sound is produced through vibrating strings, air columns, or membranes.



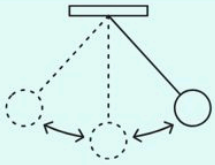
Communication: Radio waves, Wi-Fi, and mobile signals rely on electromagnetic oscillations.



Seismology: Studying earthquake waves to understand the Earth's movements.

18.1 Types of Oscillations

Types of Oscillations

	Definition	Illustration
Free Oscillations	Object oscillates on its own after being displaced, with no continuous external force. ***Eg. A pendulum swinging after being pushed.***	
Forced Oscillations	Object is kept oscillating by a continuous external force. ***Eg. A child on a swing being pushed repeatedly.***	

18.2 Oscillations in Action

Visualizing Oscillations

Here are some ways to observe oscillations when the frequency is too high for the human eye to detect:



A stroboscope flashes light at specific intervals to make fast-moving objects appear as if they are moving slowly or even stationary.



A high-speed camera can capture oscillations at an extremely high frame rate and then play them back in slow motion.

18.3 Characteristics of Oscillatory Motion

Motion in Oscillations

- An oscillating object moves back and forth around an equilibrium position.
- Characteristics of the movement:



Maximum Speed: When passing through the center (equilibrium position), the object moves at its fastest speed.



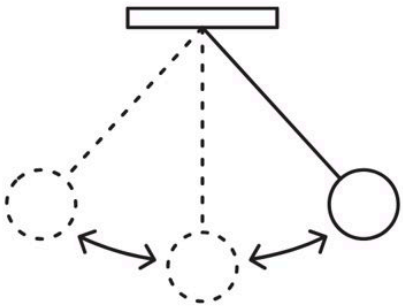
Slowing Down: As the object moves toward its extreme positions, its speed decreases.



Momentary Stop & Direction Change: At the extreme positions, the object comes to a brief stop before reversing direction.

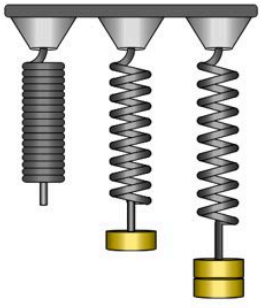
18.3 Characteristics of Oscillatory Motion

Examples



A pendulum:

- Swings **fastest** at the lowest point
- **Slows down** as it reaches the highest points.



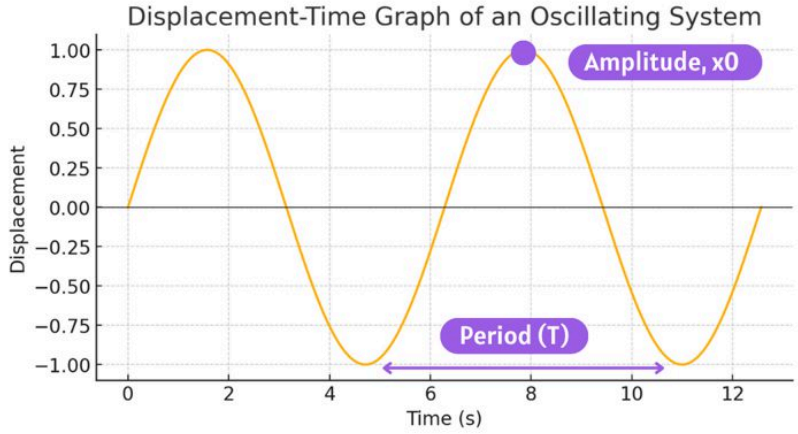
A spring-mass system:

- **Moves fastest** when passing the equilibrium point
- **Slows** at maximum compression/stretch

18.3 Characteristics of Oscillatory Motion

Displacement-Time Graph of an Oscillator

- The displacement-time graph of an oscillating system follows a **sinusoidal wave.**
- **Key Features:**



Important Formula

$$frequency = \frac{1}{period} = f = \frac{1}{T}$$

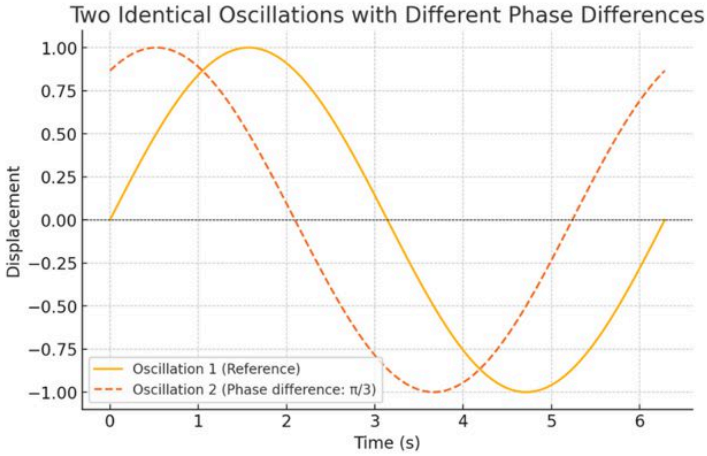
$$period = \frac{1}{frequency} = T = \frac{1}{f}$$

18.3 Characteristics of Oscillatory Motion

Phase Difference

Phase - The position of a point in an oscillation cycle at a given time. It tells us where the oscillating object is within its repetitive motion.

Phase Difference - How much one oscillation is ahead or behind another. It is the angular difference between two oscillating systems.

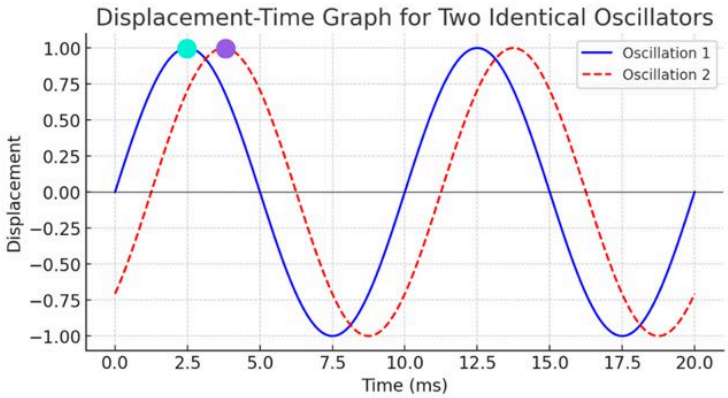


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18.3 Characteristics of Oscillatory Motion

Worked Example:

Calculate the phase difference (in degree) between the following two oscillations.



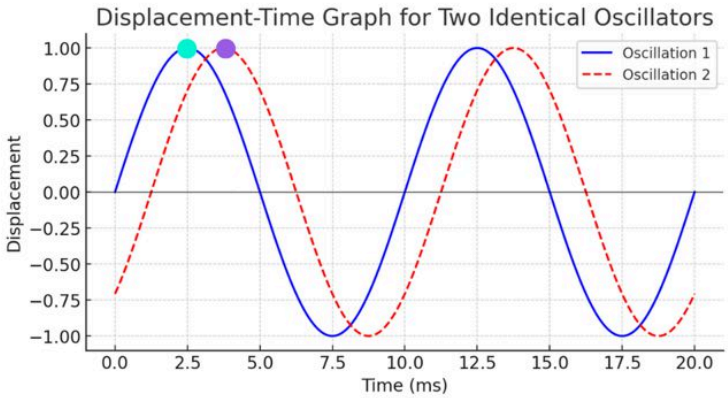
Step I: Determining the period

$$T = 10ms$$

18.3 Characteristics of Oscillatory Motion

Worked Example:

Calculate the phase difference (in degree) between the following two oscillations.



Step I: Determining the period

$$T = 10ms$$

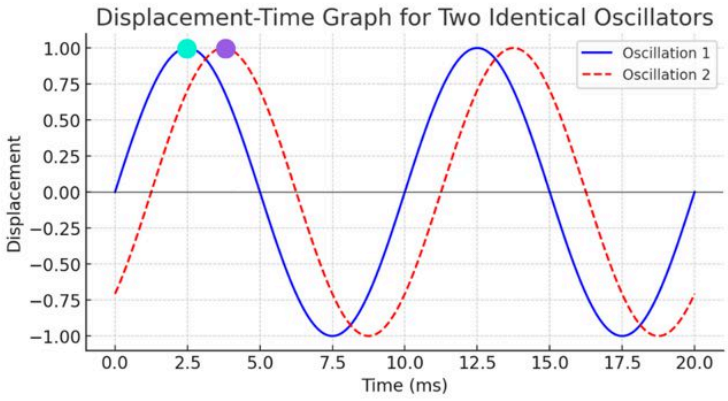
Step 2: Measure the time intervals between two points

$$\Delta t = 1.25\ ms$$

18.3 Characteristics of Oscillatory Motion

Worked Example:

Calculate the phase difference (in degree) between the following two oscillations.



Step I: Determining the period

$$T = 10ms$$

Step 2: Measure the time intervals between two points

$$\Delta t = 1.25\ ms$$

Step 3: Find the phase difference

$$\frac{\Delta t}{T} \times 360° = \frac{1.25}{10} \times 360° = 45°$$

18.4 Fundamentals of Simple Harmonic Motion

Simple Harmonic Motion

SHM is a **special type of oscillation** where the
restoring force is always directed toward the equilibrium position
and is **proportional to displacement**.

18.4 Fundamentals of Simple Harmonic Motion

Simple Harmonic Motion

Oscillation just means moving back and forth repeatedly, like a swing or a pendulum. In Simple Harmonic Motion (SHM), this movement is smooth and follows a regular pattern (like a sine wave).



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SHM is a **special type of oscillation** where the **restoring force is always directed toward the equilibrium position** and is **proportional to displacement**.

This is the force that pulls or pushes the object back to its resting position (called the equilibrium position).

18.4 Fundamentals of Simple Harmonic Motion

Simple Harmonic Motion

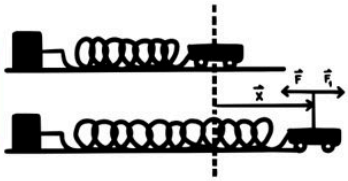
Oscillation just means moving back and forth repeatedly, like a swing or a pendulum. In Simple Harmonic Motion (SHM), this movement is smooth and follows a regular pattern (like a sine wave).



SHM is a **special type of oscillation** where the **restoring force is always directed toward the equilibrium position** and is **proportional to displacement**.

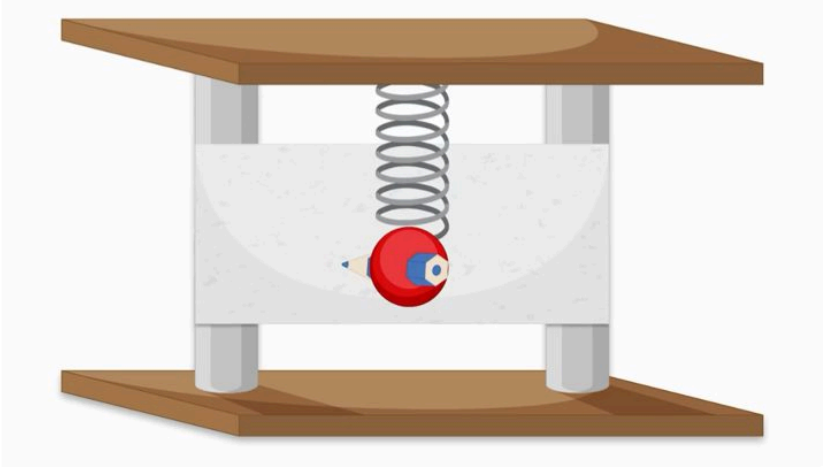
This is the force that pulls or pushes the object back to its resting position (called the equilibrium position).

In SHM, the restoring force gets stronger the further you move from the center.



18.4 Fundamentals of Simple Harmonic Motion

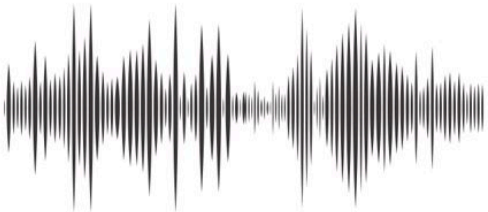
Simple Harmonic Motion



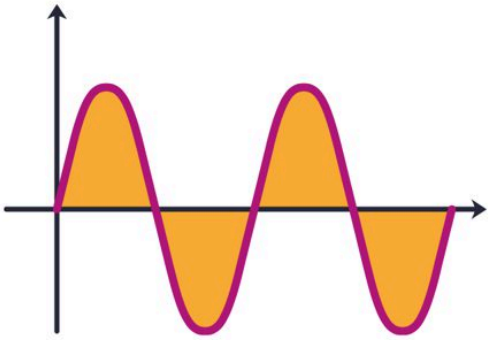
The motion is sinusoidal, meaning it follows a smooth, wave-like pattern over time.

18.4 Fundamentals of Simple Harmonic Motion

Examples



Sound Waves: Air molecules vibrate back and forth in SHM, creating pressure waves that we hear as sound.



Alternating Current (AC): In an electrical circuit, the voltage and current oscillate sinusoidally, behaving similarly to SHM.

18.4 Fundamentals of Simple Harmonic Motion

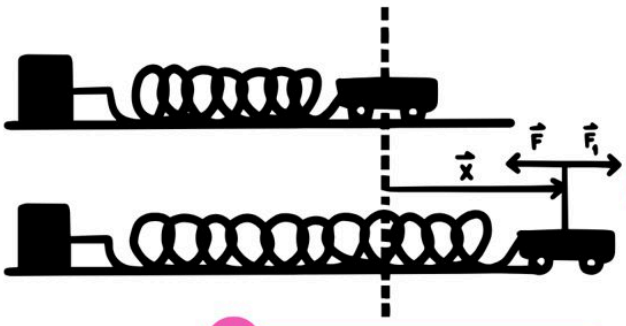
Complex Oscillations & Frequency Breakdown

- Not all oscillations are simple. A real vibrating object (e.g., a drum) can have multiple frequencies at once.
- Mathematically, any complex oscillation can be broken into simpler oscillations of different frequencies (Fourier analysis).
- This idea is important in sound processing, signal analysis, and engineering.



18.4 Fundamentals of Simple Harmonic Motion

Three Requirements for SHM in a Mechanical System



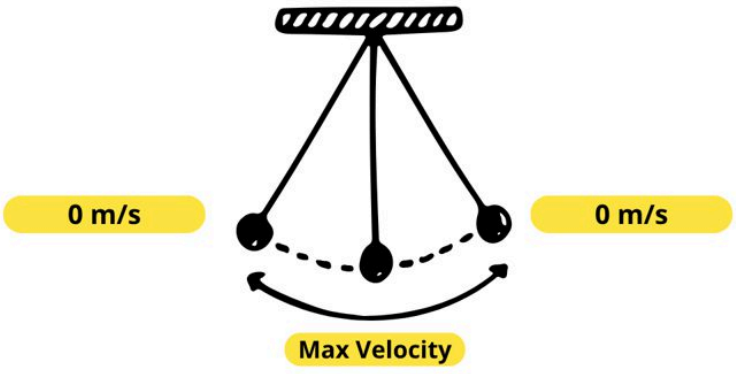
1 ***A mass that oscillates:*** Something that can move back and forth.

2 ***An equilibrium position:*** The point where the object naturally rests if undisturbed.

3 ***A restoring force that brings it back:*** A force must always act to return the object toward equilibrium.

18.4 Fundamentals of Simple Harmonic Motion

Changes in Velocity in SHM (Example: Pendulum Swing)



- **Velocity is maximum** when the object passes through equilibrium.
- **Velocity is zero** at the extreme positions (when the object momentarily stops before reversing direction).
- **Acceleration is greatest at the extreme positions** because the restoring force is strongest there.

18.5 Visualizing Simple Harmonic Motion

Experiment to Demonstrate SHM Graphically

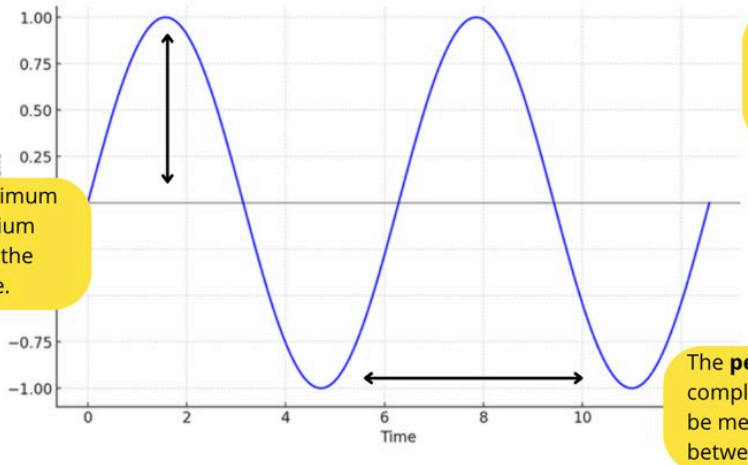


1. Attach the mass to the spring and allow it to hang at its **equilibrium position**.
2. **Pull** the mass down slightly and release it, allowing it to oscillate vertically.
3. Position the **motion sensor** below the mass to record its movement over time.
4. Use computer software to generate real-time graphs of displacement, velocity, and acceleration as functions of time.

18.5 Visualizing Simple Harmonic Motion

Displacement-Time Graph

This graph shows how the position of the mass changes over time.



It has a **sinusoidal shape**, indicating periodic motion back and forth.

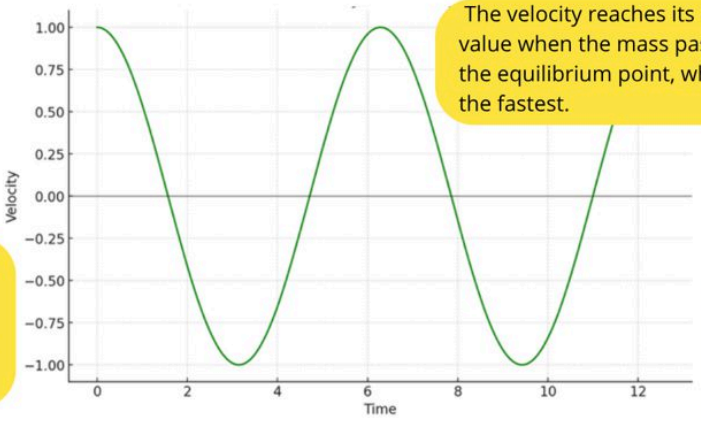
The **amplitude** is the maximum distance from the equilibrium position, showing how far the mass moves on either side.

The **period** is the time taken for one complete cycle of motion, which can be measured by finding the distance between two peaks or troughs.

18.5 Visualizing Simple Harmonic Motion

Velocity-Time Graph

The velocity graph is also sinusoidal but starts at a different point compared to displacement.



The velocity reaches its **maximum** value when the mass passes through the equilibrium point, where it moves the fastest.

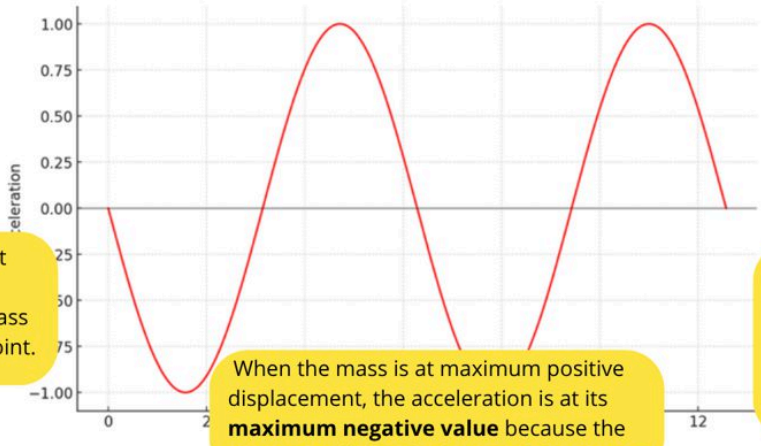
It is **90 degrees out of phase** with the displacement graph. When displacement is maximum (at the peak), velocity is zero because the mass changes direction.

You can find the velocity graph by calculating the **gradient (slope)** of the displacement-time graph, as velocity is the rate of change of displacement.

18.5 Visualizing Simple Harmonic Motion

Acceleration-Time Graph

The acceleration graph is also sinusoidal but is always in the opposite direction of the displacement graph.



It has **negative acceleration** at the beginning because the restoring force is pulling the mass back toward the equilibrium point.

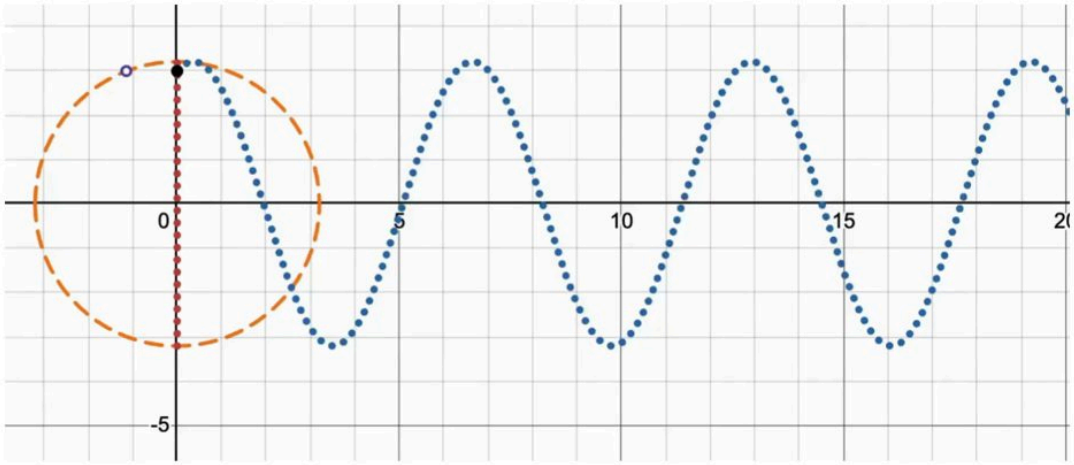
When the mass is at maximum positive displacement, the acceleration is at its **maximum negative value** because the spring is pulling it back toward the center.

The shape of this graph helps to understand how the **restoring force** works continuously to pull the mass back to the equilibrium position.

18.6 Angular Frequency

What is Angular Frequency?

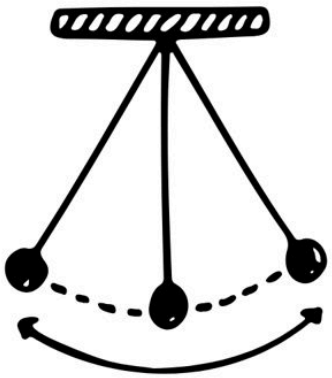
- We know frequency (f) is how many oscillations happen in one second (measured in Hz).
- But in circular motion and oscillations, we often use angular frequency instead.



18.6 Angular Frequency

Why Do We Use Angular Frequency?

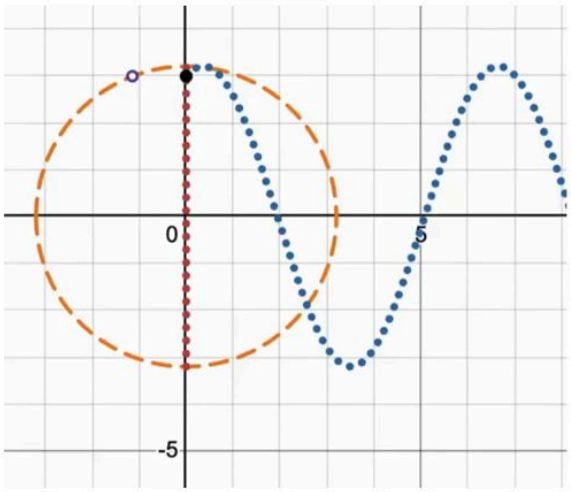
- Frequency tells us how many complete oscillations per second.
- But oscillations are often **connected to circular motion** (e.g., pendulums, springs, waves).
- Angular frequency helps describe motion in terms of angles (radians).



18.6 Angular Frequency

What is Angular Frequency?

Definition: **Angular frequency** tells us how many radians an oscillation sweeps through per second.



Formula

$$\omega = 2\pi f$$

Since $f = \frac{1}{T}$,

Formula

$$\omega = \frac{2\pi}{T} \quad \text{or} \quad T = \frac{2\pi}{\omega}$$

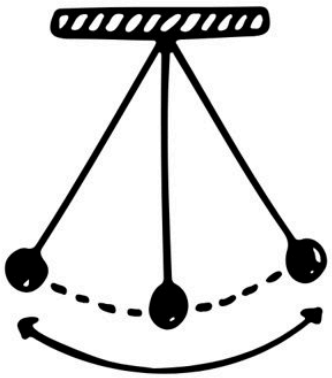
18.6 Angular Frequency

Why Do We Use Angular Frequency?

- Frequency tells us how many complete oscillations per second.
- But oscillations are often **connected to circular motion** (e.g., pendulums, springs, waves).
- Angular frequency helps describe motion in terms of angles (radians).

A pendulum swinging back and forth 2 times per second has:

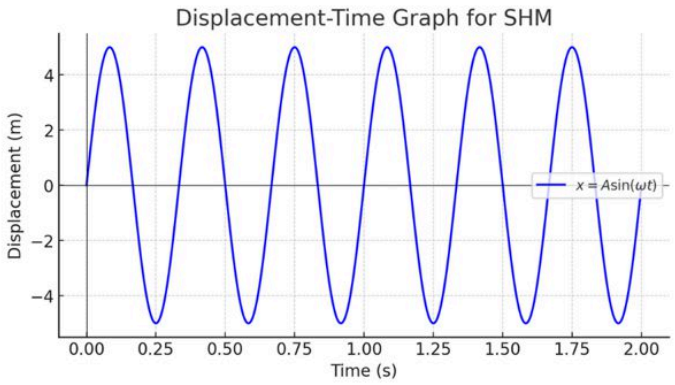
$\omega = 2\pi(2) = 4\pi$ rad / s



18.6 Angular Frequency

Worked Example

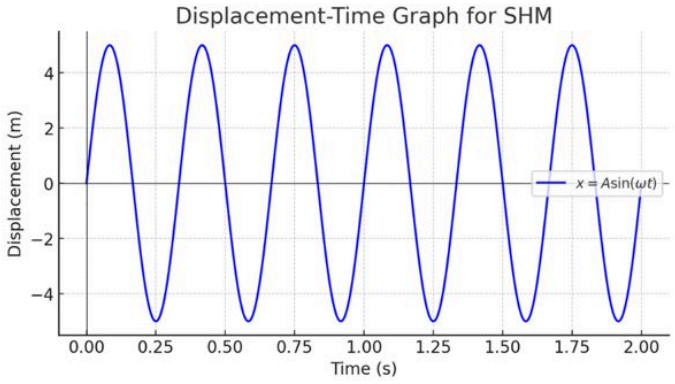
What is the angular frequency of the following displacement-time graph?



18.6 Angular Frequency

Worked Example

What is the angular frequency of the following displacement-time graph?



$$\omega = 2\pi(3) = 6\pi$$

18.7 More Equations

Equations of Harmonic Motion

Simple harmonic motion (SHM) follows a sinusoidal pattern, meaning its displacement can be described using either the sine or cosine function:

Formula

$$x = x_0 \sin \omega t$$

Formula

$$x = x_0 \cos \omega t$$

x_0 (amplitude) – The maximum displacement from the equilibrium position.

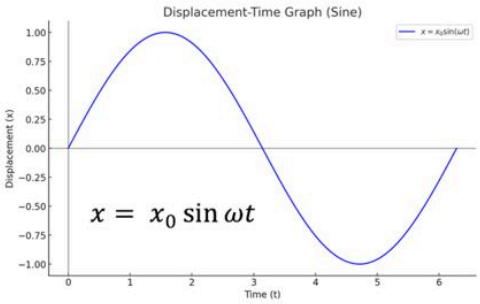
ω (angular frequency) – How fast the oscillation completes one cycle.

t (time) – The time elapsed since the motion started.

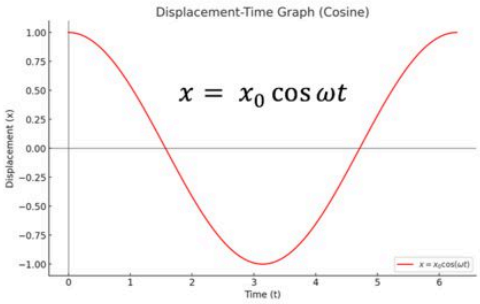
18.7 More Equations

Understanding the Difference

The two equations are similar but describe motion that starts from different points:



✅ The object starts from the equilibrium position (zero displacement) at t = 0 .
✅ Example: A pendulum released from the center moving outward.



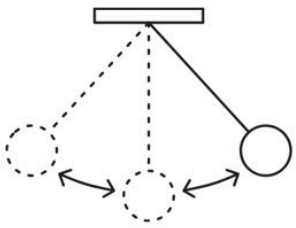
✅ The object starts from maximum displacement at t = 0 .
✅ Example: A pendulum pulled to one side and released.

18.7 More Equations

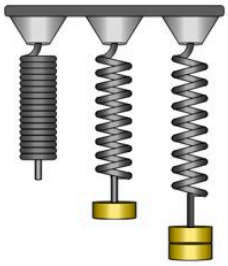
Why Do We Need These Equations?

Oscillations in SHM repeat over time, so we need a mathematical way to predict the position of an object at any given moment.

Examples of SHM systems that follow these equations:



A pendulum swinging



A mass bouncing on a spring

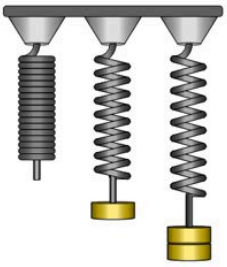
many more!

18.7 More Equations

Worked Example

A mass-spring system oscillates with an amplitude of 0.2 m and an angular frequency of π rad/s. At t = 0 , the mass is at its maximum displacement.

1. Write the equation of motion for this system.
2. Find the displacement at t = 1.5 seconds.



Q1

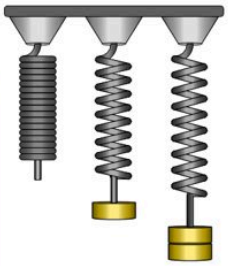
Q2

18.7 More Equations

Worked Example

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1. Write the equation of motion for this system.
2. Find the displacement at t = 1.5 seconds.



Q1

Since the mass is at its maximum displacement at t = 0 , we use:

$$x = x_0 \cos \omega t$$

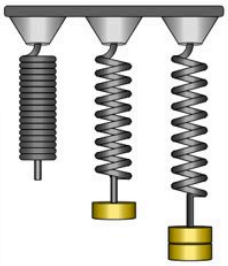
Q2

18.7 More Equations

Worked Example

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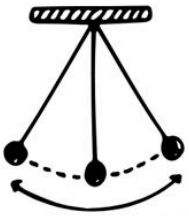
$$x = x_0 \cos \omega t$$

Q2

$$x = x_0 \cos \omega t = 0.2 \cos(\pi \times 1.5) = 0m$$

18.7 More Equations

Acceleration in Simple Harmonic Motion (SHM)



Equation

$$a = -\omega^2 x$$

Why Acceleration is Opposite to Displacement?

The negative sign means acceleration always points towards equilibrium.

Why Angular Frequency Appears in the Equation?

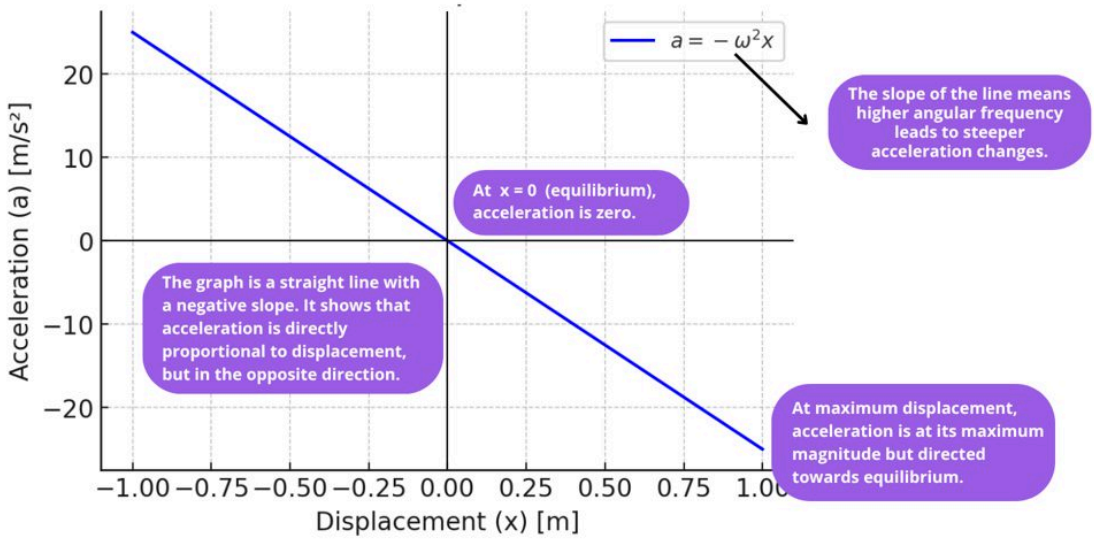
Since acceleration is the rate of change of velocity, a higher angular frequency means a greater acceleration for a given displacement.

Acceleration is directly proportional to displacement.

Eg. This means the farther the pendulum is displaced, the stronger the restoring force, and the greater the acceleration.

18.7 More Equations

Explanation of the Acceleration-Displacement Graph



18.7 More Equations

Equations for velocity

Equation that shows that v depends on x (oscillator's displacement)

$$v = \pm\omega\sqrt{x_0{}^2 - x^2}$$

Maximum speed; x = 0

Formula for maximum speed

$$v = \omega x_0$$

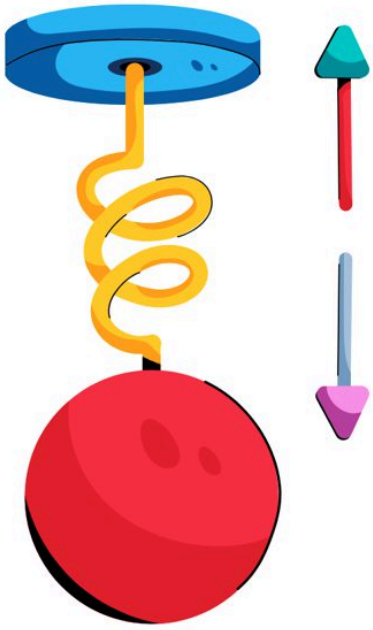
18.7 More Equations

Equations for velocity

$$v = \omega x_0$$

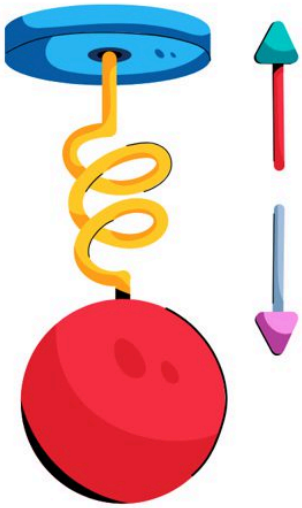
★ Greater amplitude = Greater Maximum Speed
★ The object travels a longer distance in the same time, so it must move faster

★ Greater frequency = Greater Maximum Speed
★ Higher frequency, faster oscillations, greater speed.



18.7 More Equations

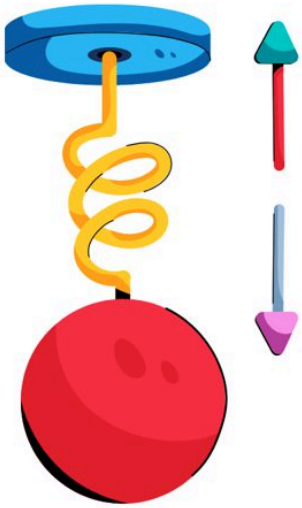
Worked Example



A mass on a spring oscillates with an amplitude of 0.2 m and a frequency of 5 Hz. Find the maximum speed of the mass.

18.7 More Equations

Worked Example



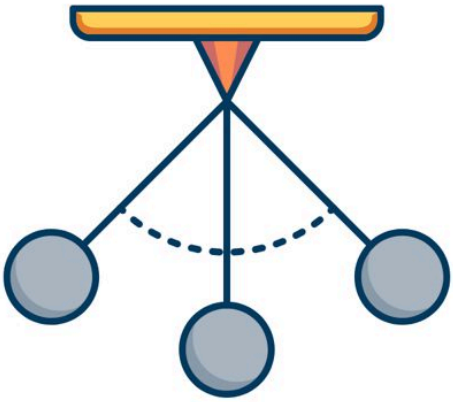
A mass on a spring oscillates with an amplitude of 0.2 m and a frequency of 5 Hz. Find the maximum speed of the mass.

$$v = \omega x_0 = (2\pi 5)(0.2) = 2\pi = 6.28\, m/s$$

18.8 Conversion of energy in SHM

Introduction to Energy in SHM

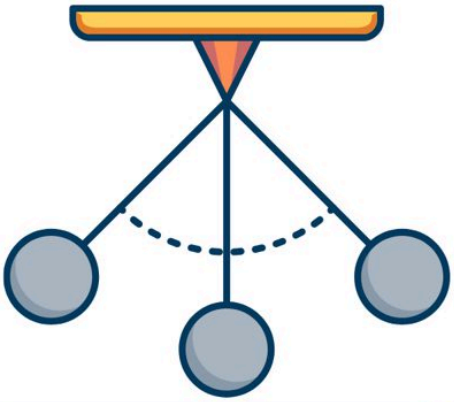
- In SHM, energy continuously transfers between potential energy (PE) and kinetic energy (KE).
- The total mechanical energy remains constant if the system is undamped (no external forces like friction).



18.8 Conversion of energy in SHM

SHM Example: Pendulum

Energy transformation occurs as follows:



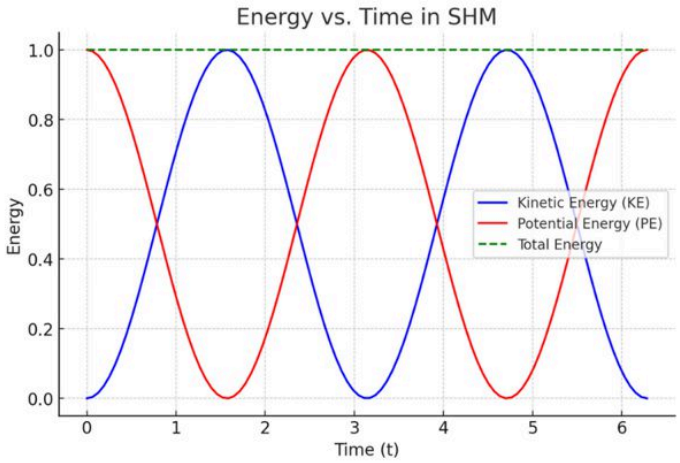
⬆️ **Highest point:**
- **PE is maximum (GPE is highest)**
- **KE is zero (no movement)**

➡️ **Between Maximum Displacement and Equilibrium**
- **As the pendulum moves down, PE decreases while KE increases.**
- **As it moves up, KE decreases while PE increases.**

⬇️ **At Equilibrium Position (x = 0, Lowest Point)**
- **KE is maximum (pendulum moves fastest).**
- **PE is zero (no height difference).**

18.8 Conversion of energy in SHM

Energy Time Graph



Potential Energy (PE) Line

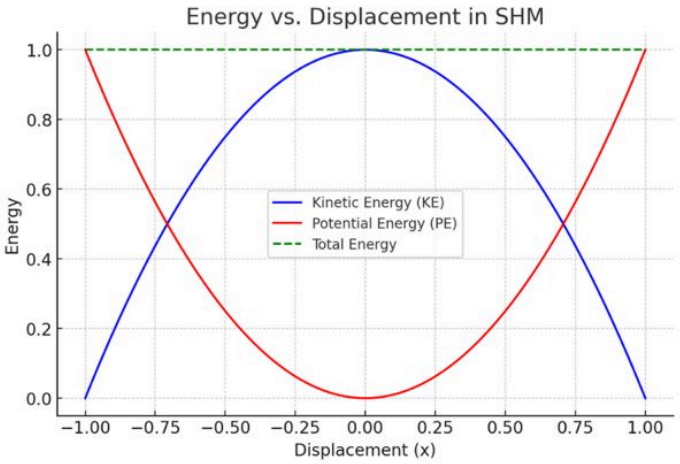
- Starts at a maximum.
- Decreases to zero at equilibrium.
- Increases again at the other extreme.

Kinetic Energy (PE) Line

- Starts at zero.
- Increases to a maximum at equilibrium.
- Decreases back to zero at maximum displacement.

18.8 Conversion of energy in SHM

Energy Displacement Graph



Potential Energy (PE) Line

- Starts at a maximum.
- Decreases to zero at equilibrium.
- Increases again at the other extreme.

18.8 Conversion of energy in SHM

Equations for Energy

Formula of Kinetic Energy

$$E_k = \frac{1}{2} mv^2$$

Formula of Maximum Velocity

$$v = \omega x_0$$

18.8 Conversion of energy in SHM

Equations for Energy

Formula of Kinetic Energy

$$E_k = \frac{1}{2} mv^2$$

Formula of Maximum Velocity

$$v = \omega x_0$$

The total energy in SHM can be calculated using the kinetic energy at the equilibrium position since, at that point, the potential energy is zero.

18.8 Conversion of energy in SHM

Equations for Energy

Formula of Kinetic Energy

$$E_k = \frac{1}{2} mv^2$$

Formula of Maximum Velocity

$$v = \omega x_0$$

The total energy in SHM can be calculated using the kinetic energy at the equilibrium position since, at that point, the potential energy is zero.

$$E_k = \frac{1}{2} m(\omega x_0)^2$$

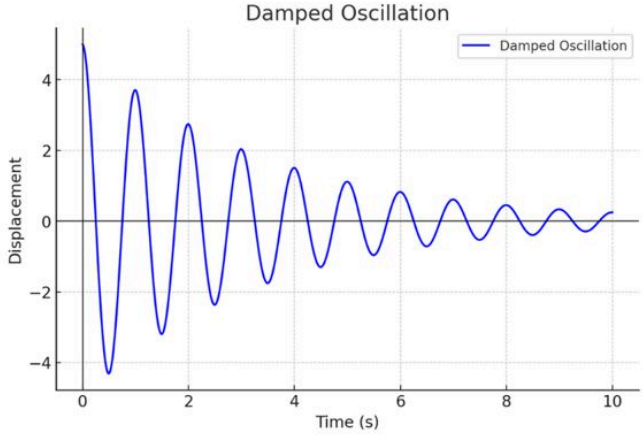
$$E_k = \frac{1}{2} m\omega^2 {x_0}^2$$

18.9 Damped Oscillations

What is Damped Oscillations?

Definition: Damped oscillations occur when the amplitude of an oscillating system gradually decreases over time due to energy loss.

Causes: Energy is dissipated due to friction, air resistance, or internal forces in the material.



18.9 Damped Oscillations

Why Does Amplitude Decrease Exponentially?

Exponential Decay: The amplitude of a damped oscillation decreases exponentially over time, meaning it reduces by a fixed proportion in each cycle.
Why? As the object slows down, the resistive forces acting on it (e.g., air resistance) also decrease, leading to a gradual reduction in amplitude rather than a sudden stop.



18.9 Damped Oscillations

Relationship Between Amplitude and Frequency

Key SHM Property: In simple harmonic motion, frequency depends only on system properties (like mass and stiffness) and is independent of amplitude.
Implication: Even as damping reduces amplitude, the frequency of oscillation remains nearly unchanged unless heavy damping occurs.



18.9 Damped Oscillations

Washing machine drum

A washing machine drum during the spin cycle is an example of **useful damping.**

When the drum spins rapidly, any imbalance in the clothes can cause vibrations. To prevent excessive shaking, washing machines have **dampers and shock absorbers** that gradually reduce the oscillations.



18.10 Resonance

Resonance

Definition

Resonance occurs when an object vibrates with **maximum amplitude** due to a matching external force at its **natural frequency.**

What is Natural Frequency?



A tuning fork produces sound at its natural frequency when struck.



A bridge sways at a certain frequency when the wind blows.

18.10 Resonance

Real-Life Example of Resonance

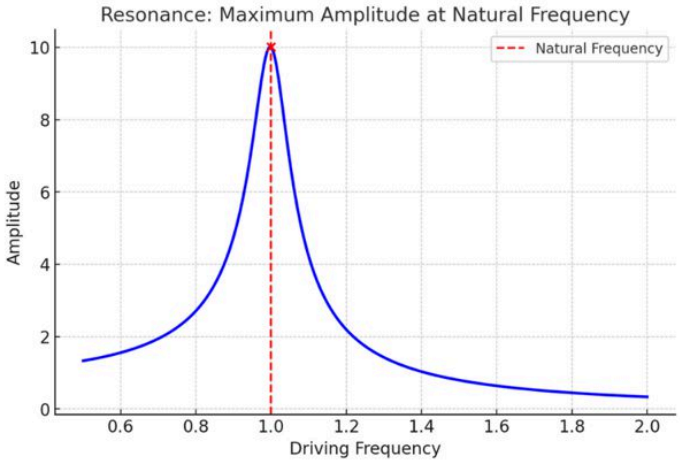
Imagine a singer breaking a glass with their voice.

- Every glass has a **natural frequency** at which it vibrates.
- If the singer produces a sound **matching** this frequency, the glass starts vibrating.
- With enough time and the right volume, the vibrations build up, causing the glass to shatter due to excessive amplitude.



18.10 Resonance

Resonance and Amplitude Growth

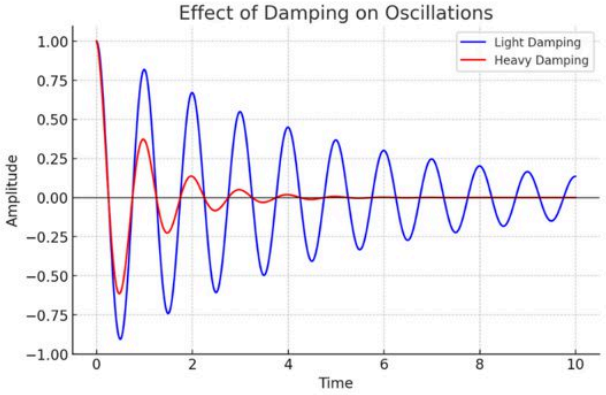


- When the forcing frequency (external driving force) matches an object's natural frequency, its amplitude increases significantly.
- This explains why bridges and buildings can collapse if external forces (wind, earthquakes) match their natural frequency.

18.10 Resonance

The Role of Damping in Resonance

- Damping reduces amplitude in resonance.
- If there is light damping, the oscillations grow large before gradually reducing.
- If there is heavy damping, resonance is minimized, preventing excessive oscillations.



18.10 Resonance

Critical Damping

Definition

Critical damping occurs when the damping force is **just enough** to bring an oscillating system back to equilibrium in the shortest possible time without overshooting or oscillating.

Car Suspension System

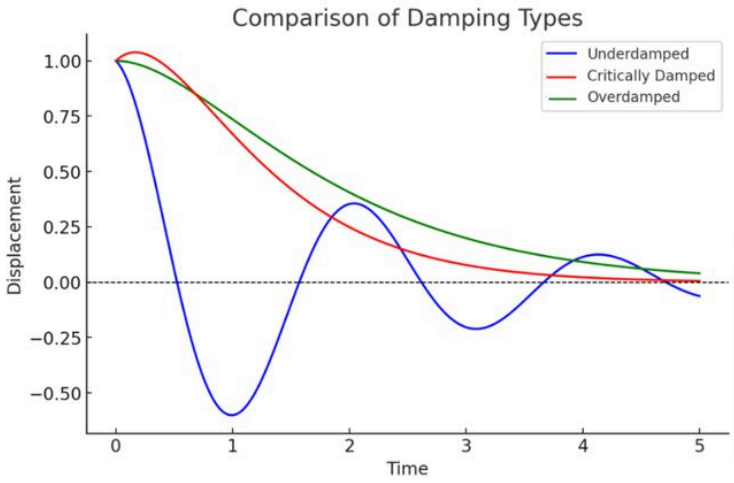


Shock absorbers help a car stabilize quickly after a speed bump. Underdamping causes bouncing, overdamping slows recovery, while critical damping ensures a smooth, fast return.

18.10 Resonance

Critical Damping Graph

Underdamped (blue): Oscillates before settling at zero.



Overdamped (green): Always above the critically damped curve, showing it takes longer to reach equilibrium.

Critically damped (red): Returns to zero quickly without oscillating.

18.10 Resonance

Useful Applications of Resonance



Microwave Ovens: Water molecules resonate with microwave radiation, heating food efficiently.



Musical instruments: Strings and air columns resonate to amplify sound.



MRI machines: Resonance is used in medical imaging to detect different tissues in the body.



Radio tuning: A radio selects the correct station by adjusting resonance with the desired frequency.

Formulae & Definitions — Topic 17 · Oscillations

Period 周期 T — the time for one complete oscillation: $T = 2\pi/\omega$ (s).

Frequency 频率 f — the number of oscillations per unit time: $f = 1/T$ (Hz).

Angular frequency 角频率 ω — $\omega = 2\pi/T = 2\pi f$ (rad s^{-1}).

Displacement 位移 x — the distance from the equilibrium position (m).

Amplitude 振幅 x_0 — the maximum displacement from the equilibrium position (m).

Phase 相位 / Phase difference 相位差 — the point reached in a complete cycle; phase difference can be given as a fraction of a cycle, in degrees or radians.

Simple harmonic motion 简谐振动 — an oscillation in which the acceleration is always proportional to the displacement and directed towards the equilibrium position: $a = -\omega^2 x$ (defining equation). ω is constant and independent of amplitude.

SHM displacement & velocity — if $x = x_0 \sin(\omega t)$ (starting at equilibrium) then $v = v_0 \cos(\omega t)$; if $x = x_0 \cos(\omega t)$ (starting at an extreme) then $v = -v_0 \sin(\omega t)$; $v_0 = \omega x_0$.

v in terms of x — $v = \pm\omega\sqrt{(x_0^2 - x^2)}$; $v = v_0$ at $x = 0$, $v = 0$ at $x = x_0$.

Energy in SHM — total energy = KE + PE = $\frac{1}{2}m\omega^2 x_0^2$ (constant); PE = $\frac{1}{2}m\omega^2 x^2$, KE = $\frac{1}{2}m\omega^2(x_0^2 - x^2)$. Energy swaps between PE and KE every half-cycle.

Damping 阻尼 — loss of energy to the surroundings; light / heavy / critical damping (critical returns to equilibrium without oscillating).

Free oscillation 自由振动 — oscillating at its own natural frequency (independent of amplitude).

Forced oscillation 受迫振动 — oscillation driven by an external periodic force; it oscillates at the driving frequency.

Resonance 共振 — maximum amplitude when the driving frequency equals the natural frequency.

Effect of damping on resonance — damping lowers the resonant frequency, reduces the peak amplitude and broadens the peak.

Uses of resonance — musical instruments, microwave cooking, MRI; damaging example: earthquakes (dampers reduce the amplitude).

点评 · Insight

重点是 S.H.M 的定义：用牛顿运动定律分析振动系统，得到 a 与 x 的关系满足 $a = -\omega^2 x$
即证明是简谐振动（否则不是）。周期/频率是固定常量（natural frequency），与振幅无关。

3 A small object of mass 24 g rests on a platform. The platform is attached to an oscillator, as shown in Fig. 3.1.

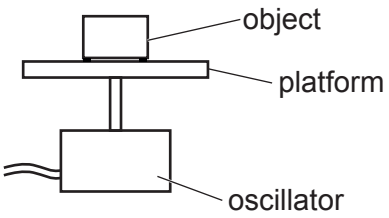


Fig. 3.1

The oscillator moves the platform up and down.

(a) The total energy of the oscillations of the object is 2.2×10^{-4} J.
In one oscillation the object travels a total distance of 14 mm.

Calculate the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [3]

(b) The frequency of the oscillator is fixed, and the amplitude of the oscillations is gradually increased.

(i) Calculate the maximum amplitude of the oscillations so the object does not lose contact with the platform.

amplitude = ... m [2]

(ii) The amplitude of the oscillations is increased so it is greater than the value in **(b)(i)**.

State and explain the position in an oscillation where the object first loses contact with the platform.

.....

.....

.....

..... [2]

[Total: 7]

4 **(a)** State what is meant by resonance.

.....

.....

..... [2]

(b) A small ball is held in place using a stretched string. One end of the string is fixed to a wall and the other end is attached to a vibration generator, as shown in Fig. 4.1.

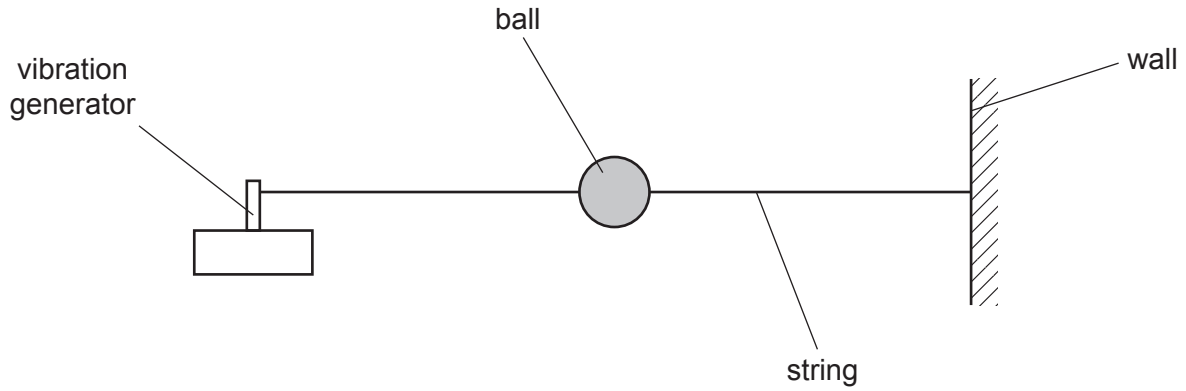


Fig. 4.1

Initially, the vibration generator is switched off.

A student displaces the ball vertically and then releases it. Fig. 4.2 shows the variation of the displacement of the ball with time after it is released.

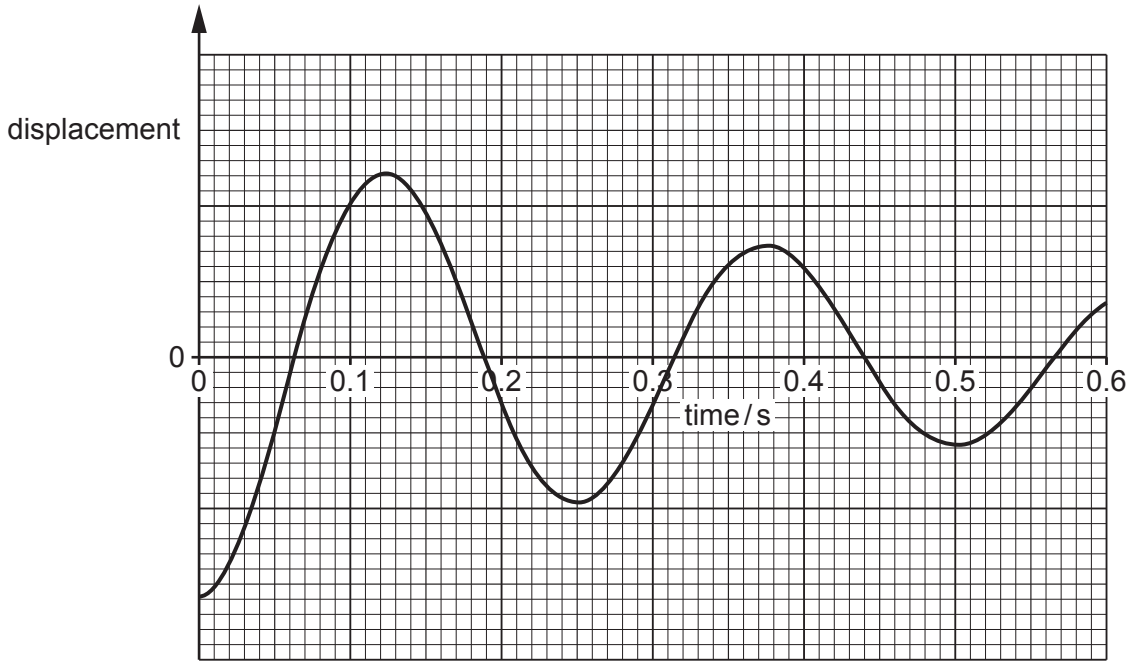


Fig. 4.2

(i) State the name of the phenomenon illustrated by the decrease in the amplitude of the oscillations in Fig. 4.2.

..... [1]

(ii) Explain the decrease with time of the amplitude of the oscillations of the ball.

.....

.....

..... [2]

(iii) Determine the frequency of the oscillations of the ball.

frequency = ... Hz [1]

(c) The vibration generator in **(b)** is switched on and its frequency *f* of vibration is gradually increased from 0 to 10 Hz.

On Fig. 4.3, sketch the variation with *f* of the amplitude of the oscillations of the ball.

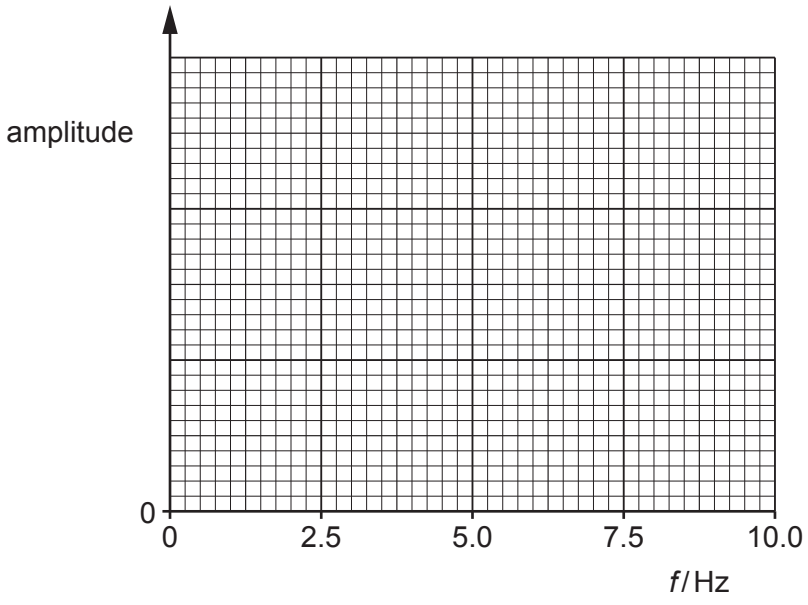


Fig. 4.3

[2]

[Total: 8]

4 A block of mass *m* oscillates vertically on a spring, as shown in Fig. 4.1.

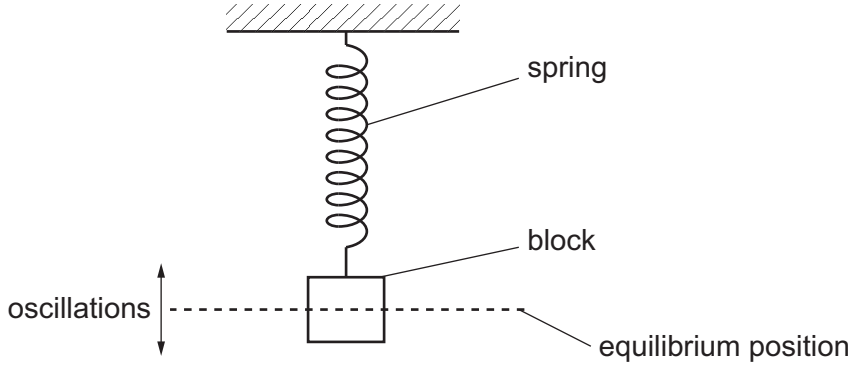


Fig. 4.1

The acceleration *a* of the block varies with displacement *x* from its equilibrium position, as shown in Fig. 4.2.

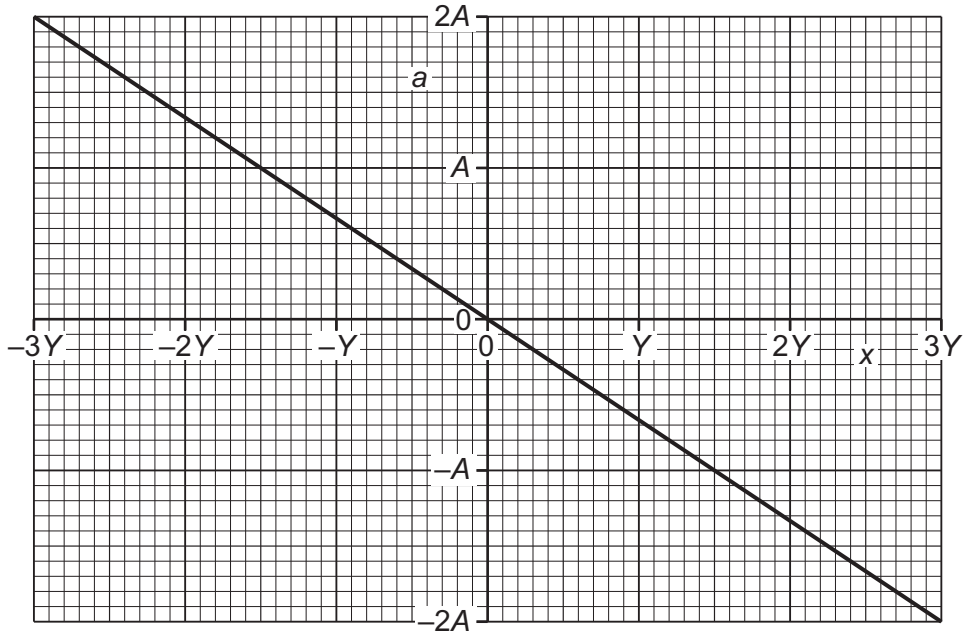


Fig. 4.2

The amplitude of the oscillations is 3*Y* and the maximum acceleration is 2*A*.

(a) Explain how Fig. 4.2 shows that the oscillations of the block are simple harmonic.

.....

.....

..... [2]

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(b) Deduce expressions, in terms of some or all of m, A and Y, for:

(i) the angular frequency ω of the oscillations

ω = ... [1]

(ii) the maximum speed v_0 of the oscillations

v_0 = ... [2]

(iii) the energy E of the oscillations.

E = ... [2]

(c) The period of the oscillations is 0.75 s and the value of $3Y$ is 1.8 cm.

Determine an expression for x in terms of time t, where x is in cm and t is in seconds.

x = ... [2]

[Total: 9]

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4 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A block is suspended from a spring, as shown in Fig. 4.1.

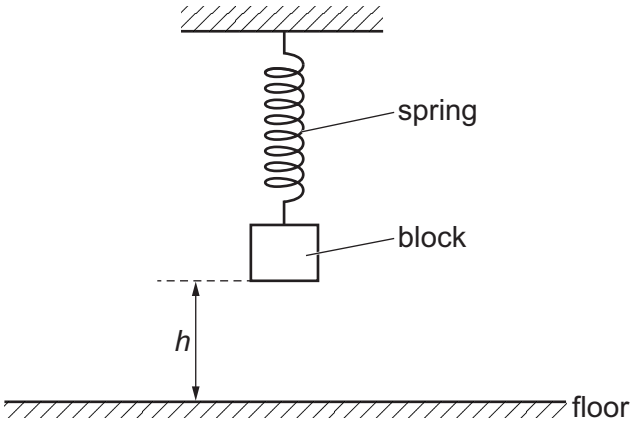


Fig. 4.1

The block is pulled down and released at time $t = 0$. It then oscillates vertically with simple harmonic motion.

Fig. 4.2 shows the variation of the velocity v of the block with height h of the base of the block above the floor.

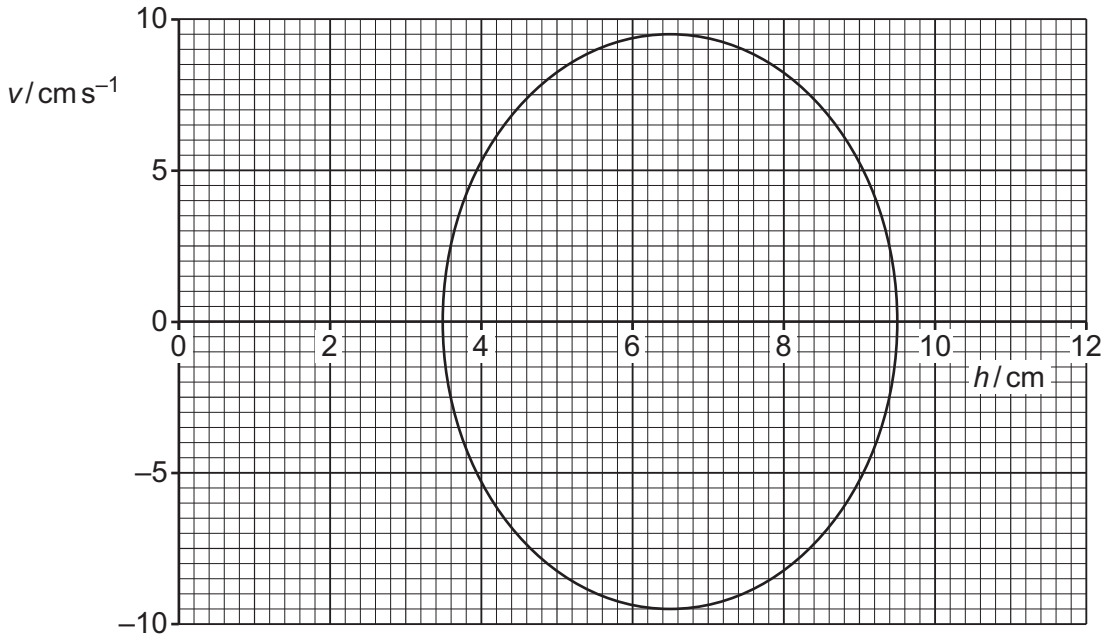


Fig. 4.2

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(i) Determine the amplitude, in cm, of the oscillations.

amplitude = ... cm [1]

(ii) Show that the angular frequency of the oscillations is $3.2\,\text{rad}\,\text{s}^{-1}$.

[2]

(iii) Calculate the period T of the oscillations.

T = ... s [2]

(iv) On Fig. 4.3, sketch the variation of h with time t from $t = 0$ to $t = 6.0\,\text{s}$.

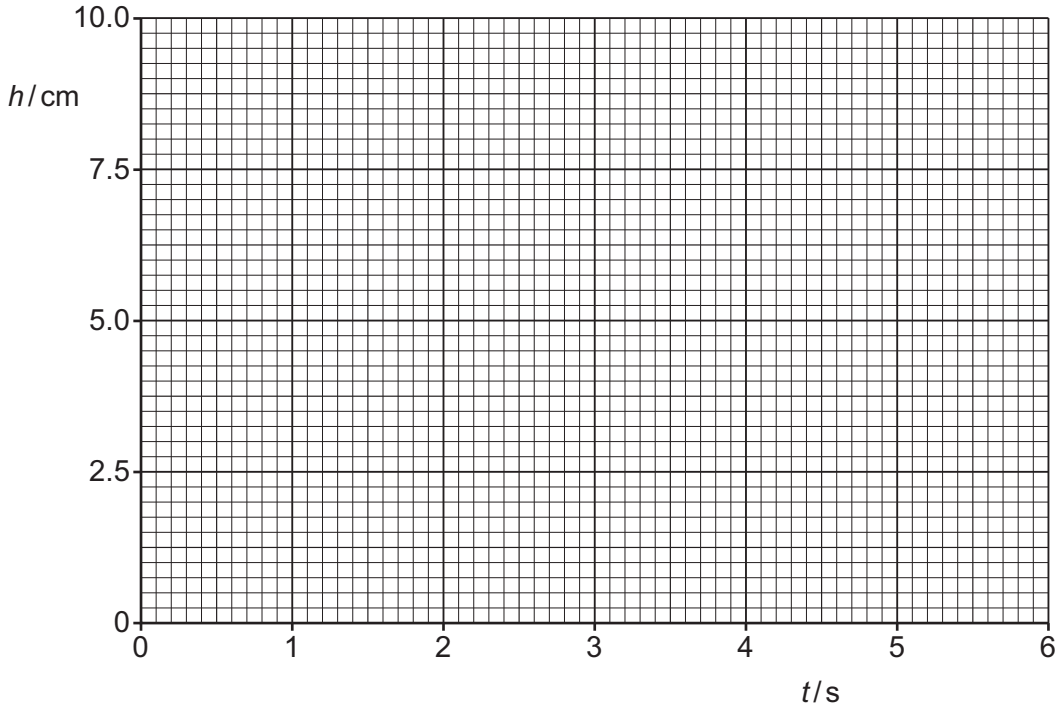


Fig. 4.3

[4]

[Total: 11]

5 Fig. 5.1 shows a pendulum consisting of a metal sphere suspended by a thin string.

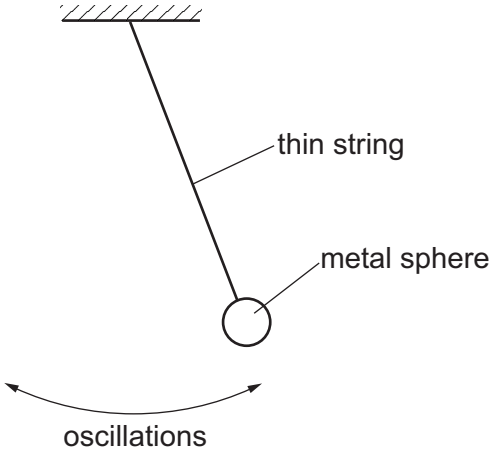


Fig. 5.1 (not to scale)

The sphere undergoes small oscillations about its equilibrium position. The oscillations may be considered to be simple harmonic.

Fig. 5.2 shows the variation with time t of the displacement x of the sphere from its equilibrium position.

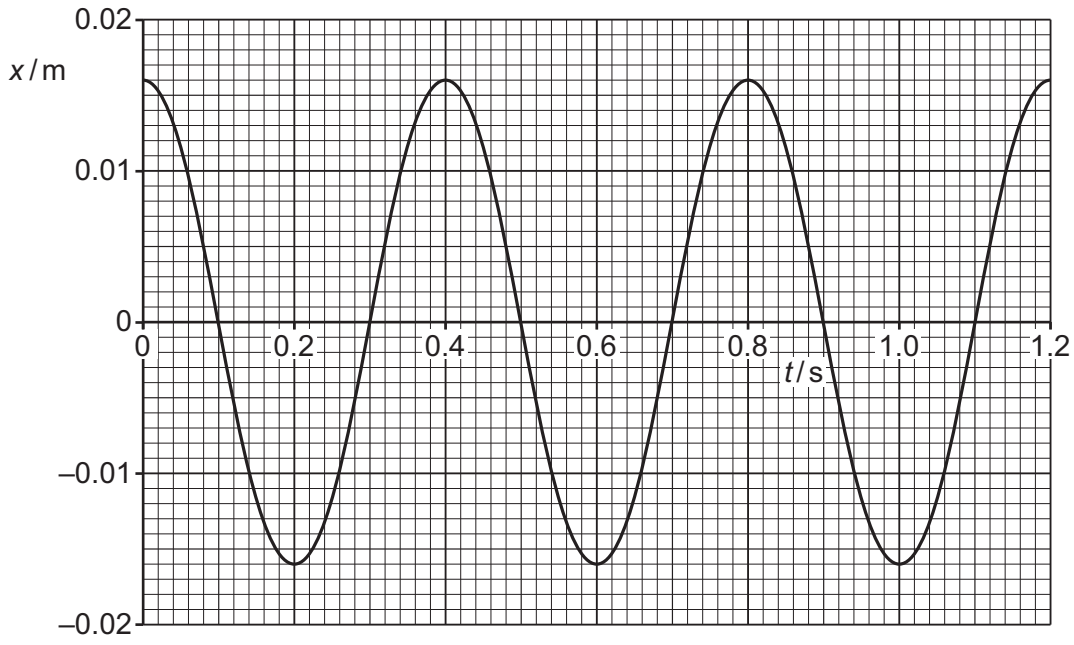


Fig. 5.2

(a) On Fig. 5.1, draw an arrow, from the centre of the sphere, to represent the direction of the resultant force acting on the sphere when it is in the position shown. [1]

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(b) The mass of the sphere is 0.15 kg.

(i) State the amplitude of the oscillations.

amplitude = ... m [1]

(ii) Determine the angular frequency of the oscillations.

angular frequency = ... rad s^{-1} [2]

(iii) Calculate the total energy of the oscillations.

total energy = ... J [2]

(c) On Fig. 5.3, sketch the variation with x of the kinetic energy E_K of the sphere.

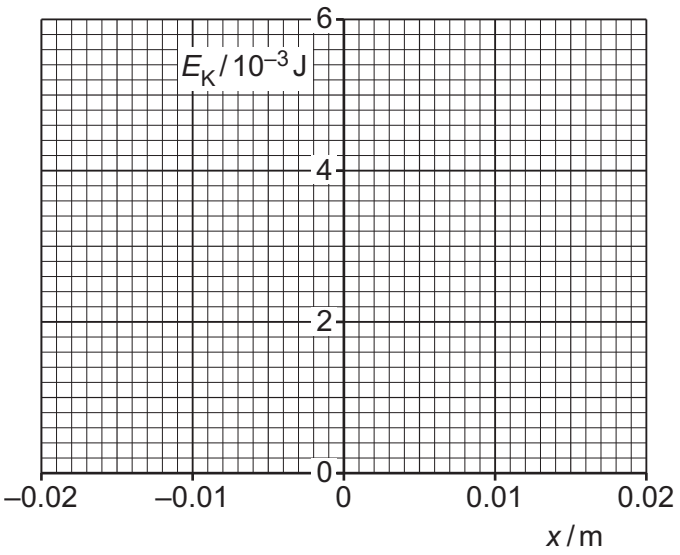


Fig. 5.3

[3]

[Total: 9]

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4 A small crystal is made to vibrate with simple harmonic motion. The variation with time t of the displacement x of one surface of the crystal from its equilibrium position is shown in Fig. 4.1.

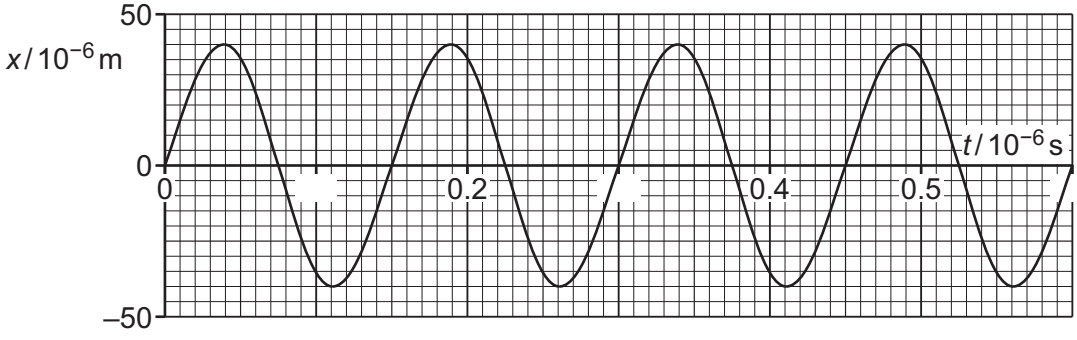


Fig. 4.1

(a) Show that the angular frequency of the vibration of the surface is 4.2×10^{7} rad s^{-1}.

[2]

(b) Determine the maximum acceleration a_0 of the vibration of the surface.

a_0 = ... m s^{-2} [2]

(c) The crystal may be modelled as a single mass of 2.4×10^{-4} kg that vibrates as shown in Fig. 4.1.

Calculate the total energy E of the vibrations.

E = ... J [3]

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(d) The crystal generates ultrasound waves that are used to obtain diagnostic information about internal structures.

(i) The crystal is made from piezoelectric material.

Explain how the crystal is made to vibrate.

.....

.....

.....

..... [2]

(ii) A parallel beam of ultrasound waves is incident on a muscle-bone boundary. Data for muscle and bone are given in Table 4.1.

Table 4.1

material	density / kg m^{-3}	speed of sound / m s^{-1}
muscle	1100	1600
bone	1900	4100

Calculate the percentage of the intensity of the ultrasound beam that is transmitted at this boundary.

percentage transmitted = ... % [3]

[Total: 12]

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4 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A small sphere is suspended from a fixed point P by a string of negligible mass, as shown in Fig. 4.1.

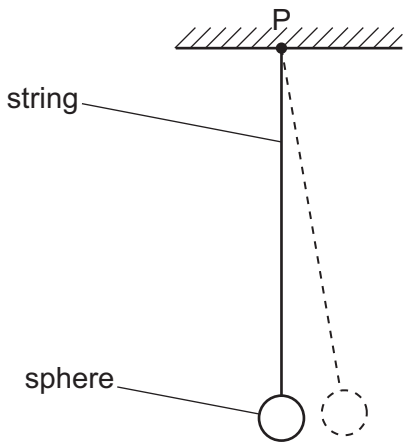


Fig. 4.1

The sphere is given a small horizontal displacement and is then released.

The variation with time of the horizontal velocity v of the sphere is shown in Fig. 4.2.

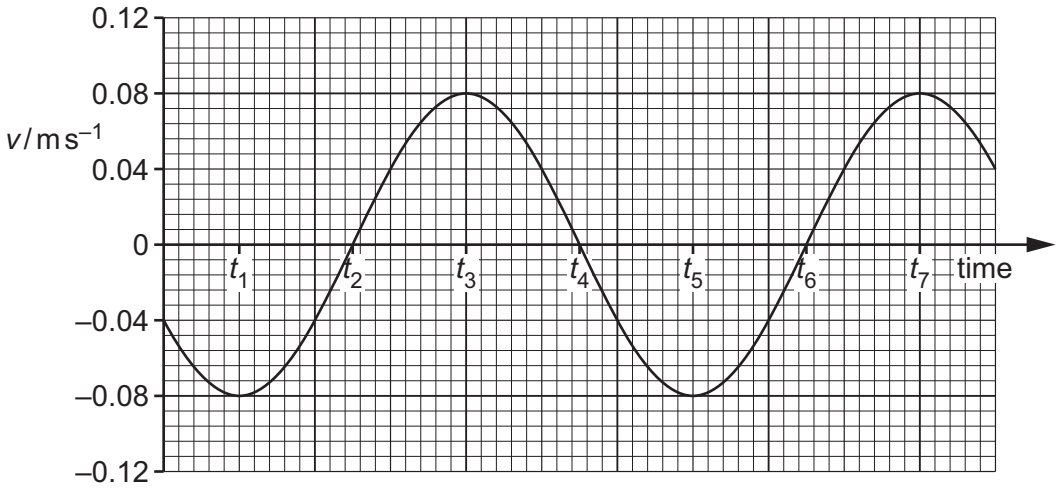


Fig. 4.2

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(i) State two times at which the sphere is passing in the same direction through the equilibrium position.

time ..... and time ..... [1]

(ii) The time interval between t_1 and t_6 is 2.2 s.

Calculate the frequency of oscillation of the sphere.

frequency = ... Hz [2]

(c) The sphere in **(b)** is undergoing simple harmonic motion.

Use your answer in **(b)(ii)** and data from Fig. 4.2 to determine the maximum displacement of the sphere from its equilibrium position.

maximum displacement = ... m [3]

[Total: 8]

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4 **(a)** State what is meant by the frequency of the oscillations of an oscillating object.

.....

..... [1]

(b) An object is oscillating.

Fig. 4.1 shows the variation of the acceleration a of the object with its displacement x from the equilibrium position.

Fig. 4.2 shows the variation of the kinetic energy E_K of the object with time t.

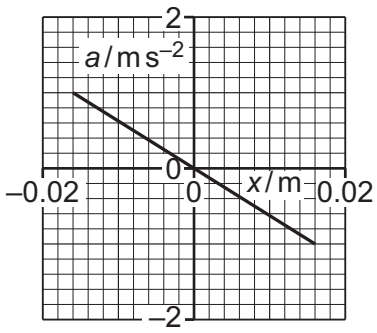


Fig. 4.1

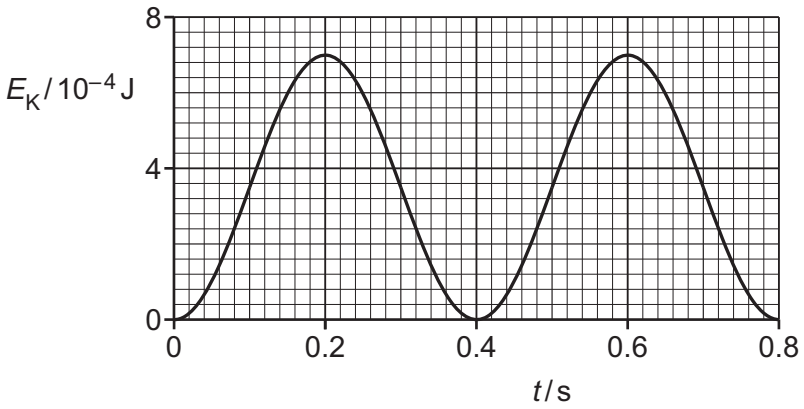


Fig. 4.2

(i) Explain how Fig. 4.2 shows that the period of the oscillations is 0.80 s.

.....

.....

..... [1]

(ii) Calculate the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [2]

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(iii) Apart from the period, frequency and angular frequency of the oscillations, determine **three** other conclusions about the object and its oscillations that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(iv) Describe the interchange between kinetic energy and potential energy during the oscillations. Numerical values are **not** required.

.....

.....

.....

.....

..... [3]

[Total: 10]

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5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

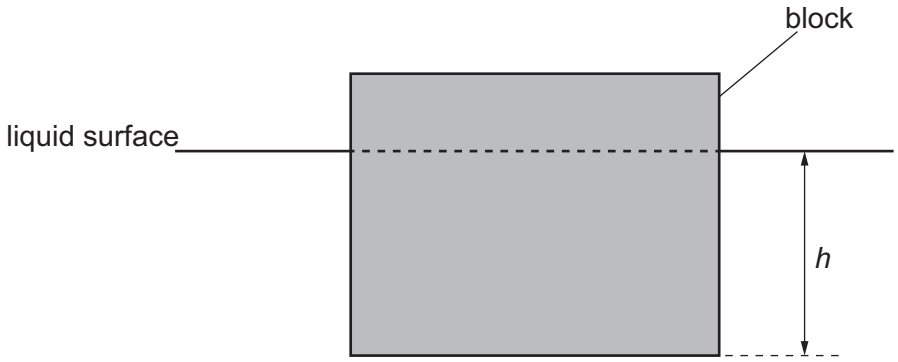


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

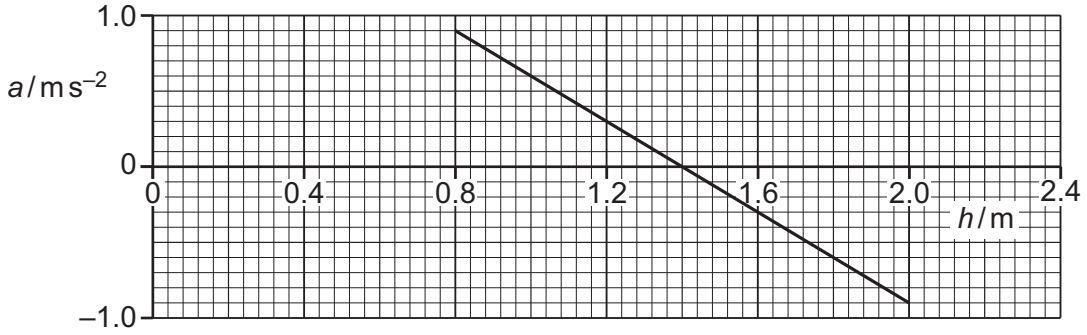


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

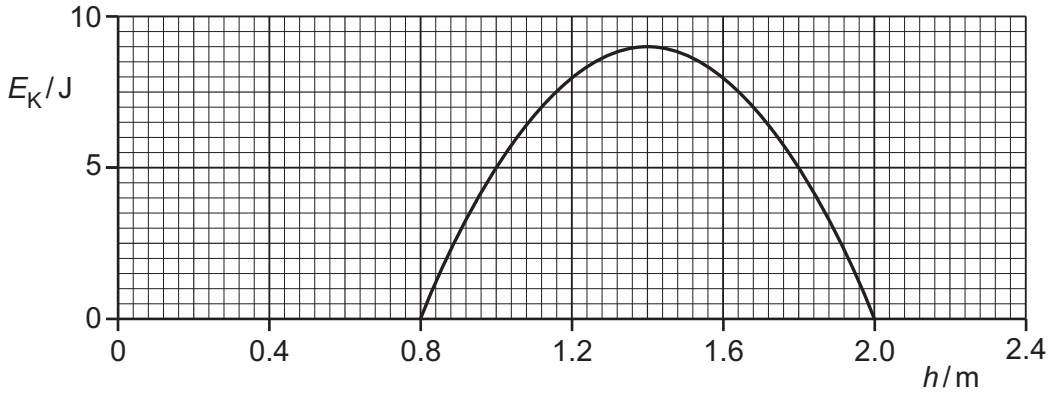


Fig. 5.3

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(a) **(i)** Determine the amplitude of the oscillations.

amplitude = ... m [1]

(ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

(b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) On Fig. 5.4, sketch the variation with h of the potential energy E_P of the oscillations.

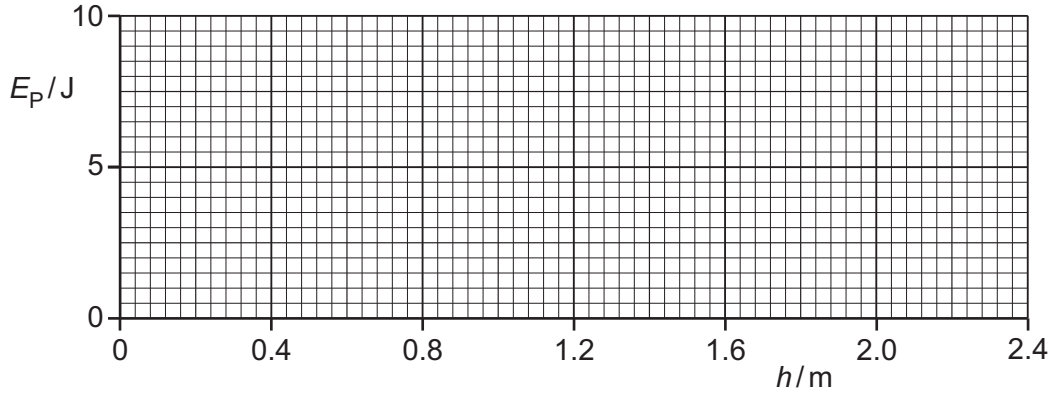


Fig. 5.4

[3]

[Total: 8]

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5 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A block is suspended by a spring. The block oscillates vertically with simple harmonic motion.

The velocity v of the block varies with time t according to

$$v = 0.56 \cos 16t$$

where v is in m s^{-1} and t is in s.

(i) Calculate the period of the oscillation.

period = ... s [1]

(ii) Determine the amplitude x_0 of the oscillation.

x_0 = ... m [2]

(iii) Use your answer in **(b)(ii)** to determine the equation for v in terms of the displacement x of the block, where v is in m s^{-1} and x is in m.

v = ... [1]

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(iv) On Fig. 5.1, sketch the variation of v with x.

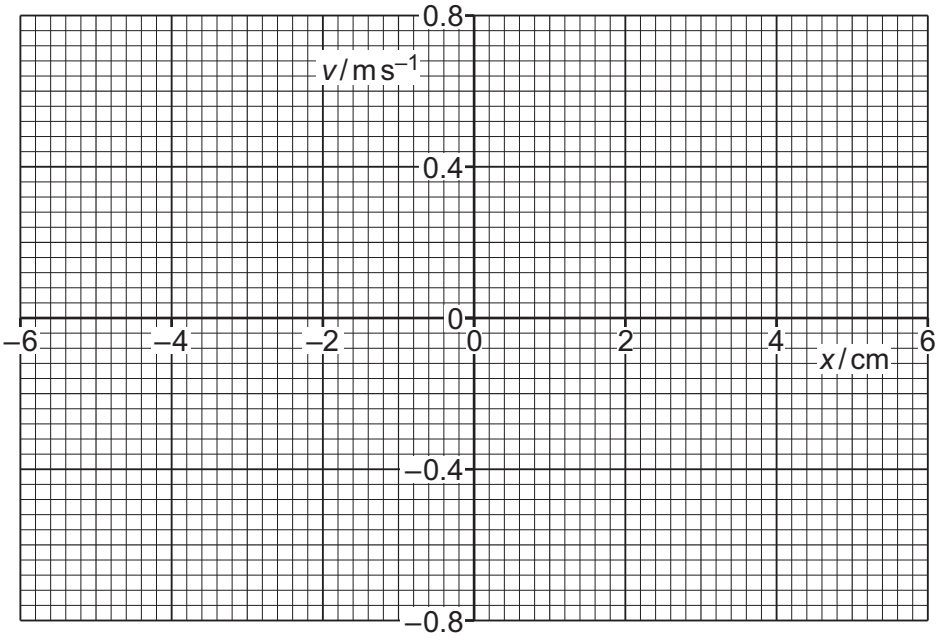


Fig. 5.1

[3]

[Total: 9]

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5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

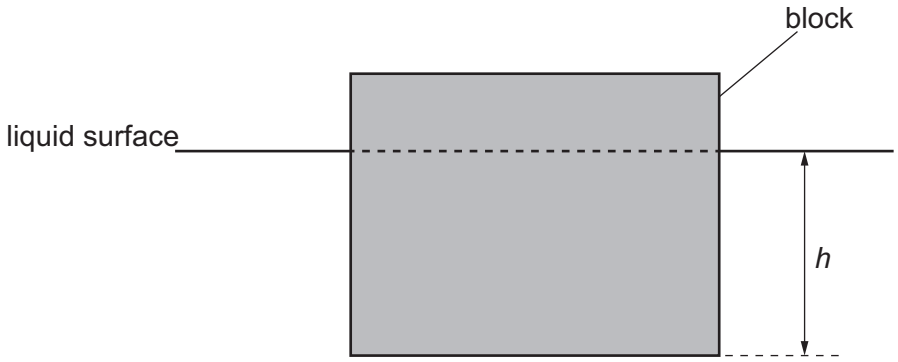


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

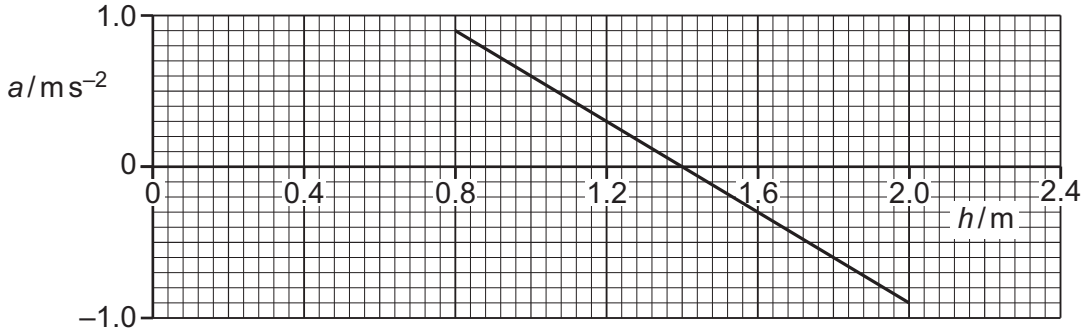


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

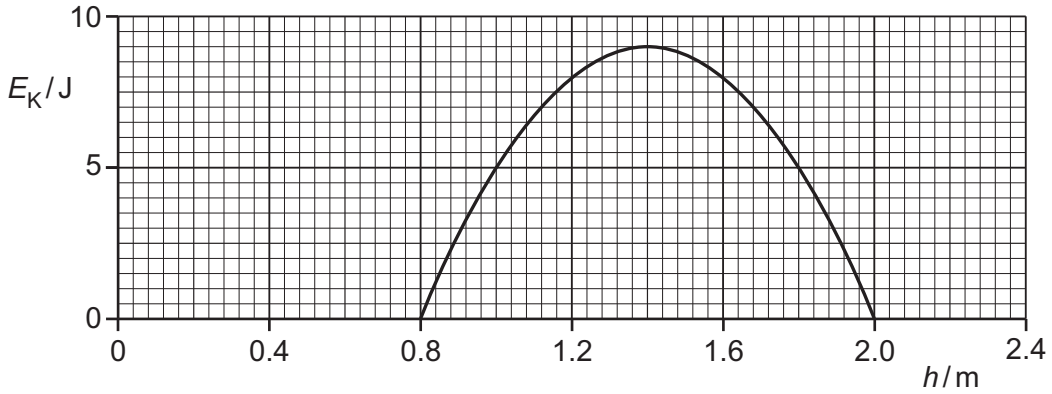


Fig. 5.3

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(a) **(i)** Determine the amplitude of the oscillations.

amplitude = ... m [1]

(ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

(b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) On Fig. 5.4, sketch the variation with h of the potential energy E_P of the oscillations.

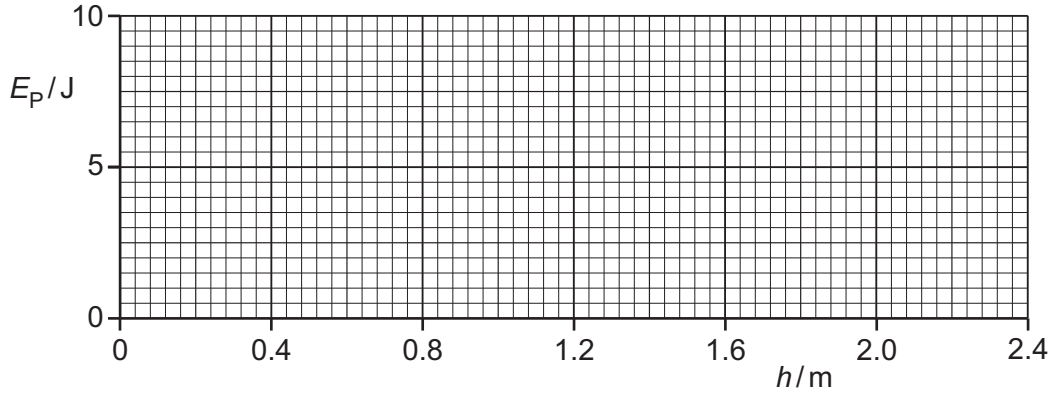


Fig. 5.4

[3]

[Total: 8]

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5 A steel ball on the end of a thin string oscillates with small oscillations, as shown in Fig. 5.1.

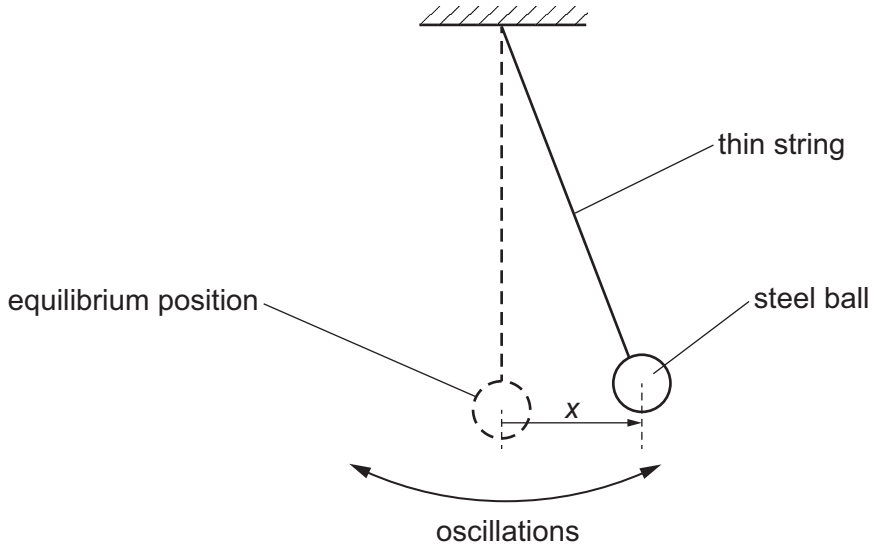


Fig. 5.1 (not to scale)

The displacement of the centre of the ball from its equilibrium position is x.

(a) Fig. 5.2 shows the variation with x of the acceleration a of the ball.

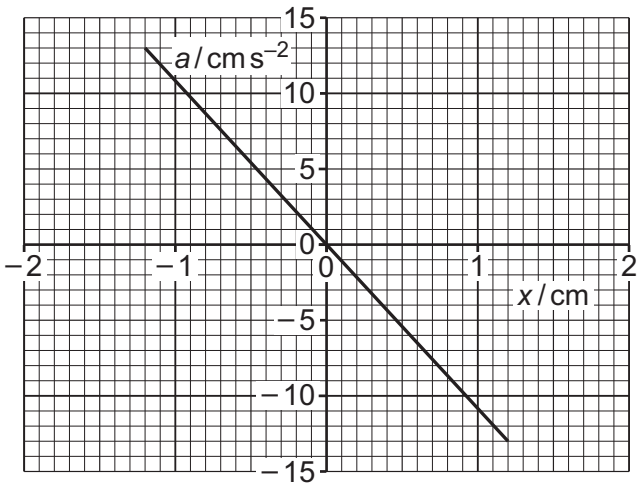


Fig. 5.2

(i) Explain how Fig. 5.2 shows that the oscillations of the ball are simple harmonic.

.....

.....

..... [2]

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(ii) Determine the period T of the oscillations.

T = ... s [3]

(b) At time $t = 0$, when the displacement of the ball has its maximum value, the ball is immersed in a trough containing thick oil so that the ball is just below the surface of the oil. This results in the subsequent motion of the ball being heavily damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) On Fig. 5.3, sketch a possible variation of the displacement x of the ball with t between $t = 0$ and $t = 2T$.

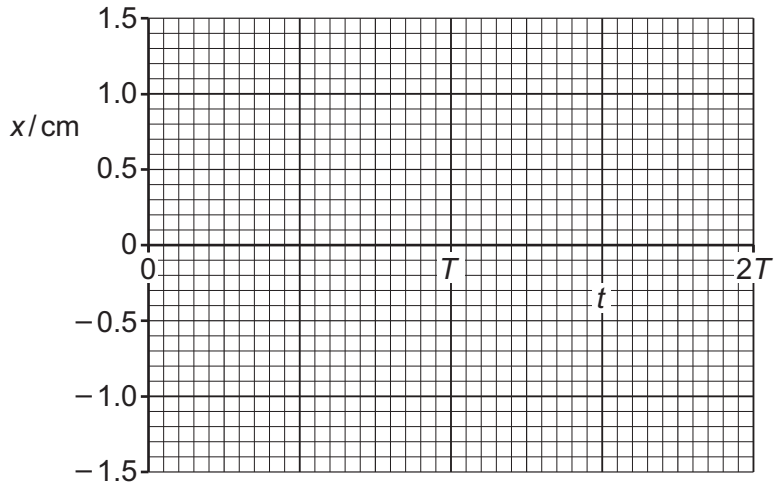


Fig. 5.3

[3]

[Total: 10]

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7 A bar magnet is suspended from a spring. One pole of the magnet oscillates freely in a coil of wire, as shown in Fig. 7.1.

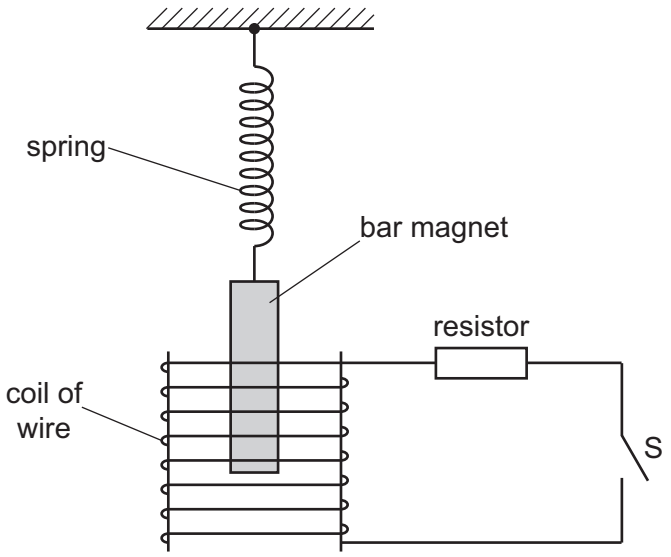


Fig. 7.1

The switch S is initially open.

(a) The switch S is now closed. As a result, the oscillations of the magnet are lightly damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) Describe what is observed to indicate that the damping is light.

.....

..... [1]

(iii) By reference to electromagnetic induction and to conservation of energy, explain why the oscillations are damped.

.....

.....

.....

..... [3]

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(b) The procedure in **(a)** is repeated after replacing the resistor with one of greater resistance.

Suggest, with a reason, the effect of this change on the oscillations.

.....

.....

..... [2]

[Total: 8]

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5 A block of mass 27 g is suspended from a spring so that it rests in equilibrium, as shown in Figure 5.1.

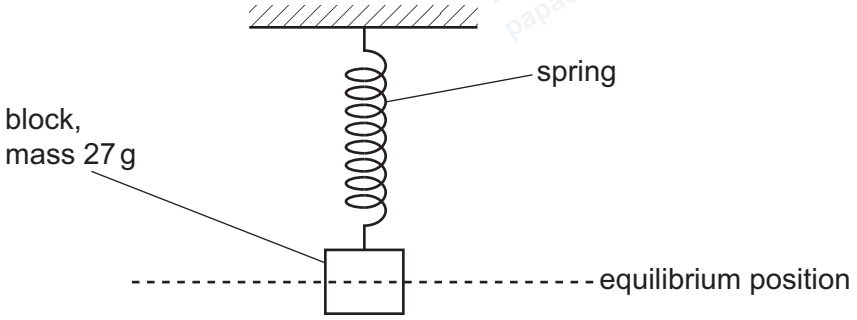


Figure 5.1

The block is displaced downwards through a small distance. The block is released and then oscillates vertically with simple harmonic motion.

The oscillations have a period of 1.7 s and an energy of 13 μJ.

(a) (i) Show that the angular frequency of the oscillations is 3.7 rad s^{-1}.

[1]

(ii) State what is meant by the amplitude of the oscillations.

.....

.....

..... [1]

(iii) Calculate the amplitude of the oscillations.

amplitude = ... m [3]

(b) On Figure 5.2, sketch the variation of the kinetic energy E_K of the block with displacement x from the equilibrium position for one complete oscillation of the block. Add a suitable scale to the x-axis.

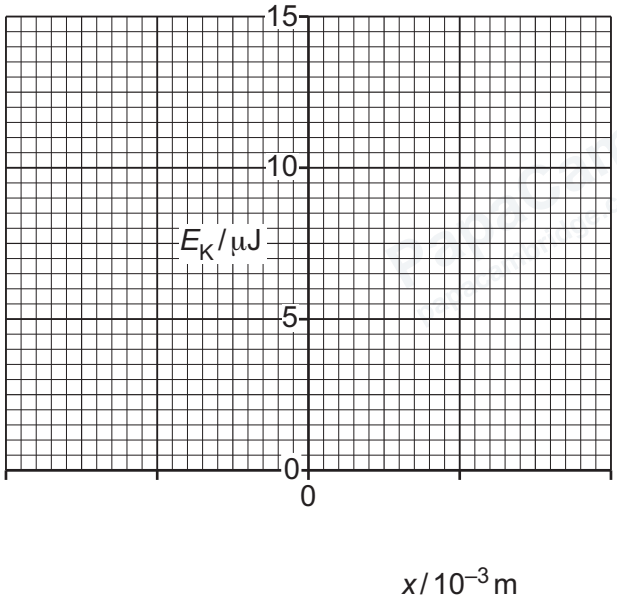


Figure 5.2

[3]

[Total: 8]

5 A metal ball is attached to the lower end of a spring, as shown in Figure 5.1.

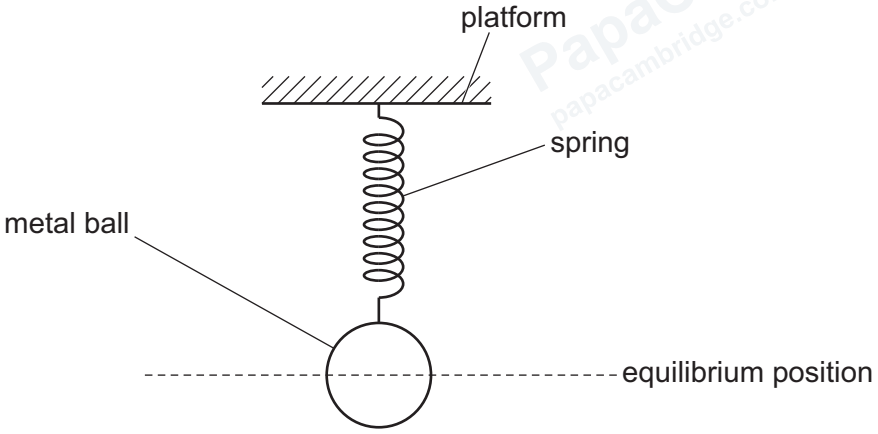


Figure 5.1

The upper end of the spring is attached to a platform that is initially stationary.

(a) The ball is pulled down by a small distance and released so that it oscillates with simple harmonic motion. The oscillations have an amplitude of 2.5 cm and a maximum speed of 4.4 cm s^{-1}.

(i) On Figure 5.2, sketch the variation of the velocity v of the ball with the displacement x from its equilibrium position as the ball undergoes one full oscillation.

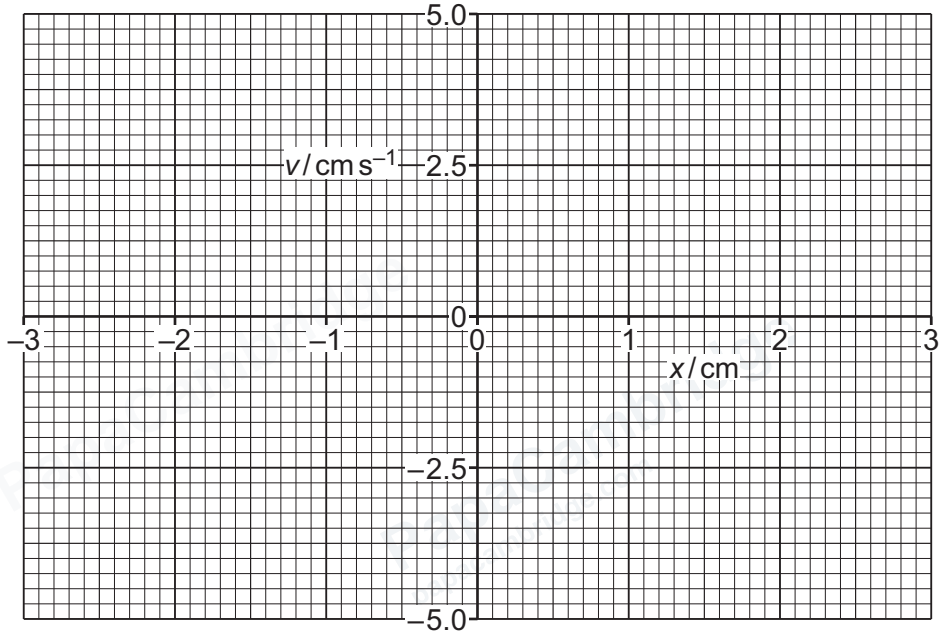


Figure 5.2

[3]

(ii) Show that the frequency of the oscillations is 0.28 Hz.

[2]

(b) The ball is returned to rest at its equilibrium position.

The platform is now moved up and down by a motor so that it oscillates vertically with simple harmonic motion of constant amplitude but variable frequency. The oscillating platform causes the ball to oscillate.

(i) State what is meant by resonance.

.....

.....

..... [2]

(ii) The frequency f of the oscillations of the platform slowly increases from 0 to 0.40 Hz.

On Figure 5.3, sketch the variation with f of the amplitude of the oscillations of the **ball**.

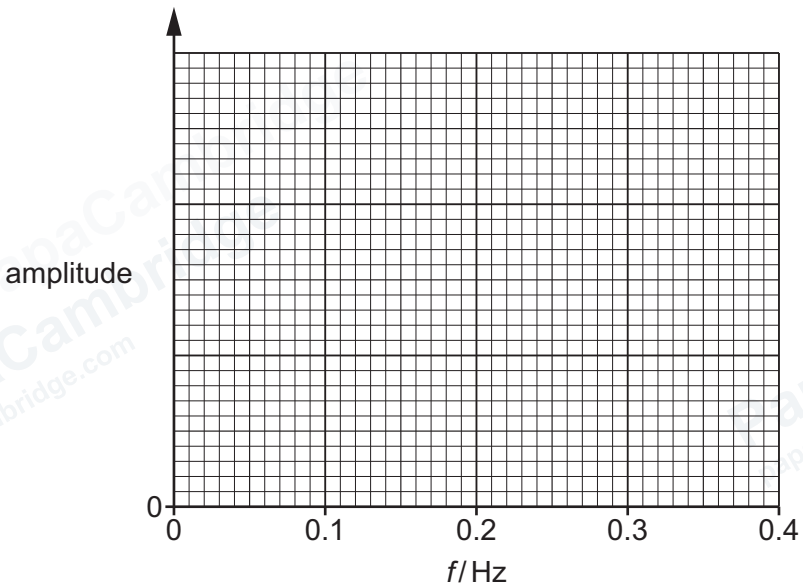


Figure 5.3

[2]

[Total: 9]